Hands-On Lab

Create an Elastic Kubernetes Service (EKS) Cluster on AWS

Test this lab if you're interested in working with AWS Kubernetes services and exploring container orchestration using Amazon EKS. Ideal for Kubernetes Admin/ Architect/ Developer, Cloud Engineers, DevOps Engineers, and Solutions Architects.

Scenario:

Imagine a fintech company that has developed a microservices-based application for real-time financial analytics. The application needs to be highly available, scalable, and secure to handle large volumes of data and traffic. The company chooses AWS EKS for its Kubernetes needs due to AWS's reliability, scalability, and robust ecosystem.

Description:

This guide explains the process of creating and managing a production-ready AWS Elastic Kubernetes Service (EKS) cluster using both GUI (AWS Console) and CLI (AWS CLI & EKSCTL). It is rooted in a practical scenario involving a fintech company aiming to optimize a real-time financial analytics application running on top of Managed Kubernetes on AWS i.e. EKS.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

K21Academy
Learn Cloud, Data & AI from Experts

Contents

1. Introduction....4
1.1. EKS Deployment Options....4
1.2. EKS Installation Options....5
2. Documentation....6
3. Prerequisite....7
4. Cost to Perform This Lab....8
5. GUI Method: Set up EKS Cluster using AWS Console....9
5.1. Log in to the AWS Portal....9
5.2. Create EKS Cluster Master Node (Control Plane)....9
5.3. Creating an EKS role....12
5.4. Back to Cluster Configuration page....15
5.5. Verify the status of the Cluster....21
6. Create EKS Cluster Worker Nodes (Data plane)....22
6.1. Set up Worker Nodes in EKS....22
6.2. Verify Worker Nodes in EKS....30
7. Connect & Manage the Amazon EKS cluster....32
7.1. Method 1: Connect & Manage EKS Cluster on AWS Cloud Shell (Easy & Recommended)....33
7.2. Method 2: Connect & Manage EKS Cluster on Client Machine [Optional]....35
7.3. [Non-Amazon Linux AMI] Install AWS CLI....40
7.4. Configure AWS CLI....41
7.5. Install Kubectl on an Amazon Linux Machine....43
7.6. Set Cluster Context & Verify the Status of the Cluster on the Client Machine....43
8. Upgrade Amazon EKS Cluster....46
9. CLI Method: Setup EKS Cluster (Master & Worker Nodes) Using EKSCTL....52
9.1. Install EKSCTL on an Amazon Linux Machine....52
9.2. Create EKS Master & Worker Nodes....53
9.3. Verify Master & Worker Node from Console....55
9.4. Set up the kubectl configuration....56
9.5. Delete EKS Cluster using EKSCTL....58
10. Share your learnings....59
10.1. Share Hands-On Lab on LinkedIn to Build Authority & Job Opportunities....60
10.2. Share the win on the Community via Progress Diary....62
11. Delete EKS Cluster from the Console....64
11.1. Delete Worker Nodes....64
11.2. Delete Master Node....66
11.3. Delete IAM Roles....66
11.4. Stop/Terminate EC2 instance....67
12. Troubleshooting....70
12.1. Error on kubectl 'Unable to connect to the server: dial tcp: lookup *. eks.amazonaws.com on IP:53: no such host....70

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

12.1.1. Issue..... 70
12.1.2. Cause:..... 70
12.1.3. Fix:..... 70
12.2. CloudFormation Stack Already Exists..... 71
12.2.1. Issue:..... 71
12.2.2. Cause:..... 71
12.2.3. Fix:..... 71
12.3. Not able to create the Cluster..... 71
12.3.1. Issue:..... 71
12.3.2. Cause:..... 72
12.3.3. Fix:..... 72
12.4. Not able to delete the Cluster..... 72
12.4.1. Error:..... 72
12.4.2. Cause:..... 73
12.4.3. Fix:..... 73
12.5. Couldn't get the current server API group list..... 73
12.5.1. Error:..... 73
12.5.2. Cause:..... 73
12.5.3. Fix:..... 73
12.6. Cannot create cluster 'k21eks01' because EKS does not support creating control plane instances in us-east-1e, the targeted availability zone..... 74
12.6.1. Issue..... 74
12.6.2. Reason:..... 74
12.6.3. Fix:..... 75
12.7. EKS Node Group Creation Failure..... 75
12.7.1. Issue:..... 75
12.7.2. Cause:..... 77
12.7.3. Fix:..... 78
13. Summary..... 80

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

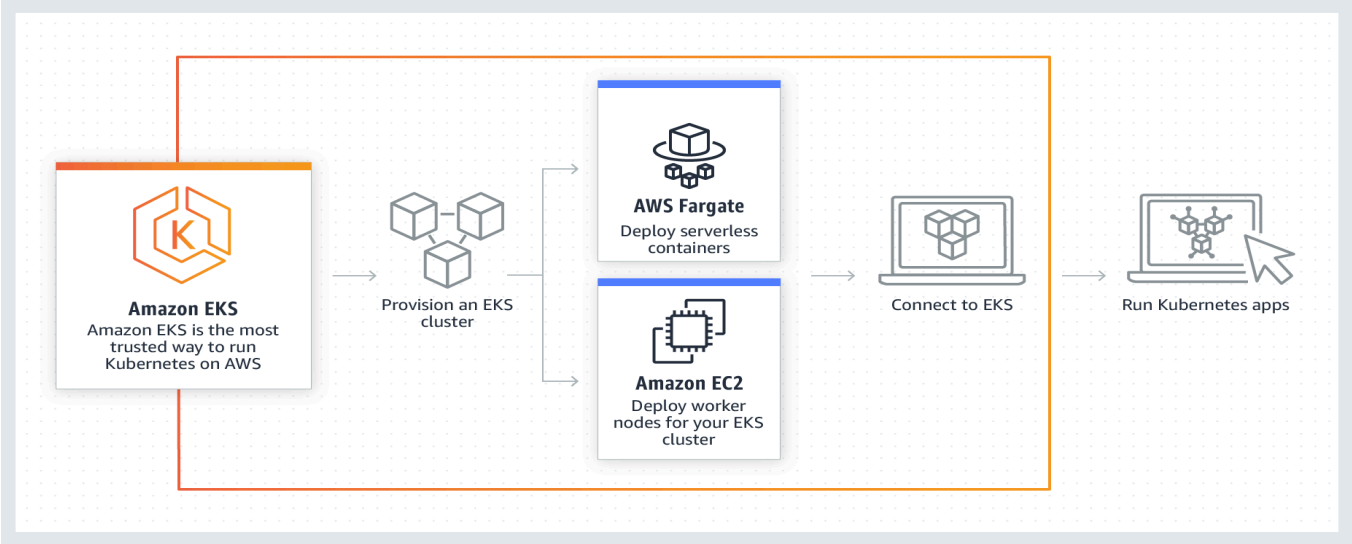
1. Introduction

EKS (Elastic Container Service for Kubernetes) is a managed Kubernetes service that allows you to run Kubernetes on AWS without the hassle of managing the Kubernetes control plane (Master Node).

The Kubernetes control plane plays a crucial role in a Kubernetes deployment as it is responsible for how Kubernetes communicates with your cluster — starting and stopping new containers, scheduling containers, performing health checks, and many more management tasks.

The big benefit of EKS, and other similar hosted Kubernetes services, is taking away the operational burden involved in running this control plane.

You deploy cluster worker nodes using AWS Console or CLI, and EKS will provision, scale and manage the Kubernetes control plane for you to ensure high availability, security and scalability.



1.1. EKS Deployment Options

Here are EKS deployment options, in this lab we are using the Amazon EKS Deployment option. To know more about these deployment options, refer to https://docs.aws.amazon.com/eks/latest/userguide/eks-deployment-options.html

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

Feature	Amazon EKS	Amazon EKS on AWS Outposts	Amazon EKS Anywhere	Amazon EKS Distro
Hardware	AWS-supplied	AWS-supplied	Supplied by you	Supplied by you
Deployment location	AWS cloud	Your data center	Your data center	Your data center
Kubernetes control plane location	AWS cloud	AWS cloud or your data center	Your data center	Your data center
Kubernetes data plane location	AWS cloud	Your data center	Your data center	Your data center
Support	AWS support	AWS support	AWS support	OSS community support

1.2. EKS Installation Options

1. There are two EKS Installation Options available
   - Using AWS Console (GUI)
   - Using EKSCTL (CLI)

   We will explore both options in this guide.

K21Academy

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

2. Documentation

1. Create EKS Cluster using Console or CLI
   https://docs.aws.amazon.com/eks/latest/userguide/create-cluster.html

2. What is EKS
   https://docs.aws.amazon.com/eks/latest/userguide/what-is-eks.html

3. Installing the AWS CLI
   https://docs.aws.amazon.com/cli/latest/userguide/cli-chap-install.html

4. Installing Kubectl
   https://docs.aws.amazon.com/eks/latest/userguide/install-kubectl.html

5. Installing the eksctl
   https://docs.aws.amazon.com/eks/latest/userguide/eksctl.html

6. AWS EC2:
   https://docs.aws.amazon.com/ec2/index.html

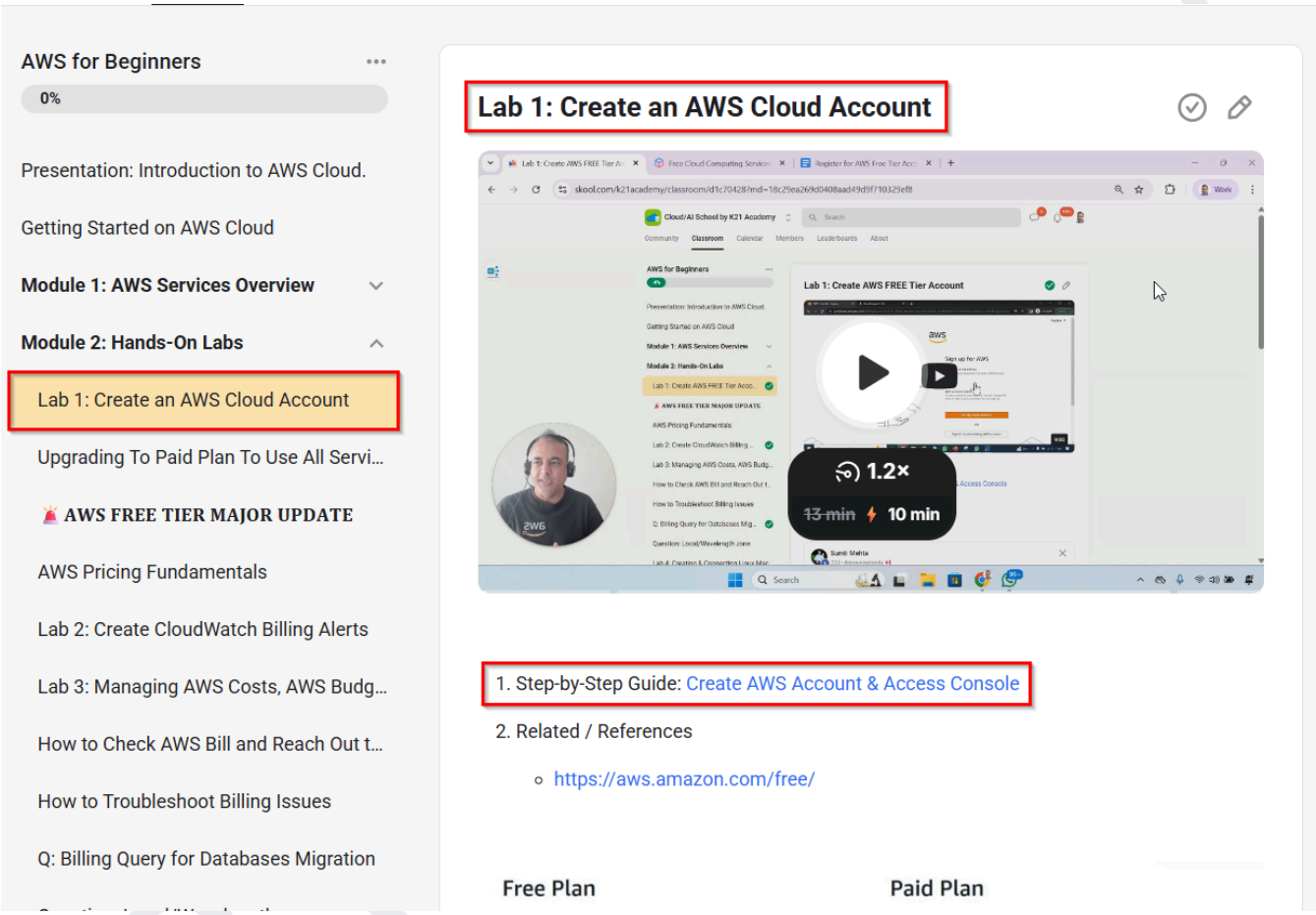
7. IAM Roles:
   https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles.html

K21Academy

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

3. PREREQUISITE

1. AWS account: If you don't have an AWS account, then please check Activity ***Guide: Create AWS Cloud Account Here***.



Note: If you decide that you no longer need an instance, stop/terminate the EKS Cluster after using it. If you keep that cluster long, you will be highly charged for that usage.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

4. Cost to Perform This Lab

The estimated cost for this lab performed in the AWS free tier is $0, as long as usage stays within the free tier limits.

If you exceed the free tier limits, charges may be incurred. Please refer to the pricing details using the links below:

📌 **Important Note:** Any charges will be deducted from the free credits you received when you created your AWS account. As long as you have remaining credits, you won't need to pay out of pocket for these charges. (if you have a free tier with credits)

AWS Free Tier Credits Breakdown

$100 Available Immediately: When you sign up for an AWS Free Tier account, you receive $100 in credits that are available for use right away.

$100 After Completing Specific Activities: After fulfilling certain conditions, such as completing AWS labs or other specific tasks, you'll receive an additional $100 in credits.

K21Academy

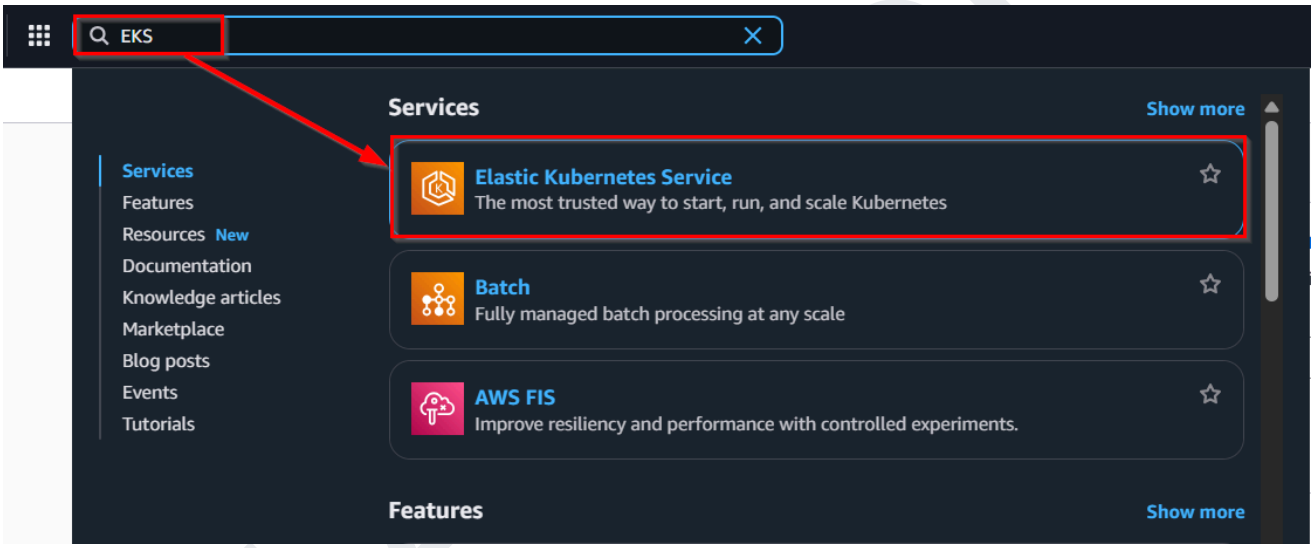
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

5. GUI Method: Set up EKS Cluster using AWS Console

Setting up an Amazon EKS (Elastic Kubernetes Service) Cluster involves a series of steps, and the GUI method provides a graphical interface to simplify the process. This method is great for those who prefer a visual approach or are new to the command line.

5.1. Log in to the AWS Portal

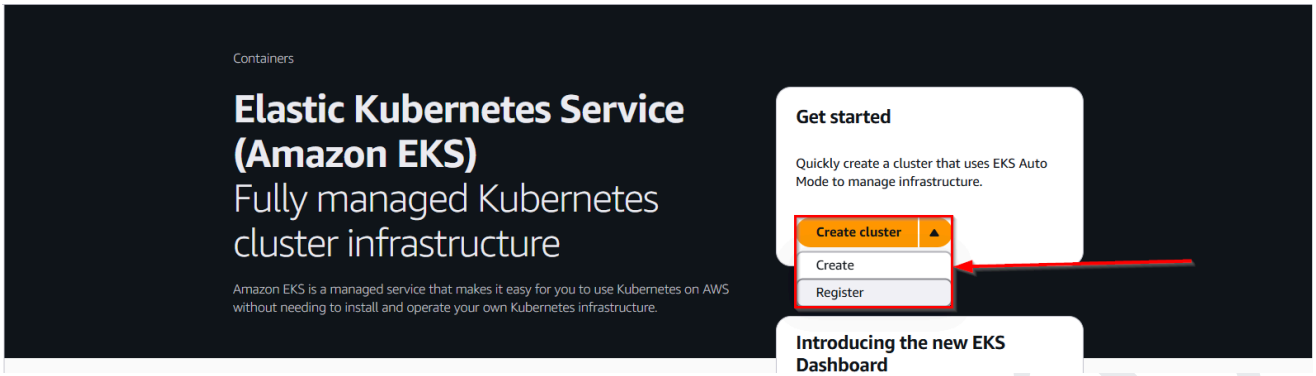
Log in to the AWS Portal using your account credentials. In the search bar, type in **EKS** and select **Elastic Kubernetes Service**.



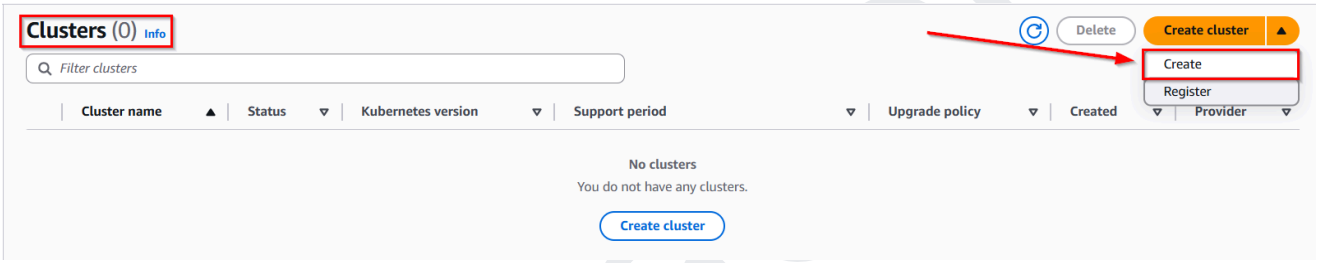
5.2. Create EKS Cluster Master Node (Control Plane)

1. **In AWS EKS**, we have two options: **"Create Cluster"** and **"Register Cluster"**. "Create Cluster" is a management task that you perform to set up a new, independent EKS cluster, while "Register Cluster" is a management task used to add an existing Kubernetes cluster to AWS management.
   https://docs.aws.amazon.com/cli/latest/reference/eks/register-cluster.html

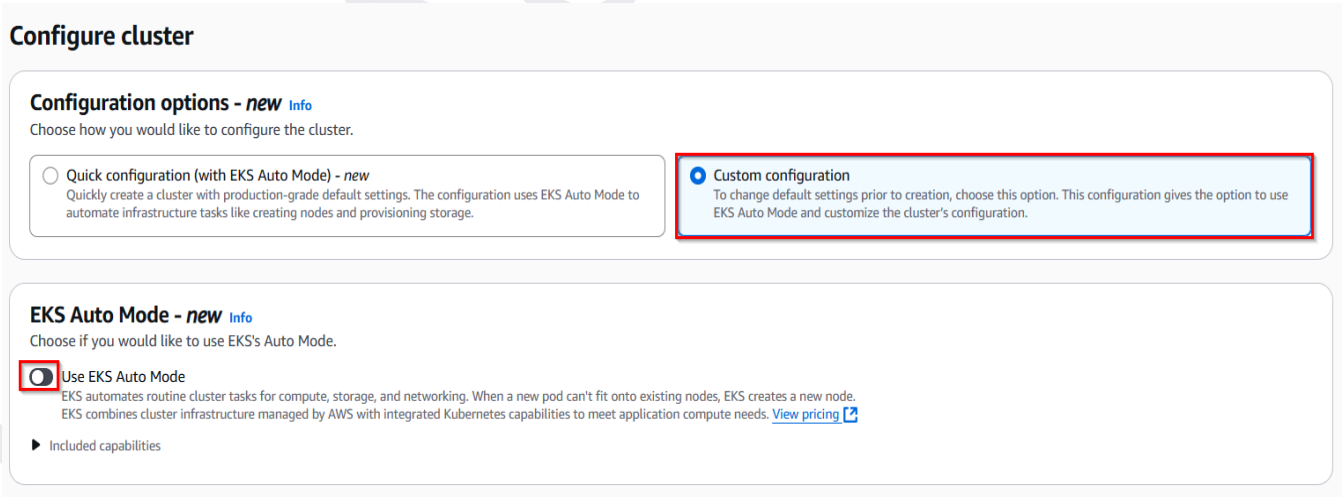
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



2. Here we are creating a new cluster. So, click on **Create.**

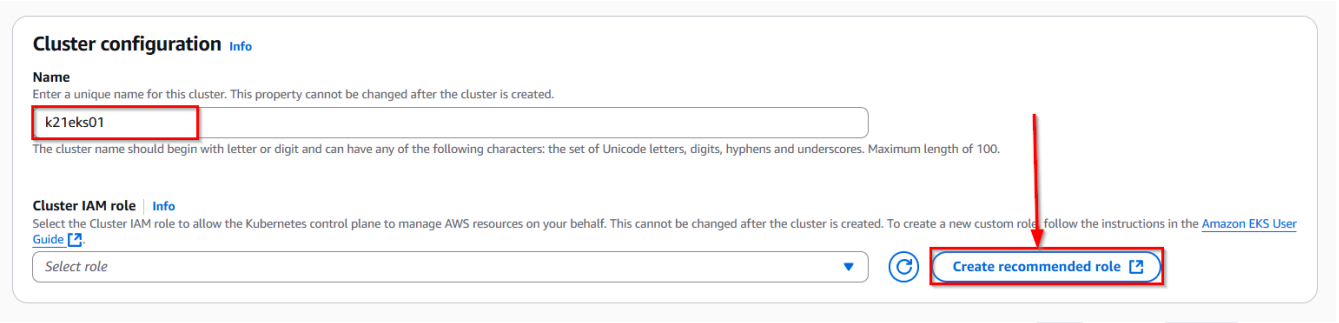


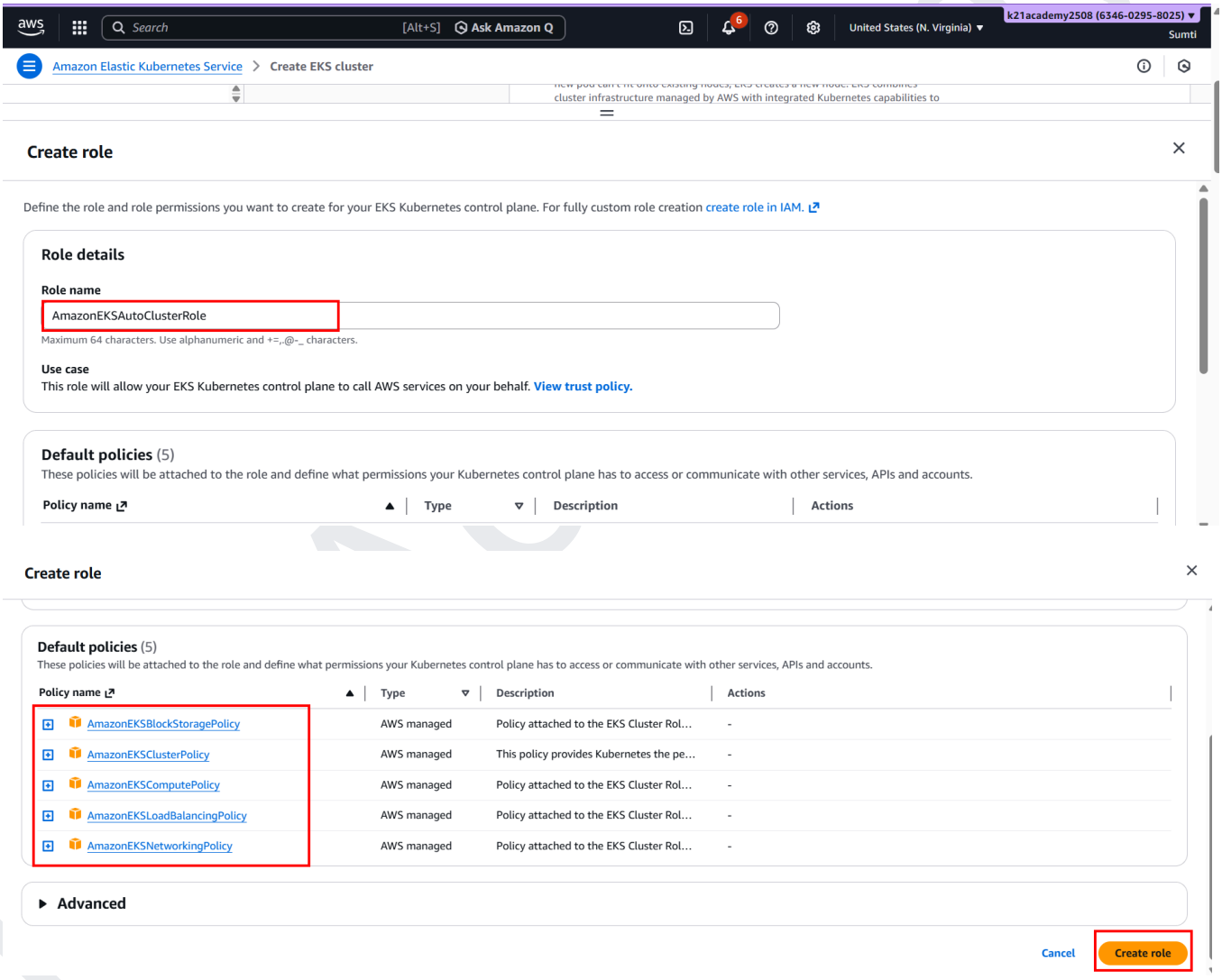
3. In Cluster Configuration settings, provide the cluster name as **k21eks01** and in the cluster services role we don't have any role so we are **creating a new role here.**



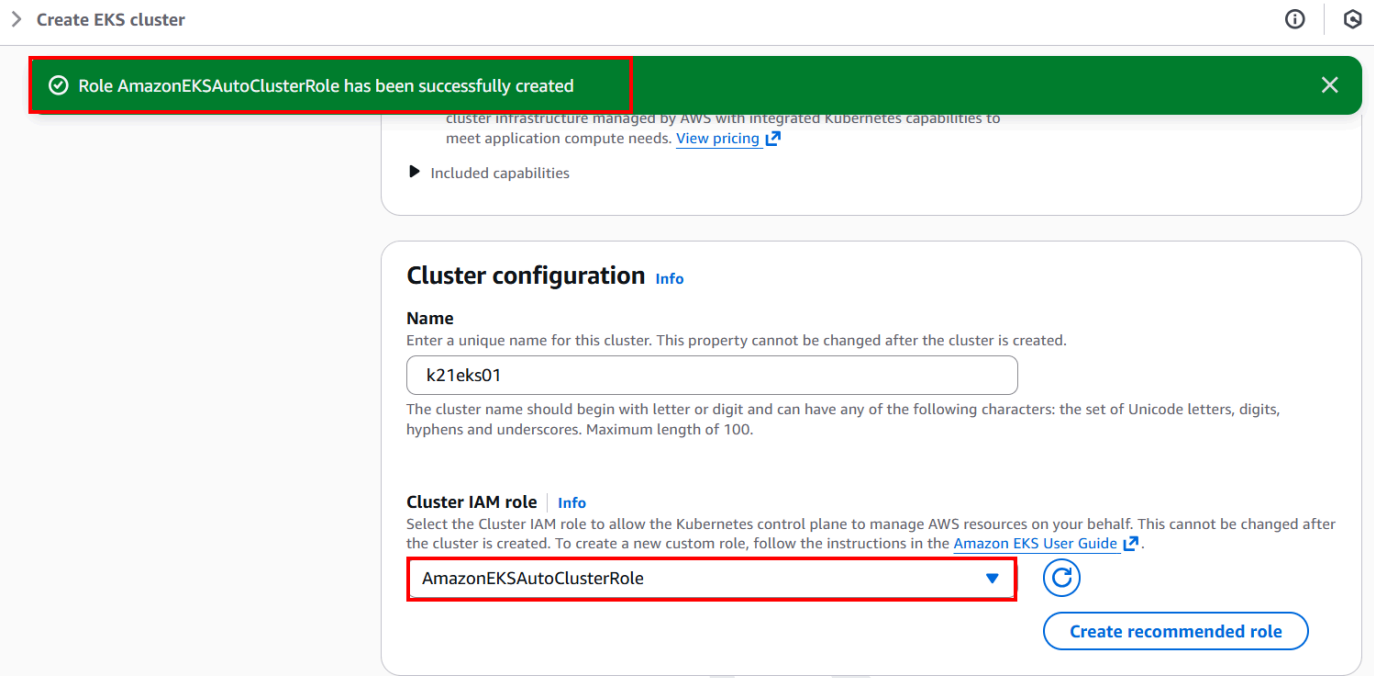
EKS Auto Mode automates routine cluster tasks like compute, storage, and networking. When a new pod can't fit on existing nodes, EKS automatically creates a new node to accommodate it. This mode integrates AWS-managed infrastructure with Kubernetes, providing a streamlined experience for scaling applications.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



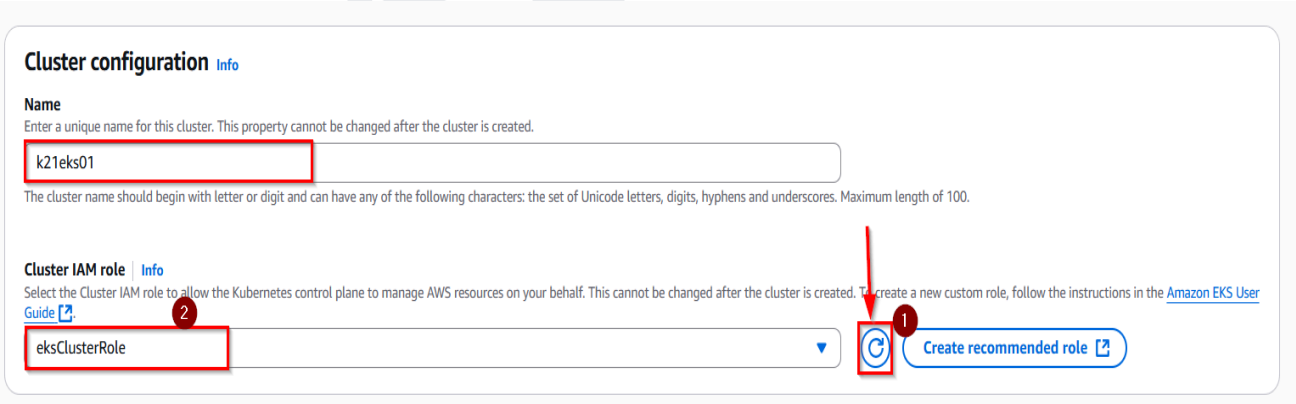


Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

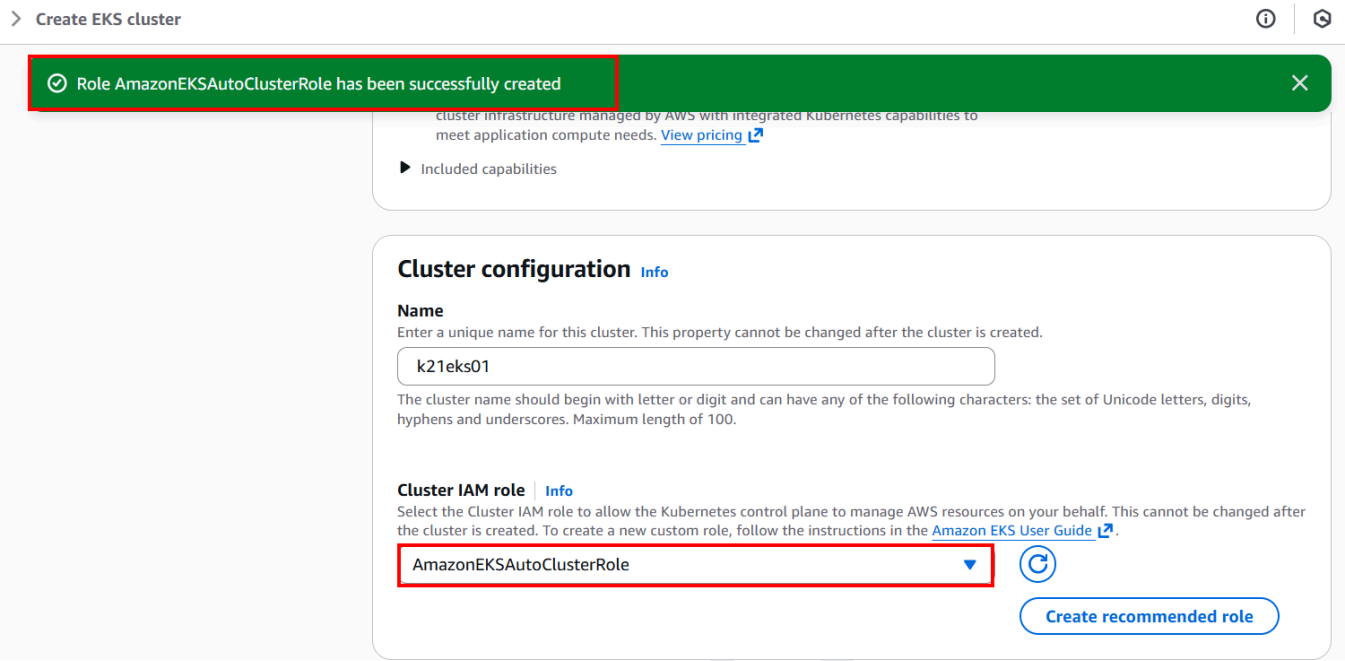


5.3. Back to Cluster Configuration page

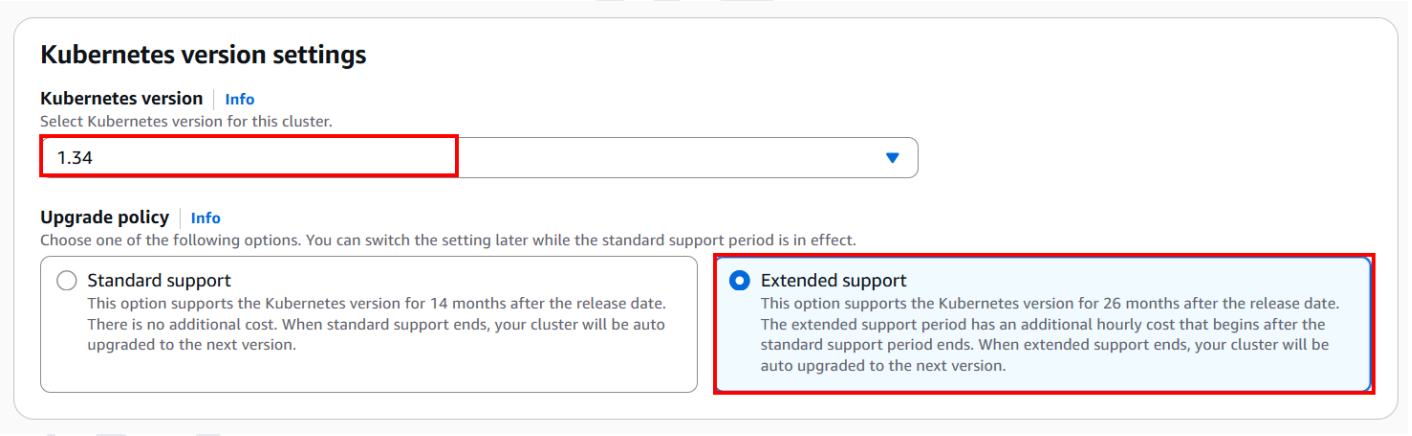
1. Click on the **Refresh button**, then select the recently **created EKS role** from the dropdown and click on **Next.**



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



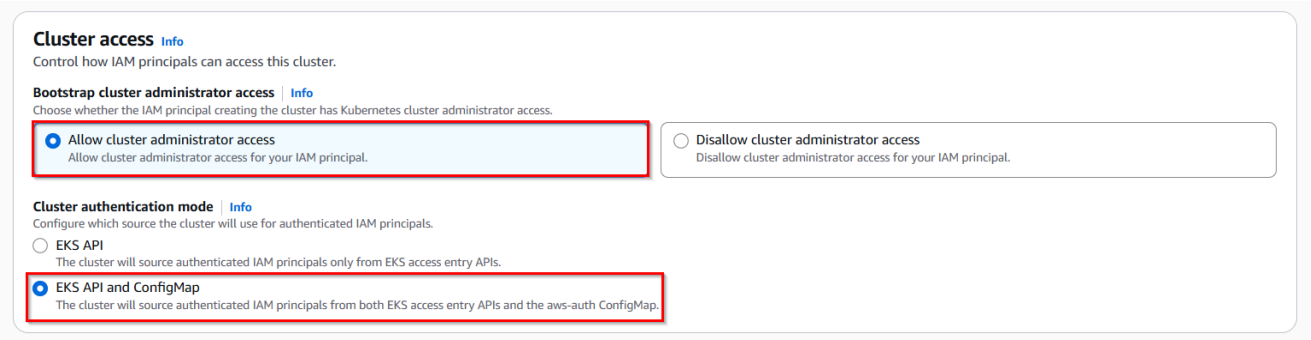
2. Select the Kubernetes Version to 1.34

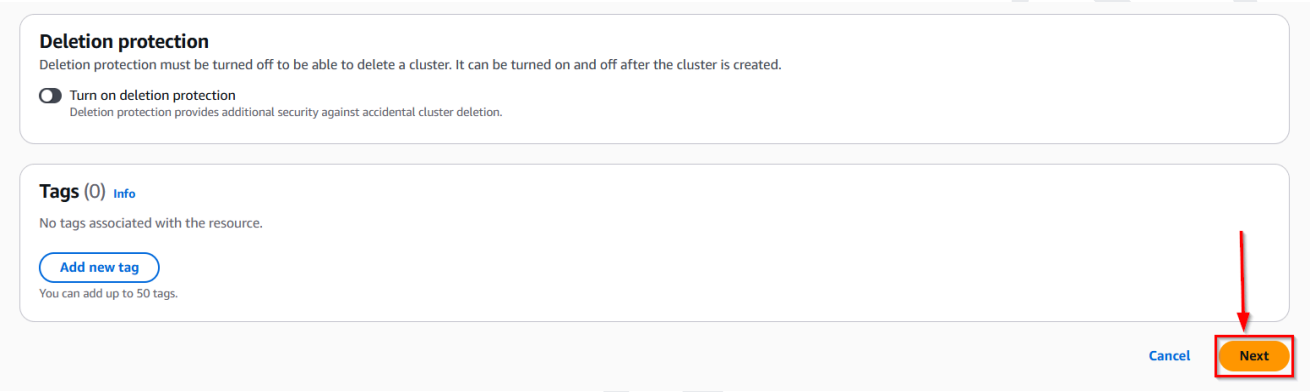


Note: As Kubernetes versions are updated rapidly, always select the **Kubernetes (n-1) version.** In this case, version 1.34 is the latest version, so I was selecting version 1.34. In your case, if it's the highest available, please select the appropriate version.

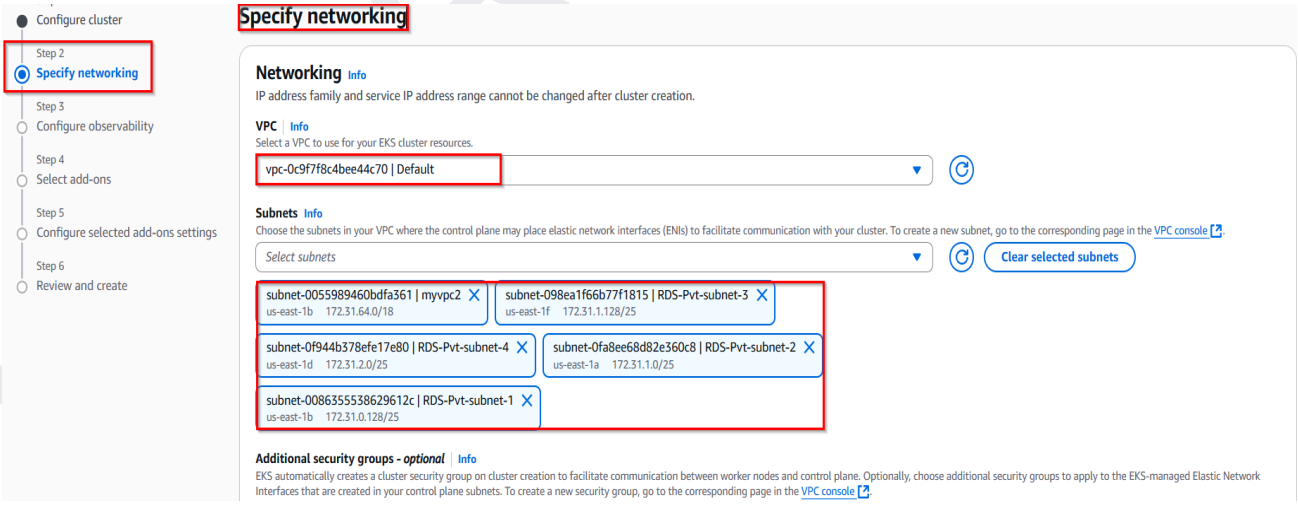
3. In the **Cluster Access** section, do not make any changes. Leave the settings as default, then click **Next**.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

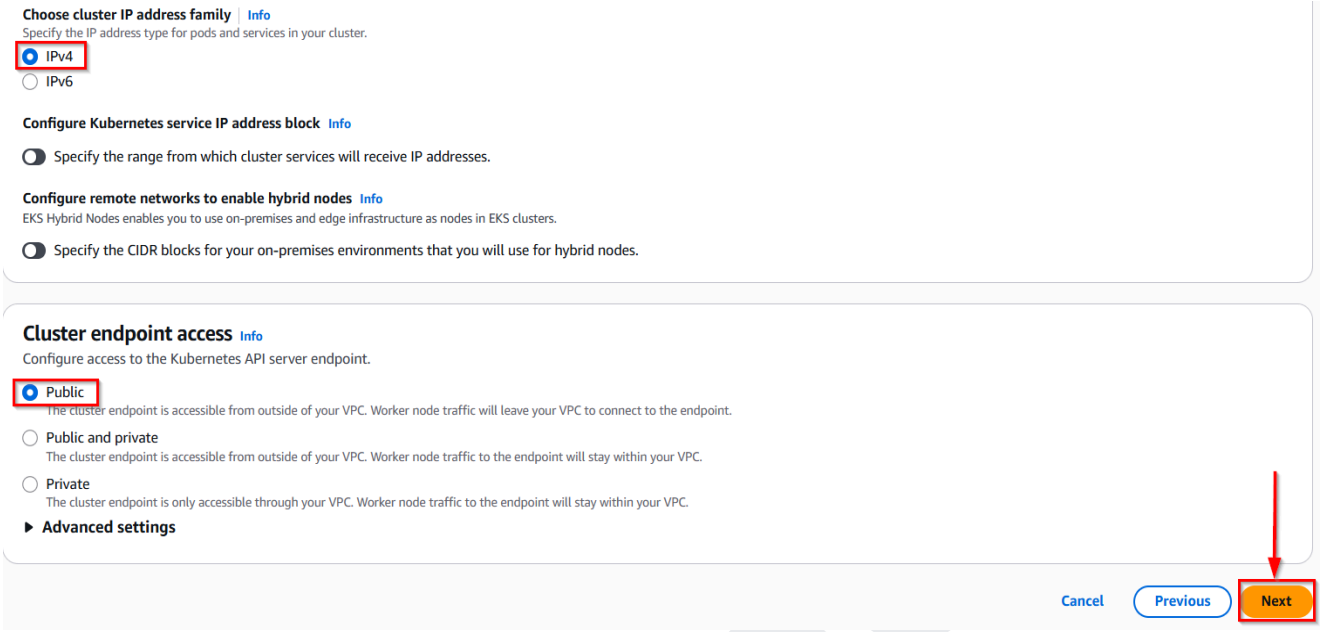




4. In the **Step 2: Specify Networking** section, **do not make any changes to the VPC, subnet, or security groups; leave them as the default.** In the **Cluster Endpoint Access** section, select **Public** and click on **Next.**

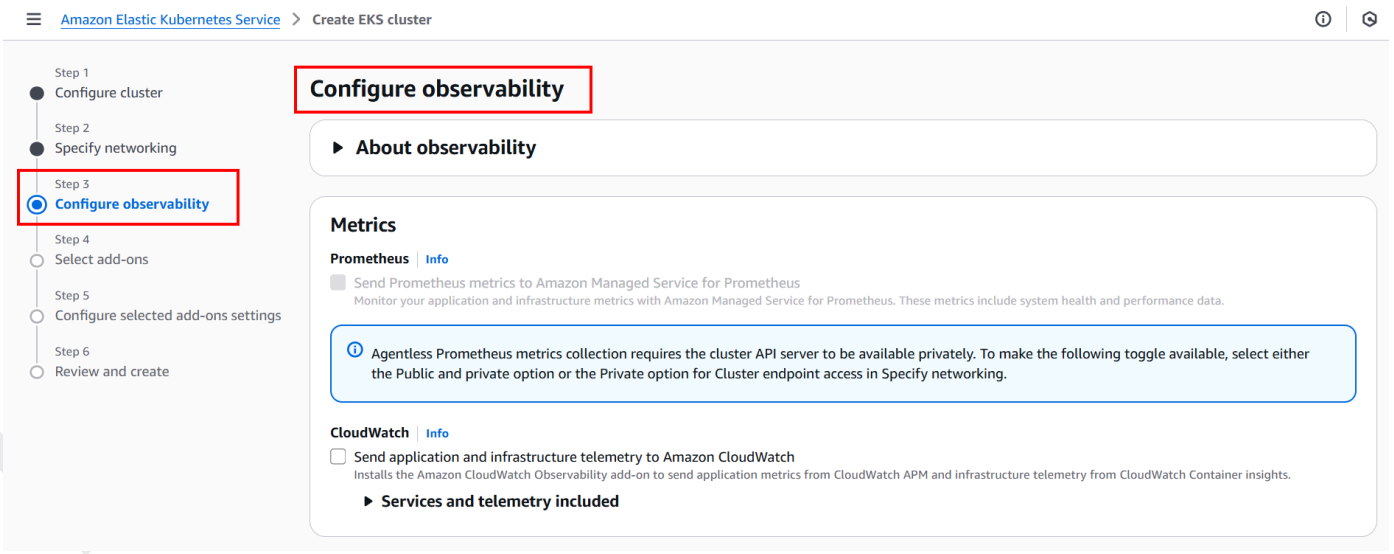


Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

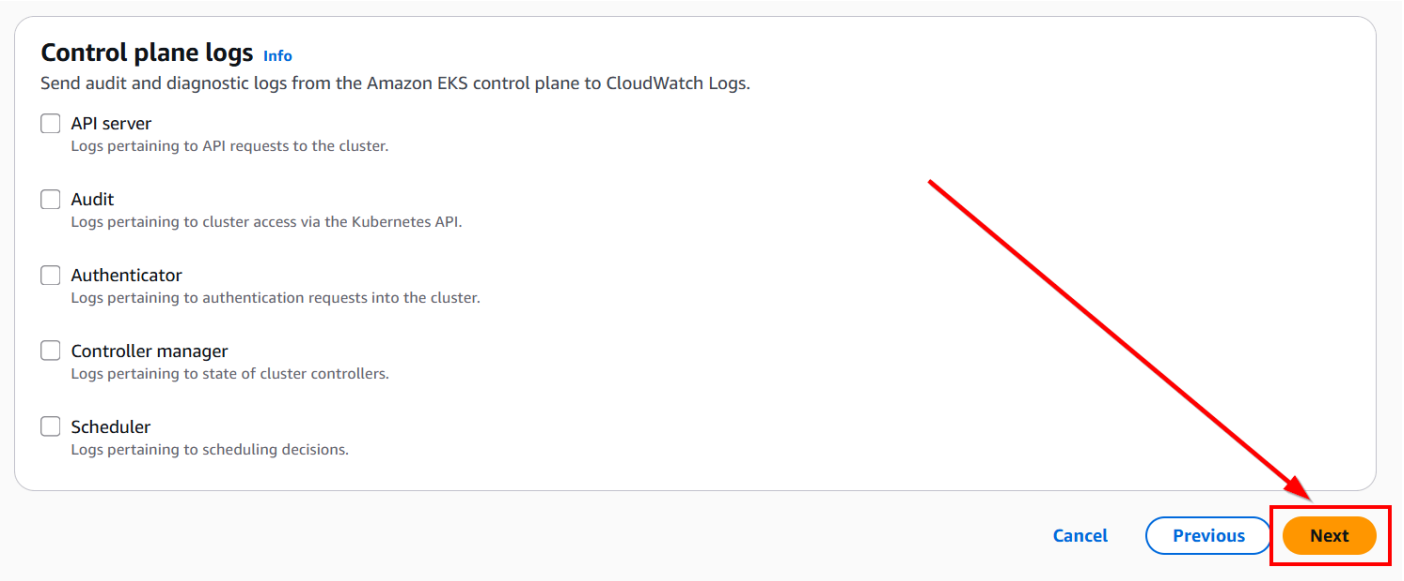


Note: Make sure that you selected **the Public** Option; you will encounter an error while connecting to the EKS cluster on the Client Machine

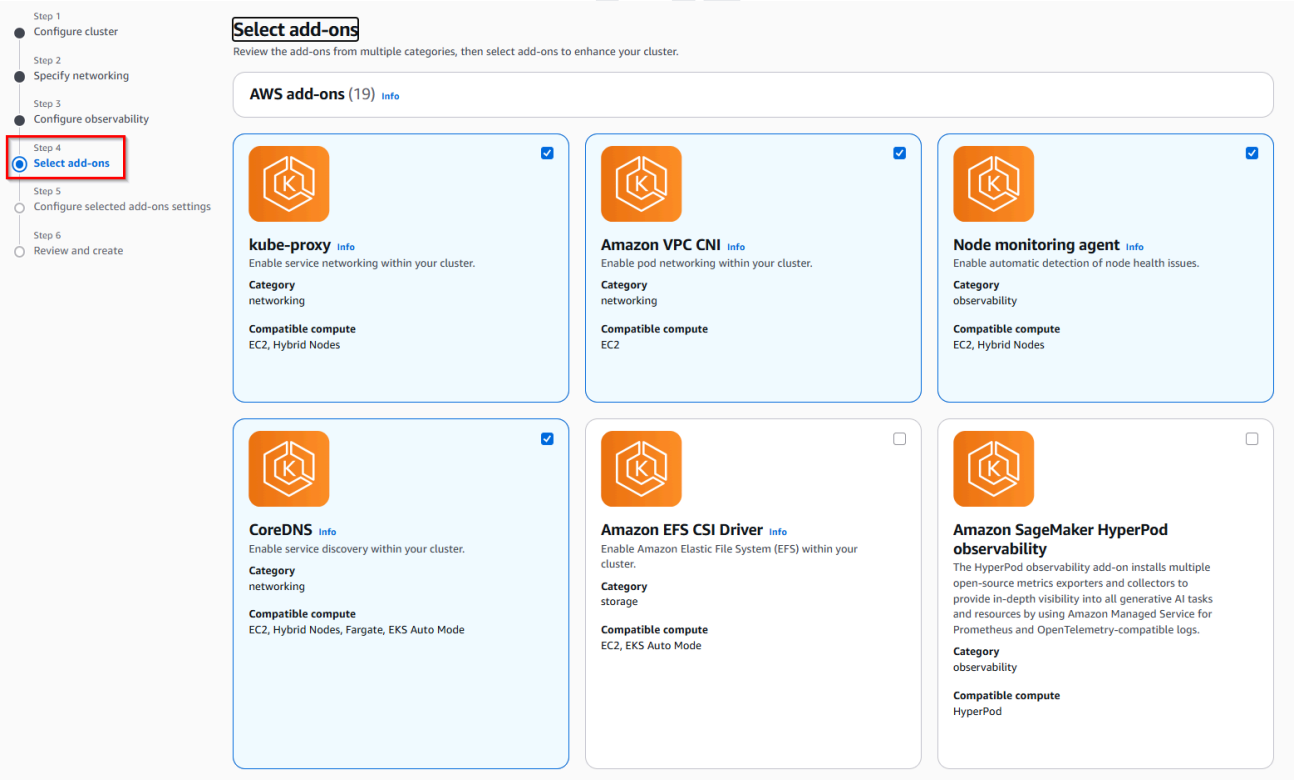
5. In the Step 3 **Configure Observability section**, do not make any changes, **leave it as the default**, and click on **Next.**



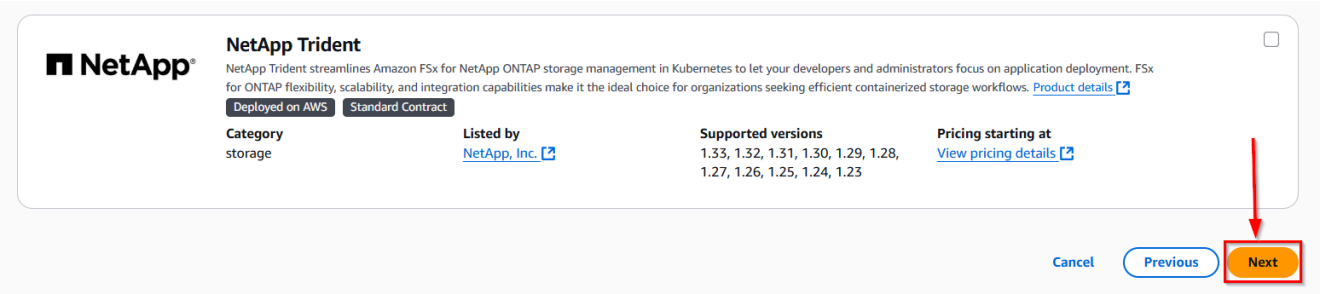
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



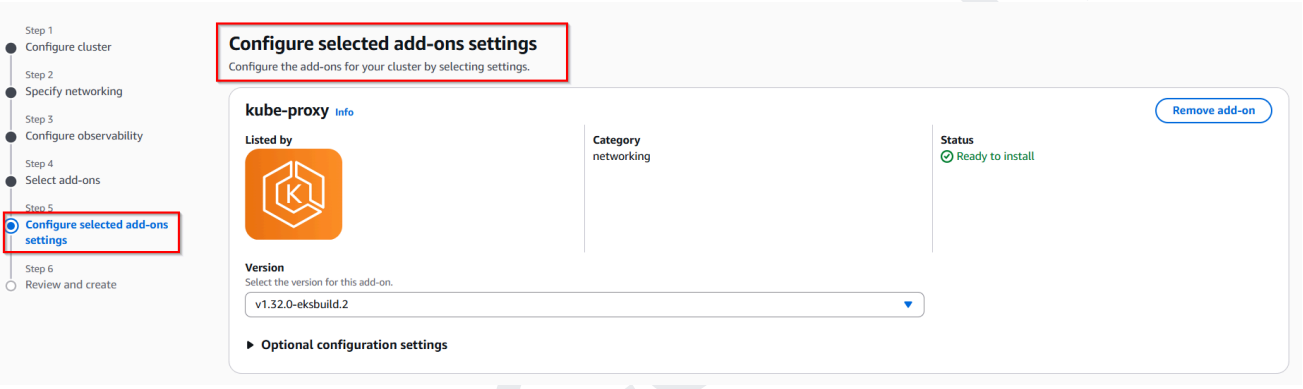
6. In Step 4, **Select add-ons section**, do not make any changes, **leave it as default**, and click on **Next.**

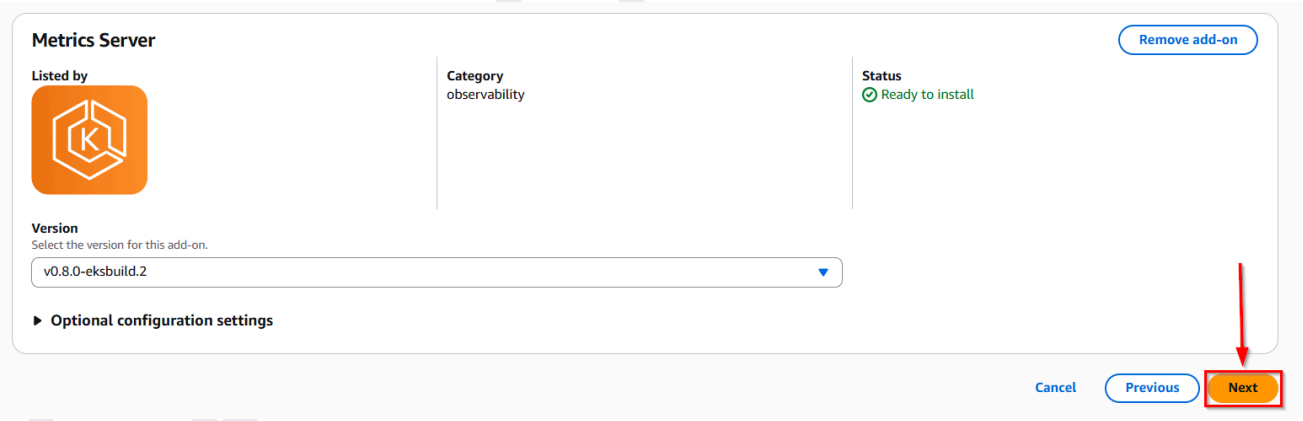


Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



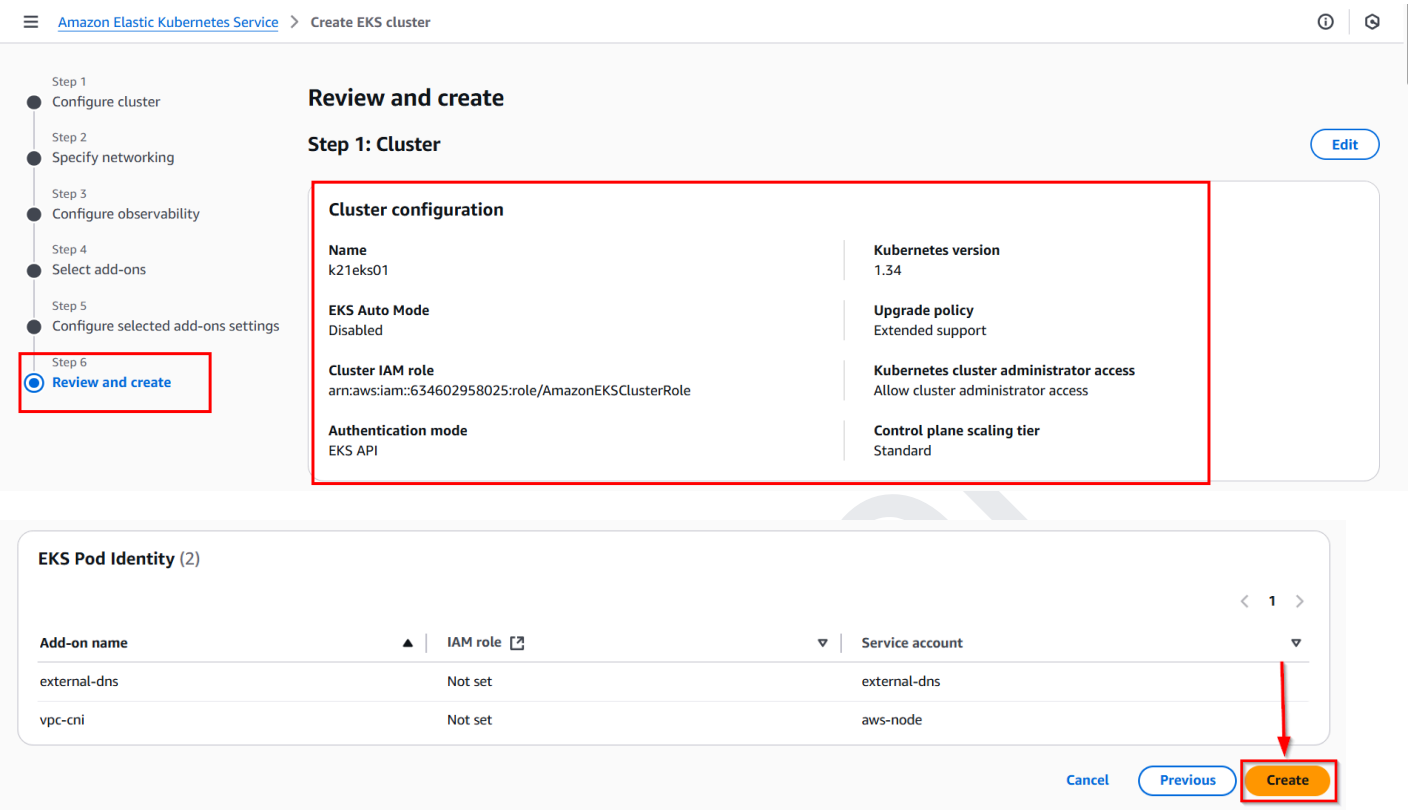
7. In Step 4, **Configure Selected Addons settings section**, do not make any changes, **leave it as default**, and click on **Next.**





8. In Step 6, in the **Review and Create section**, review all the settings and click on **Create.**

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



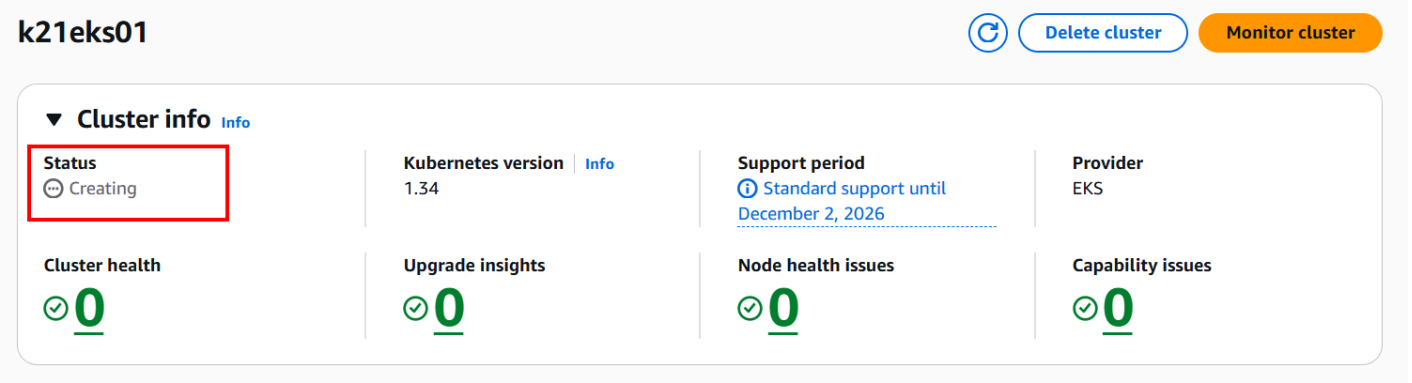
Note 1: If you encounter any issues related to **not creating the Cluster,** please refer to the **troubleshooting section**.

Note 2: If you encounter any issues related to the **Cannot create cluster 'k21eks01' because EKS does not support creating control plane instances in us-east-1e, the targeted availability zone,** please refer to the **troubleshooting section**.

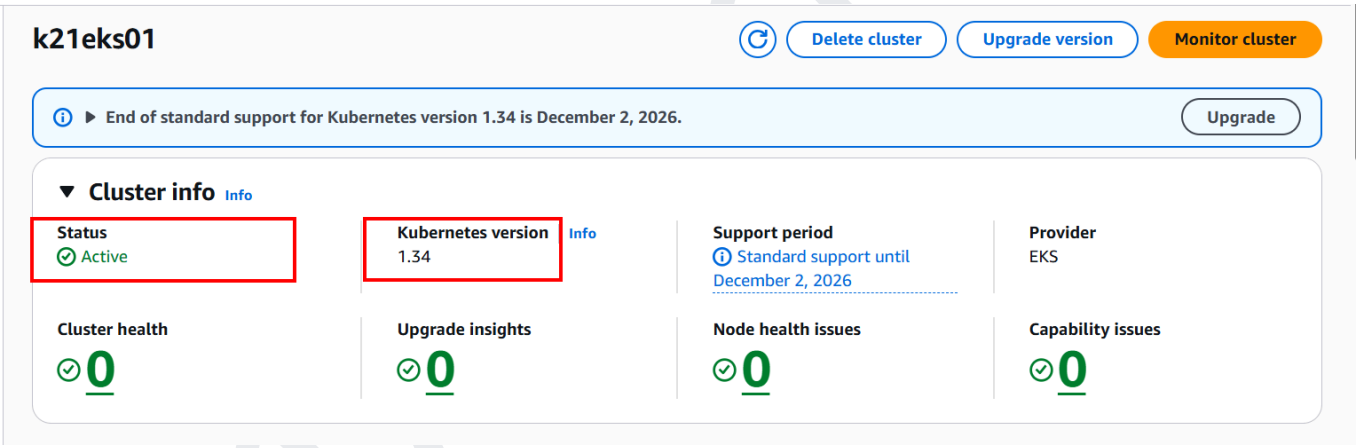
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

5.4. Verify the status of the Cluster

1. The cluster creation is in progress. It will take **15-30 minutes** for the cluster to come up.



2. Cluster is created with the name ***k21eks01*** and is showing as status Active.



Note: This step will create a master node on the Kubernetes cluster. In the next section, we will create Worker nodes.

Note: If you no longer need the cluster, stop or terminate the EKS cluster after usage. Keeping the cluster for an extended period will result in high charges.

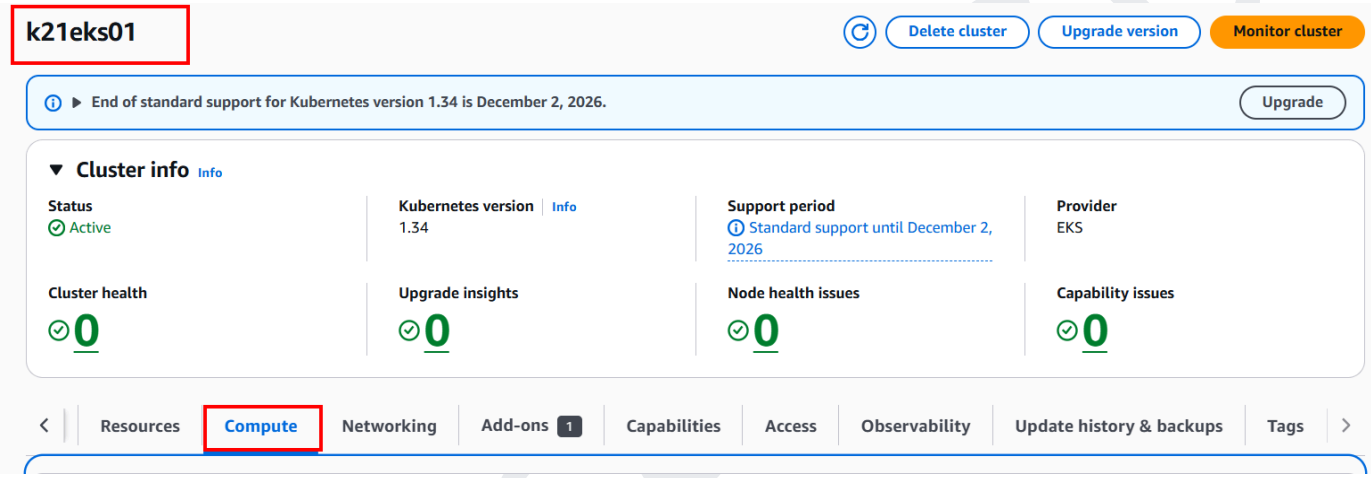
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

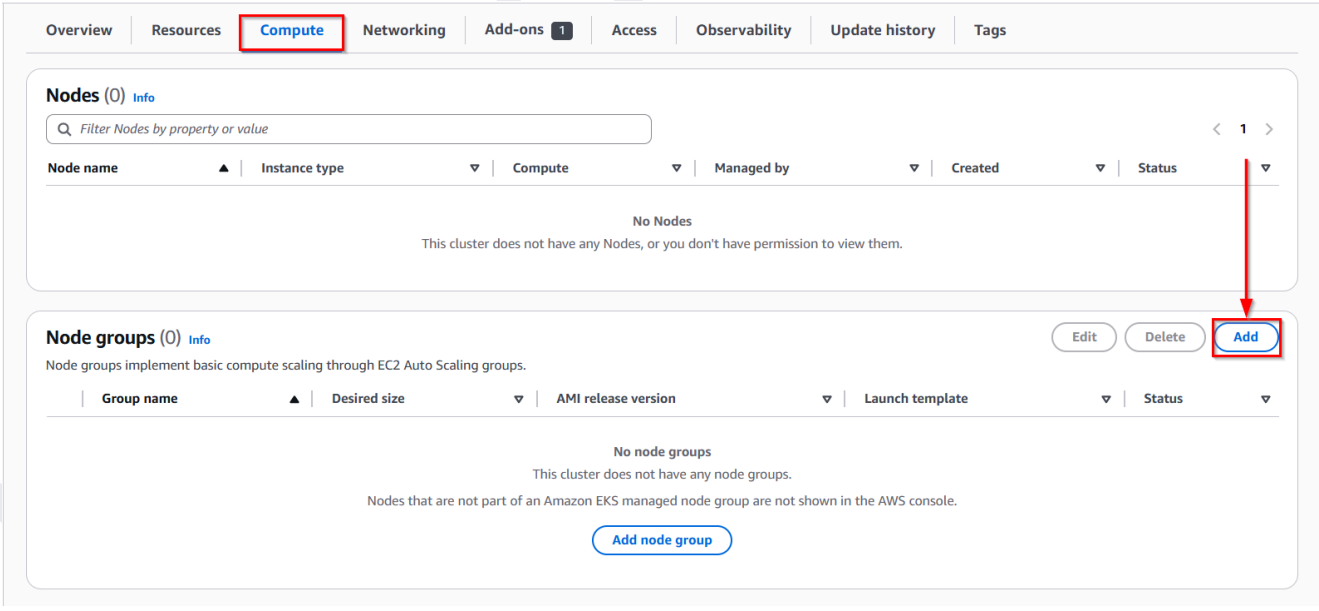
6. Create EKS Cluster Worker Nodes (Data plane)

We can configure worker nodes on EC2 instances (virtual machines) or Fargate (serverless). Here, we are going to configure a worker node on a virtual machine (EC2).

6.1. Set up Worker Nodes in EKS

1. On the cluster page, select the **Compute** tab, and then choose **Add Node Group.**





2. On the Configure node group page, fill out the parameters accordingly, and then choose Next.
   - **Name** – Enter a unique name for your managed node group.

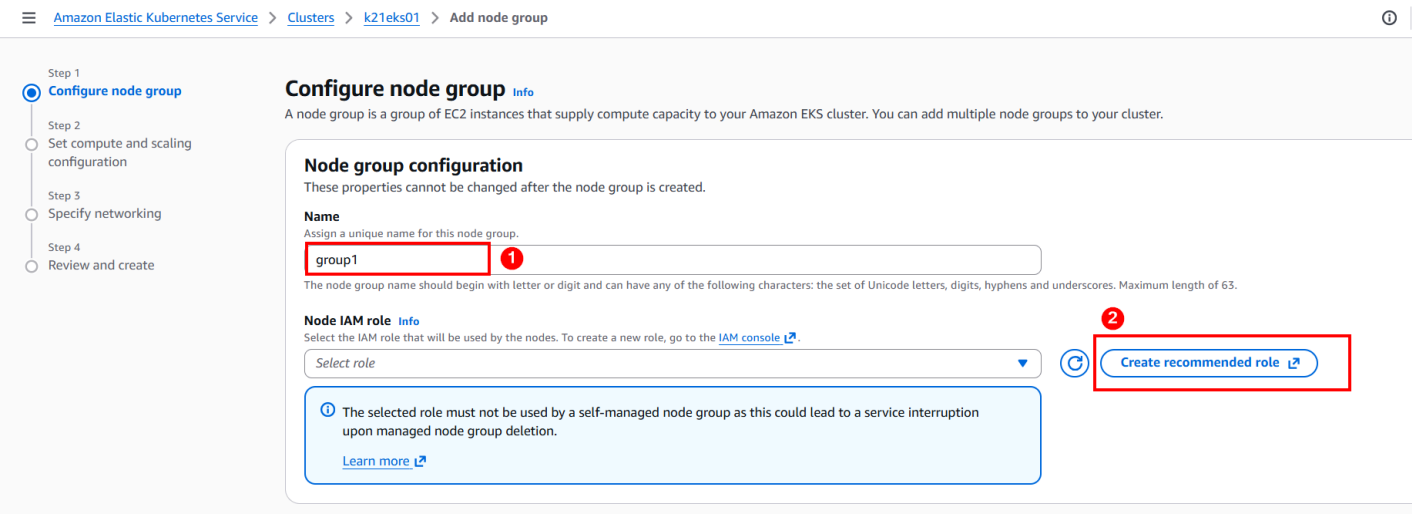
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

- **Node IAM role name** – Choose the node instance role to use with your node group.

  For more information, see Amazon EKS worker node IAM role.

To create your Amazon EKS worker node role in the IAM console

1. Open the IAM console at https://console.aws.amazon.com/iam/.
2. Choose **Roles**, then **Create role**.
3. Choose **EC2** from the list of **Common use cases** under **Choose a use case,** then choose **Next: Permissions**.
4. In the **Filter policies** box, enter `AmazonEKSWorkerNodePolicy`. Check the box to the left of **AmazonEKSWorkerNodePolicy.**
5. In the **Filter policies** box, enter `AmazonEKS_CNI_Policy`. Check the box to the left of **AmazonEKS_CNI_Policy.**
6. In the **Filter policies** box, enter `AmazonEC2ContainerRegistryReadOnly`. Check the box to the left of **AmazonEC2ContainerRegistryReadOnly.**
7. Choose **Next: Tags**.
8. (Optional) Add metadata to the role by attaching tags as key–value pairs. For more information about using tags in IAM, see Tagging IAM Entities in the *IAM User Guide*.
9. Choose **Next: Review**.
10. For **Role name**, enter a unique name for your role, such as `NodeInstanceRole`. For **Role description**, replace the current text with descriptive text such as `Amazon EKS - Node Group Role`, then choose **Create role**.

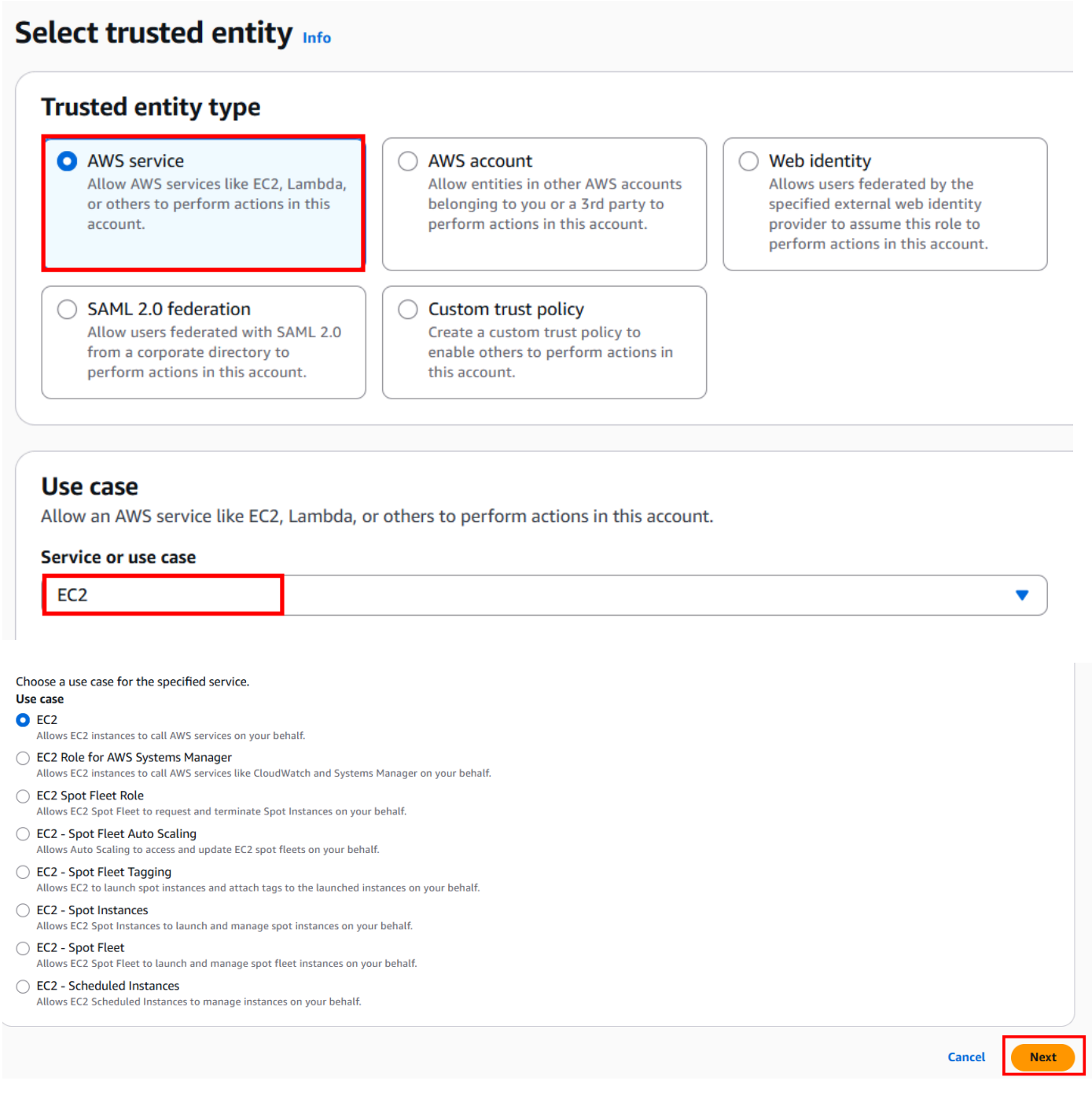


3. Create a New Role by following the steps below

   Click on: **Create role**

   Select **AWS services** as **EC2**: Click **Next.**

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



4. Under permission search for Amazon eks and Amazon EC2 container & select the following permissions:

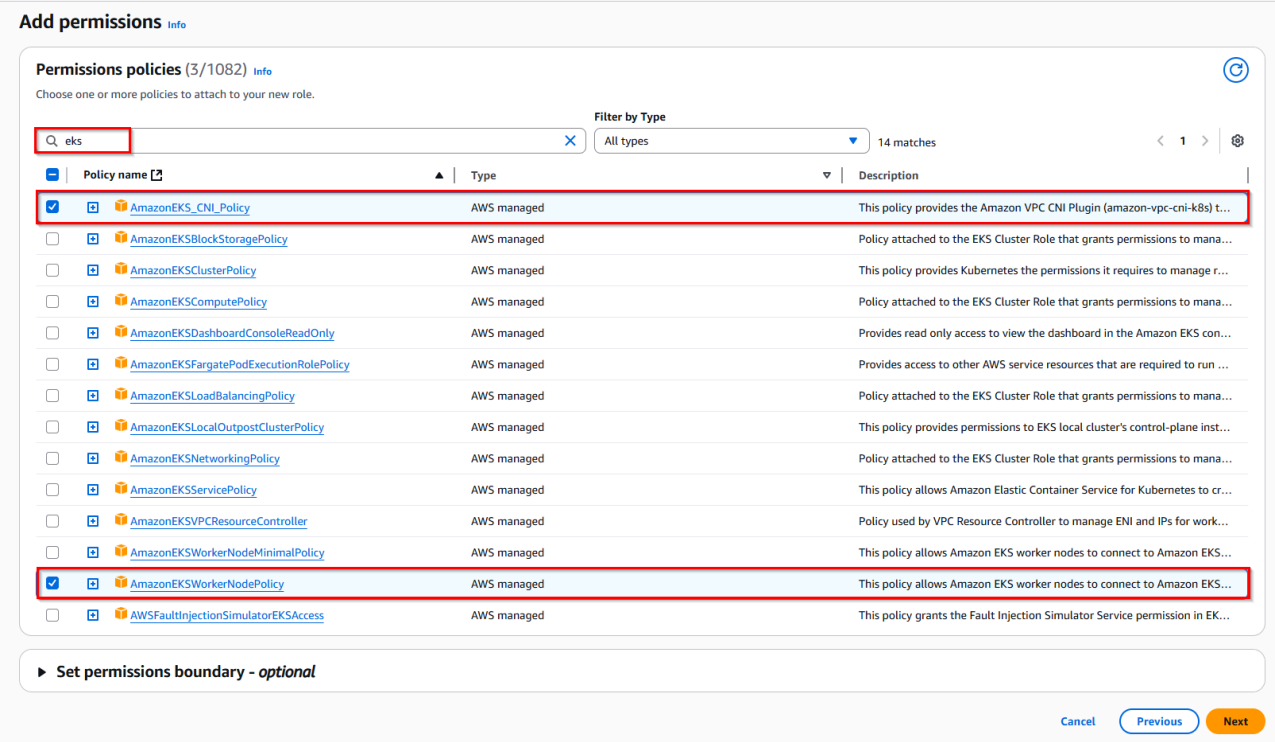
   AmazonEKSWorkerNodePolicy

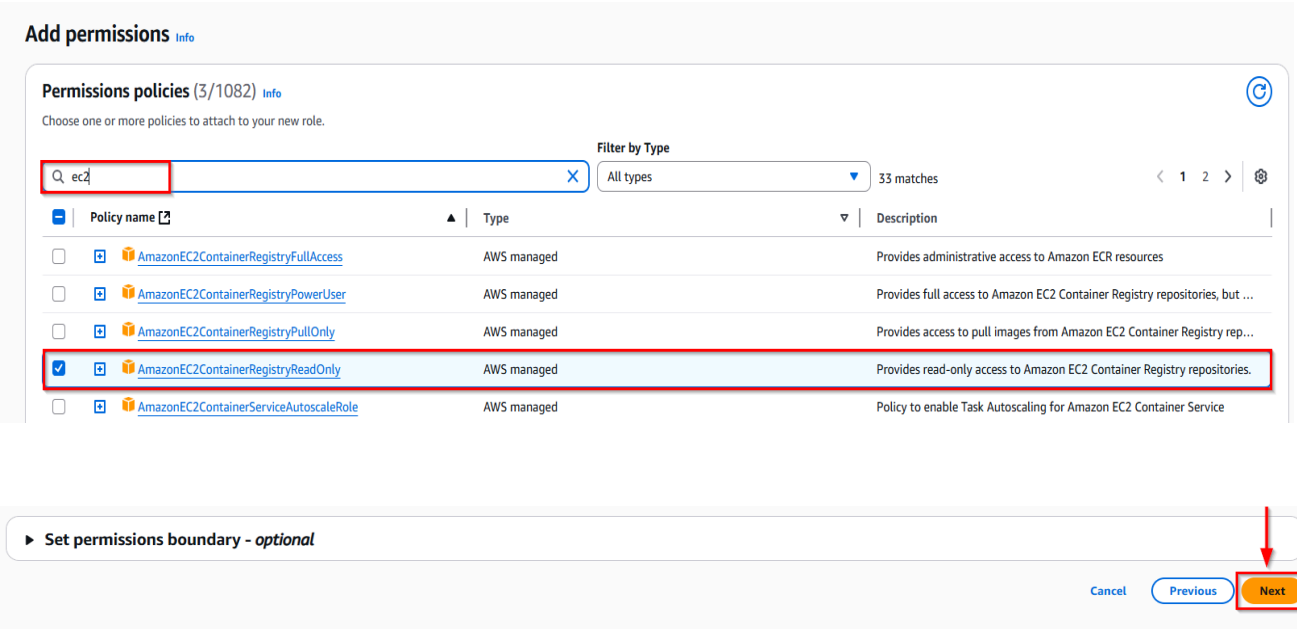
   AmazonEKS_CNI_Policy

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

AmazonEC2ContainerRegistryReadOnly

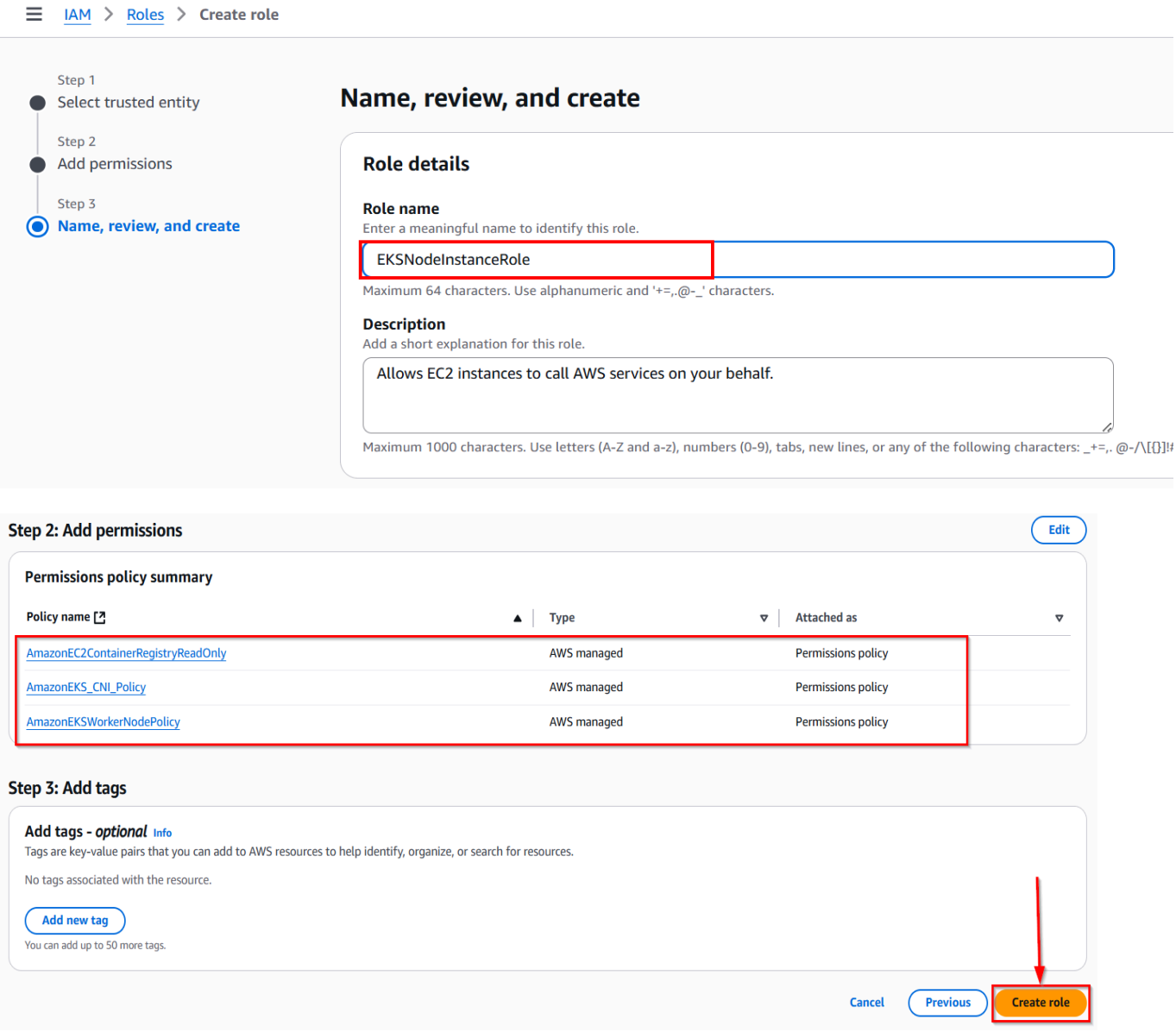
And then click on **Next.**





5. Provide the name to the role as **EKSNodeInstanceRole** and click on **Create role.**

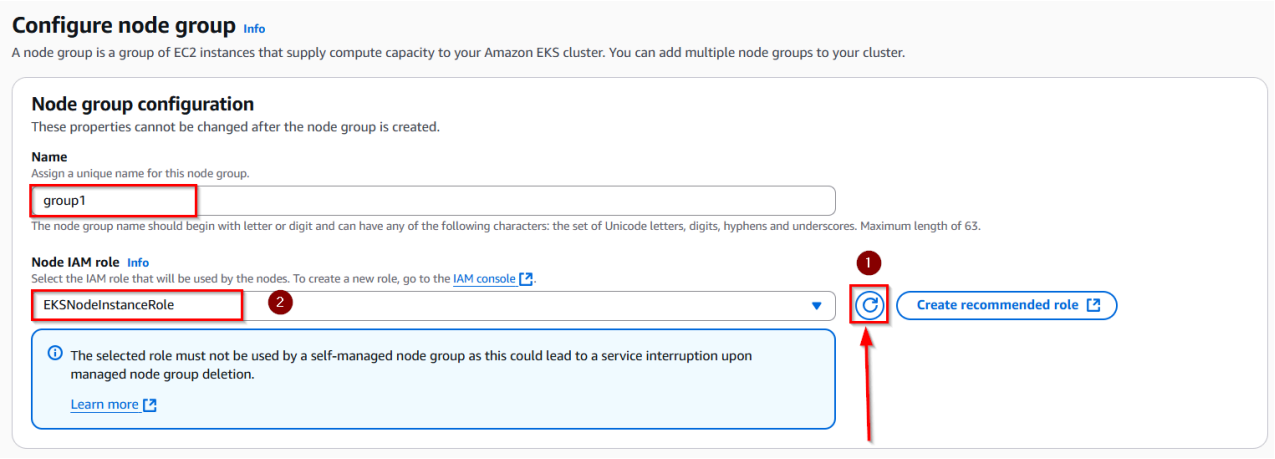
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

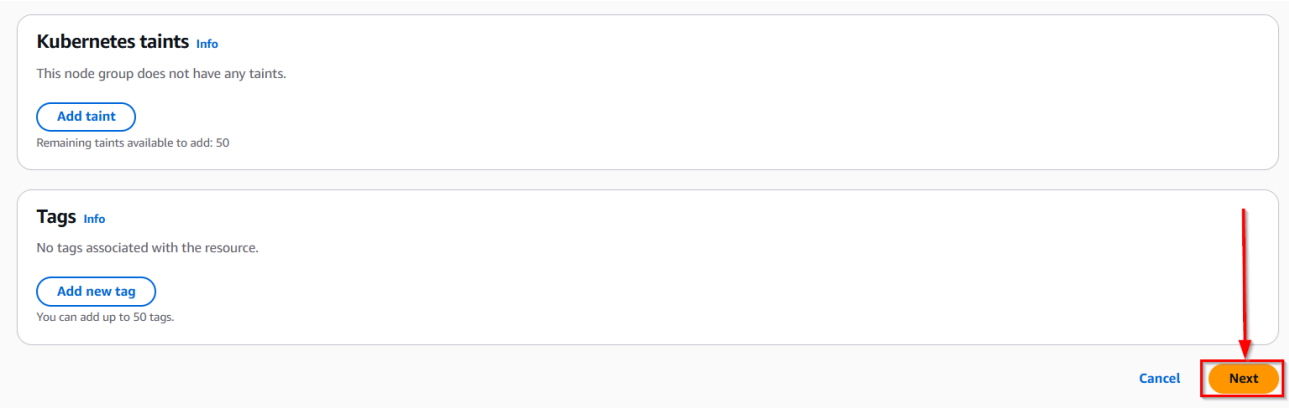


Now **the role** is successfully created.

6. Now, Go Back to your EKS Dashboard and refresh once, then select the role we just created and click on Next.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.





7. Now Under the Compute and scaling configuration Select the following

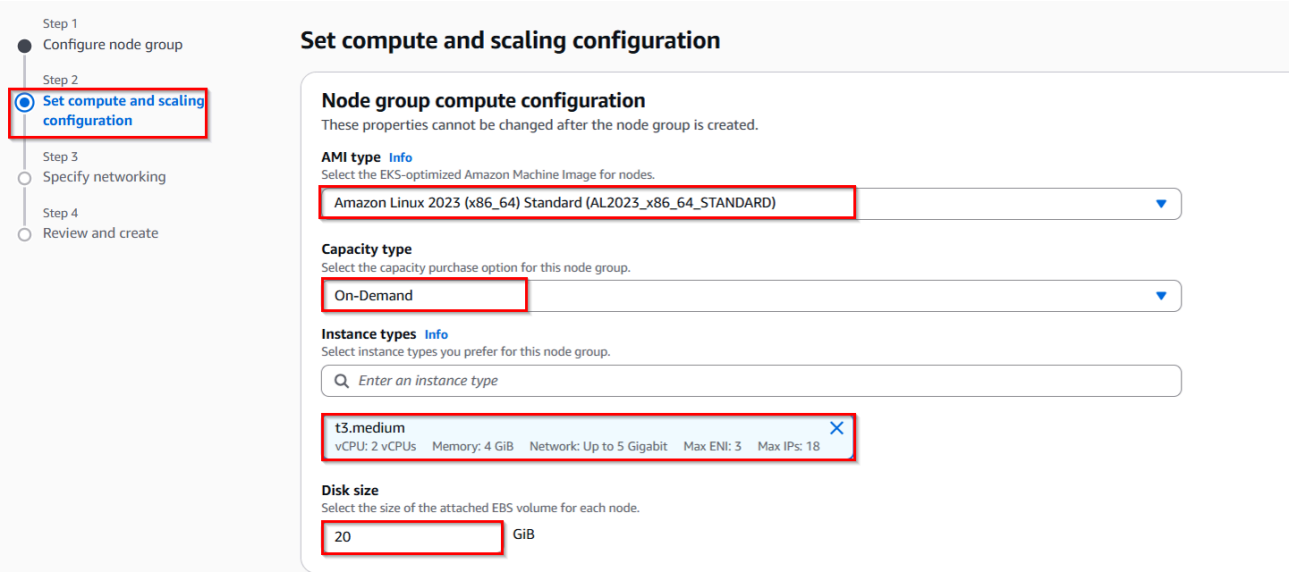
   AMI type: Amazon Linux 2023 (x86_64) Standard (AL2023_x86_64_STANDARD)

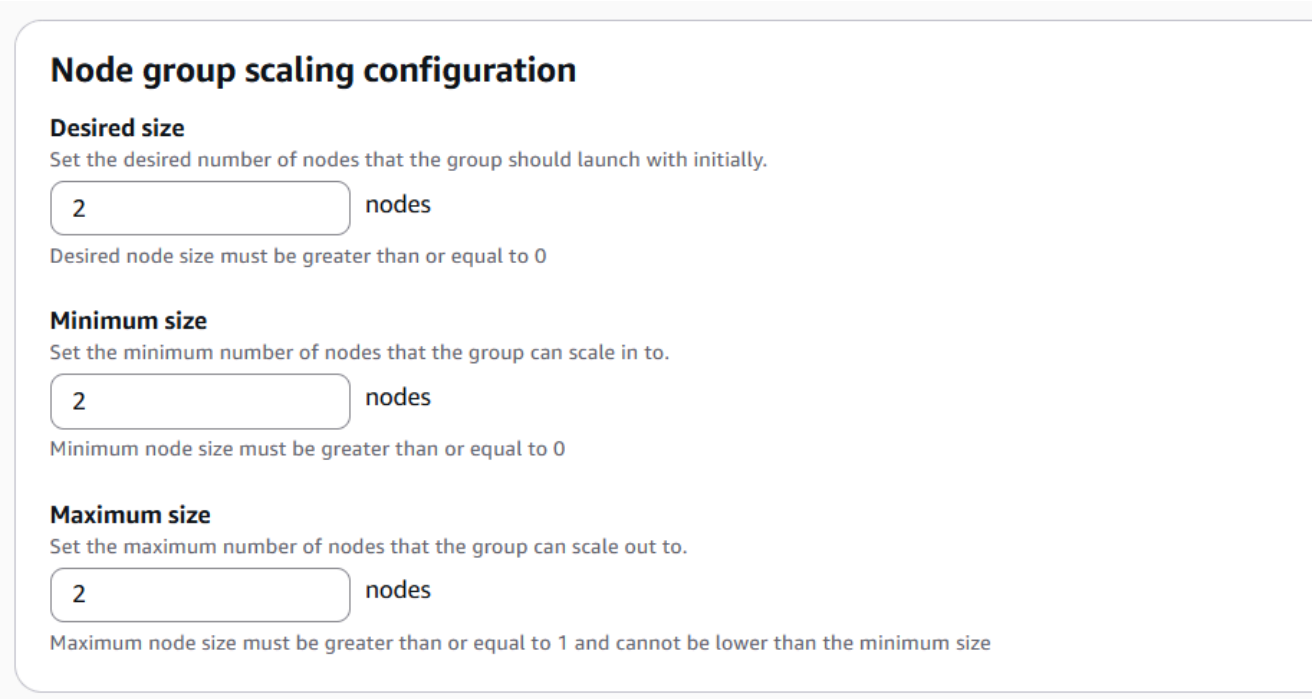
   Capacity Type: On-Demand

   Instance type: t3.medium

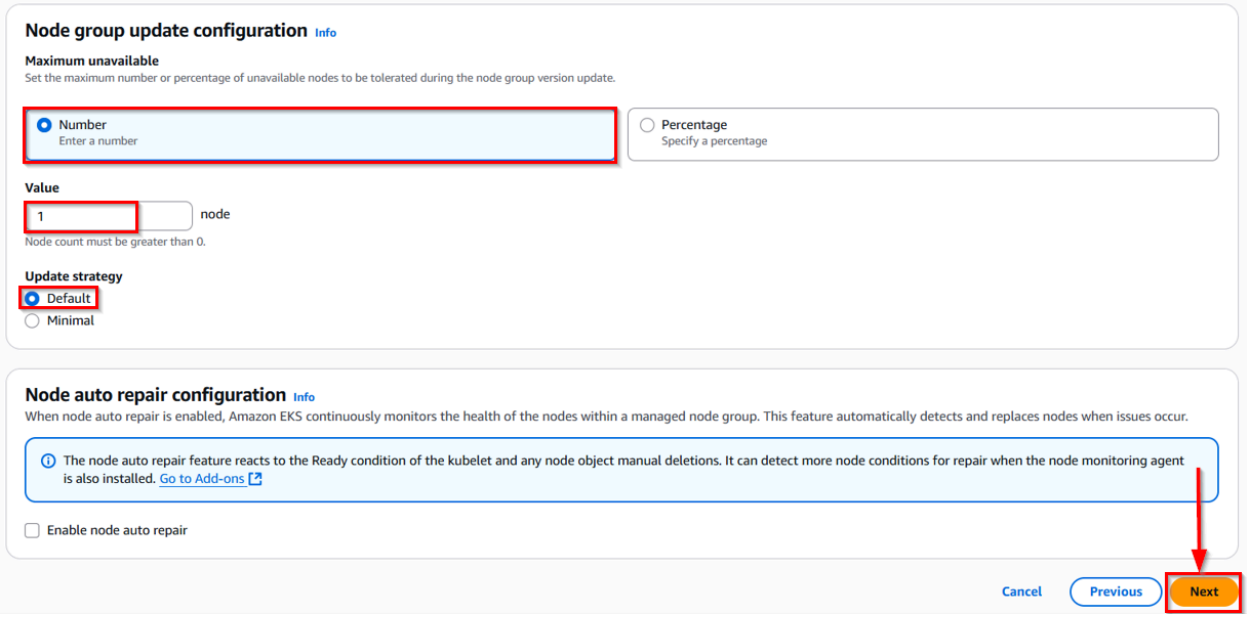
   Disk size: 20 GiB

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

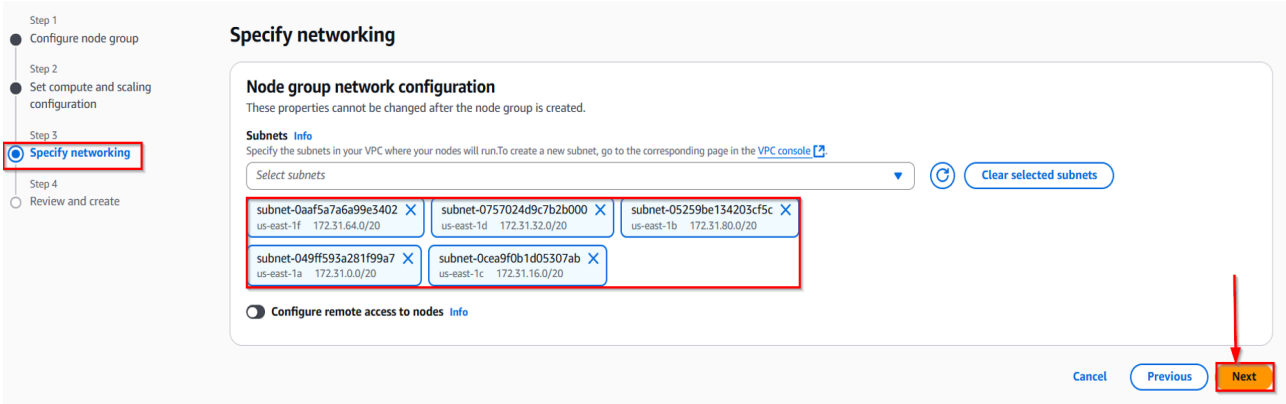




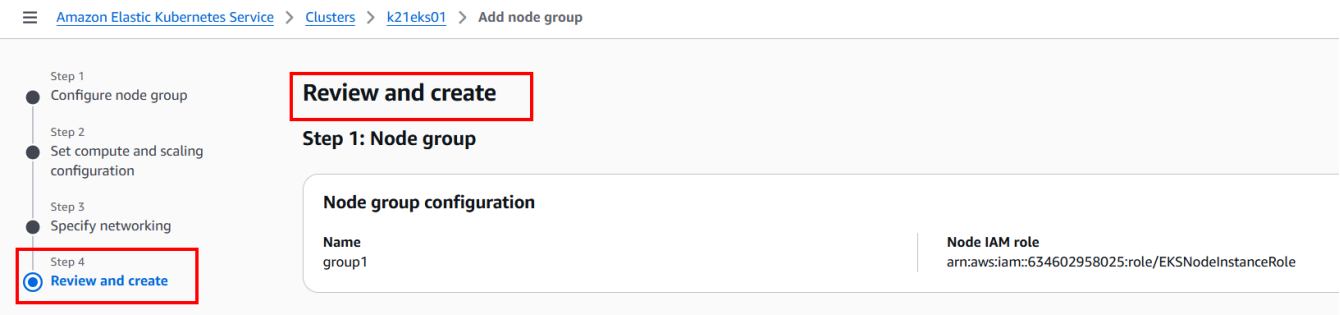
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



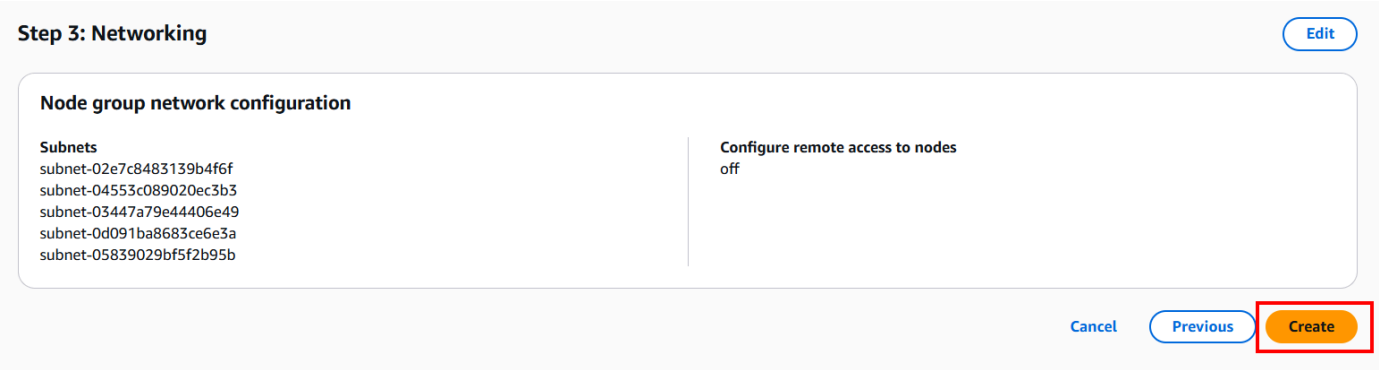
8. Under the Networking Keep subnets as default and click on next.



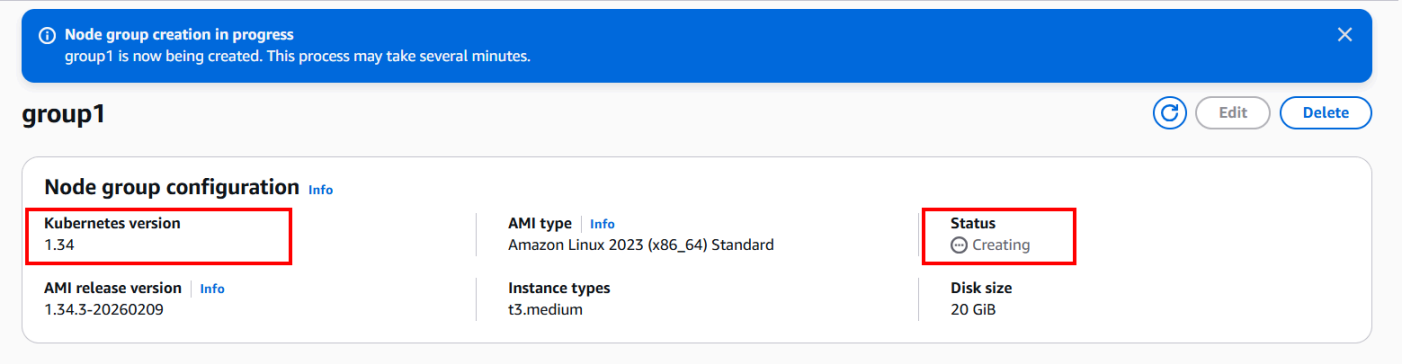
9. Now Rewire all the configurations and click on Next.



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

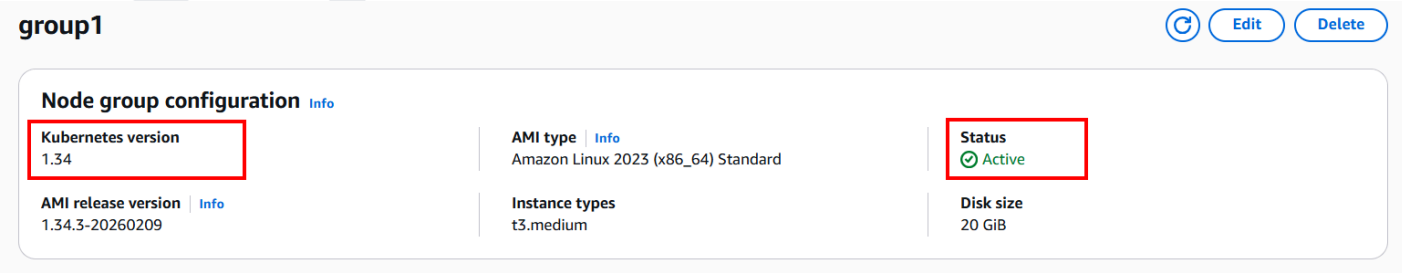


10. Now Worker Node Group is under creation, so wait for 2-3 minutes for worker nodes to be up and working.



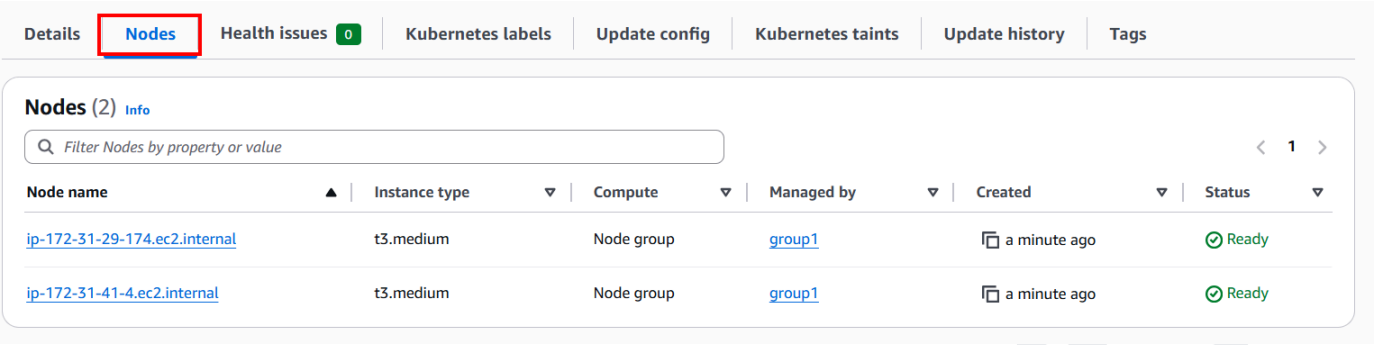
6.2. Verify Worker Nodes in EKS

1. The Node group is created & status is changed to **Active**



2. Check that the worker nodes are created in the Cluster under the **Compute** section

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



Successfully, we have created **master and worker nodes**. In the next section, we will check how to connect to this EKS cluster.

Note: If you decide that you no longer need an instance, stop/terminate the EKS Cluster after usage. If you keep that cluster for a long time, you will be highly charged for that usage.

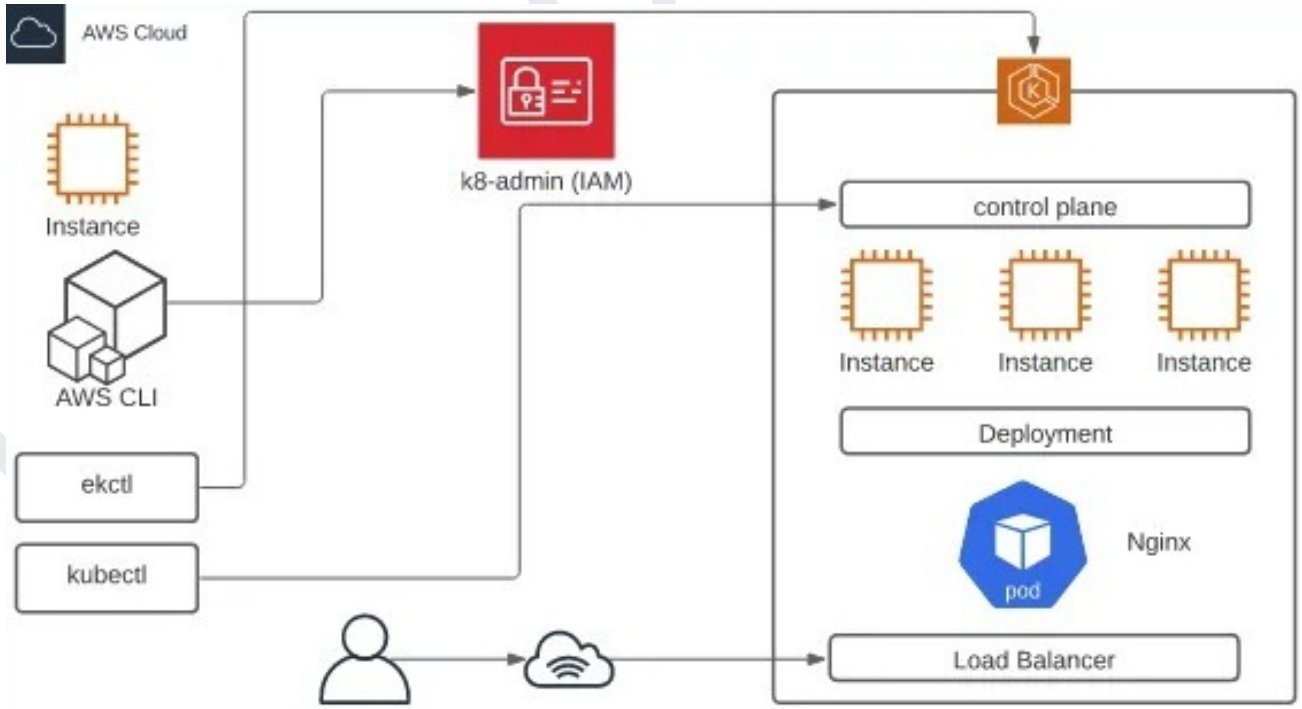
Note: if you hit the error **creation fail,** check this **troubleshooting section**

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

7. Connect & Manage the Amazon EKS cluster

Connecting to an Amazon Elastic Kubernetes Service (Amazon EKS) cluster involves several steps to ensure secure access and smooth operation within the Kubernetes environment.

1. **Install AWS CLI (Command Line Interface):** This tool allows you to interact with AWS services from the command line. You'll need it for various AWS-related tasks.
2. **Configure AWS CLI:** After installation, configure it with your AWS credentials using **aws configure** command.
3. **Download and Install kubectl:** kubectl is the Kubernetes command-line tool used to manage Kubernetes clusters. You can download it from the Kubernetes release page and install it on your local machine.
4. **Configure kubectl:**
   - **Set Cluster Context:** Use kubectl to configure access to your EKS cluster. You can do this by running **aws eks --region <region> update-kubeconfig --name <cluster_name>.**
   - **Verify Configuration:** Run kubectl get svc or any other kubectl command to ensure that you can communicate with your EKS cluster.



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

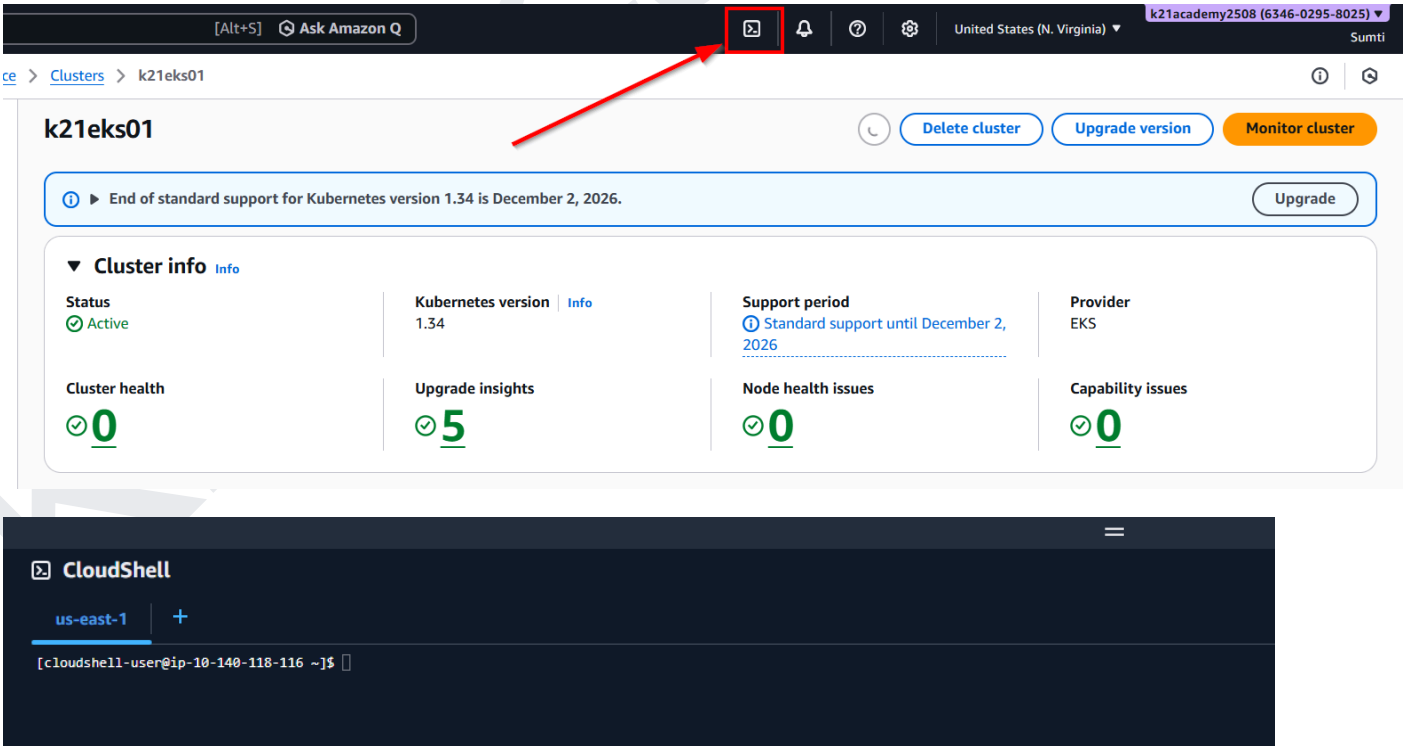
7.1. Method 1: Connect & Manage EKS Cluster on AWS Cloud Shell (Easy & Recommended)

AWS Cloud Shell is a browser-based shell provided by Amazon Web Services that enables AWS customers to manage and interact with **AWS services directly from their web browser.**

This tool is particularly useful for running AWS CLI commands and scripts without having to install or configure the AWS CLI on your local machine.

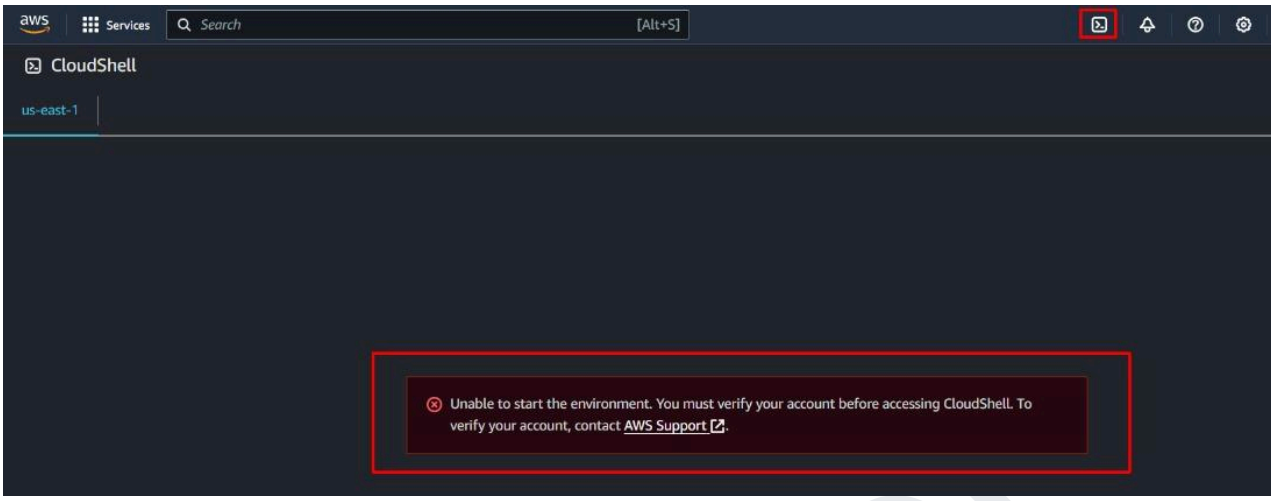
In AWS Cloud Shell, we don't need to install AWS CLI and Kubectl as they are preinstalled. We just need to open AWS Cloud Shell, set the cluster context, and then you can easily access your Kubernetes cluster and perform your Kubernetes-related tasks.

1. Open AWS Cloud Shell by clicking on the icon shown below(marked in red).

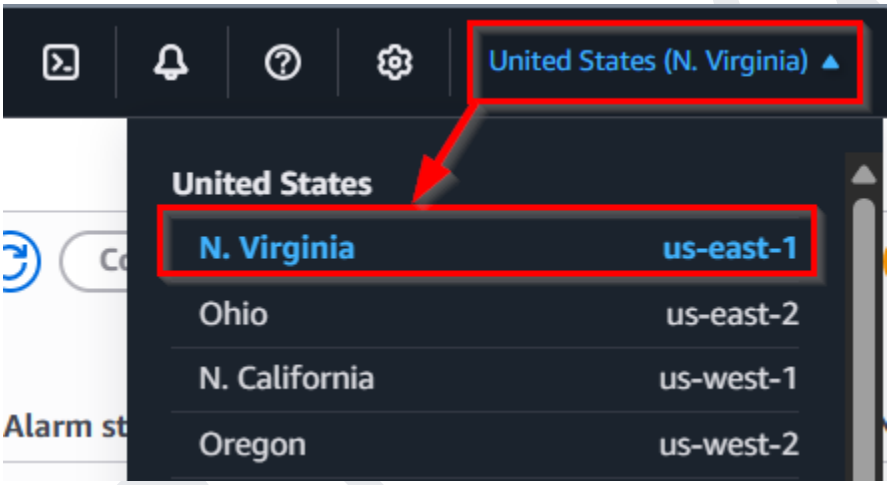


Note: If you encounter any issues while opening Cloud Shell (as shown below), please proceed with the [second method to connect to the EKS cluster.]

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



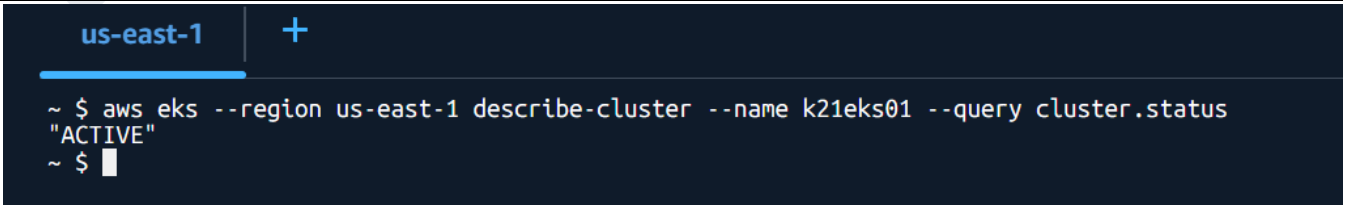
2. Check your **Cluster Name** & **Region Name** where the EKS Master node is running from the console



3. Check the status of Cluster as Active

Syntax: aws eks --region [EKS_Region] describe-cluster --name [EKS_Cluster_Name] --query cluster.status

aws eks --region us-east-1 describe-cluster --name k21eks01 --query cluster.status



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

4. Configure kubectl with EKS API Server credential

Syntax: aws eks --region [EKS_Region] update-kubeconfig --name [EKS_Cluster_Name]

aws eks --region us-east-1 update-kubeconfig --name k21eks01

```
~ $
~ $ aws eks --region us-east-1 update-kubeconfig --name k21eks01
Added new context arn:aws:eks:us-east-1:350903502296:cluster/k21eks01 to /home/cloudshell-user/.kube/config
~ $
```

5. Verify the kubectl version and check the status of nodes.

```
kubectl version --client
kubectl get nodes
```

```
~ $
~ $ kubectl version --client
Client Version: v1.33.0-eks-802817d
Kustomize Version: v5.6.0
~ $
~ $ kubectl get nodes
NAME                                STATUS   ROLES    AGE     VERSION
ip-172-31-32-163.ec2.internal   Ready    <none>   9m48s   v1.32.8-eks-99d6cc0
ip-172-31-85-187.ec2.internal   Ready    <none>   9m48s   v1.32.8-eks-99d6cc0
~ $
```

Note: if you encounter any issues related to **Couldn't get the current server API group list,** please refer to the **troubleshooting section**.

Congratulations! You have successfully connected to the EKS cluster.

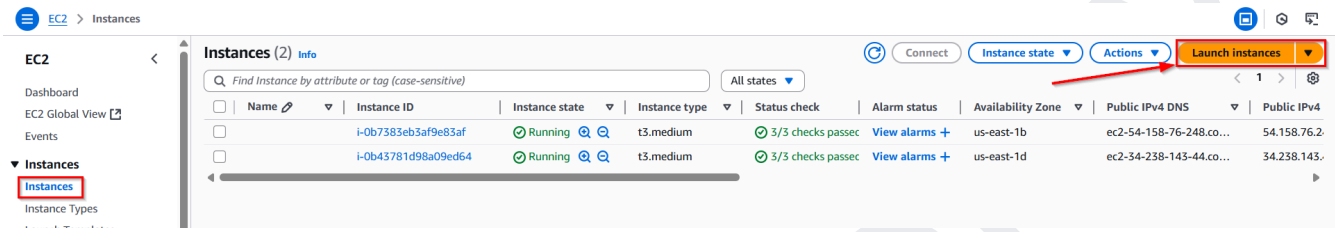
7.2. Method 2: Connect & Manage EKS Cluster on Client Machine [Optional]

Note: If you have **connected to the EKS cluster using AWS Cloud Shell,** this section is **optional.**

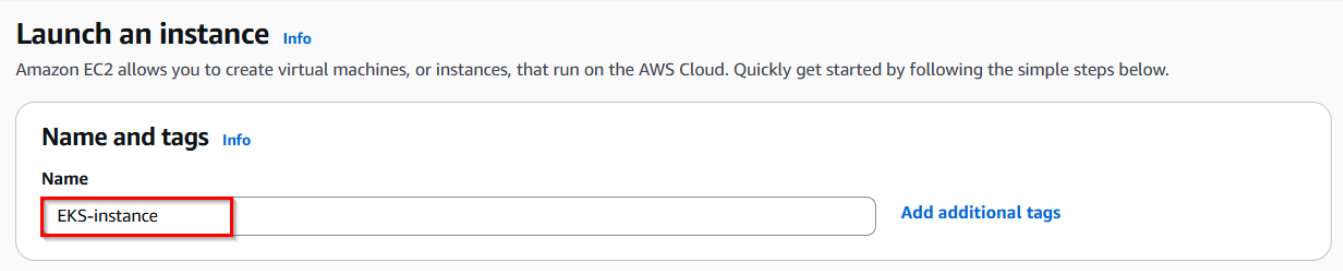
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

Connecting to and managing an Amazon Elastic Kubernetes Service (EKS) cluster on a client machine typically involves installing the **AWS CLI, kubectl, and setting up the cluster context.**

1. Go to **EC2** and create a Linux machine with images as AWS Machine Image

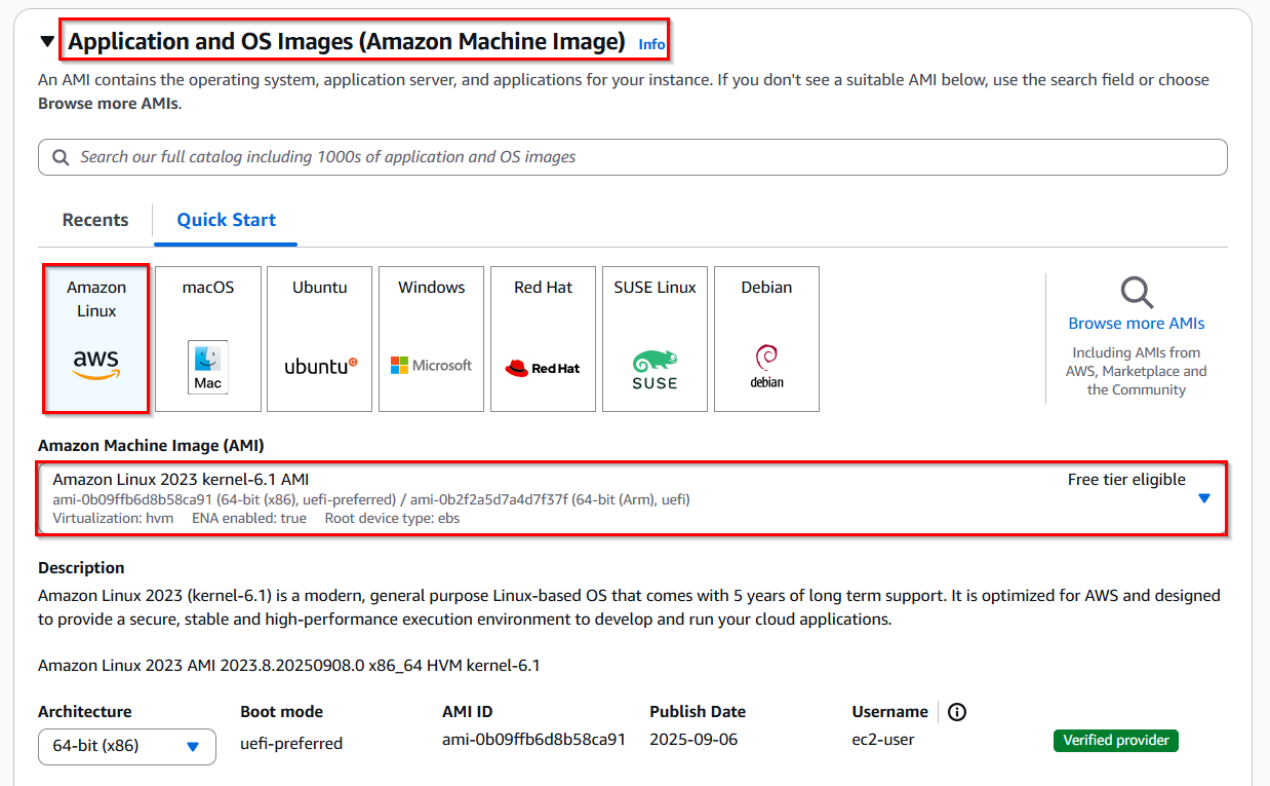


2. Provide a name and a tag to the instance - **EKS-instance.**

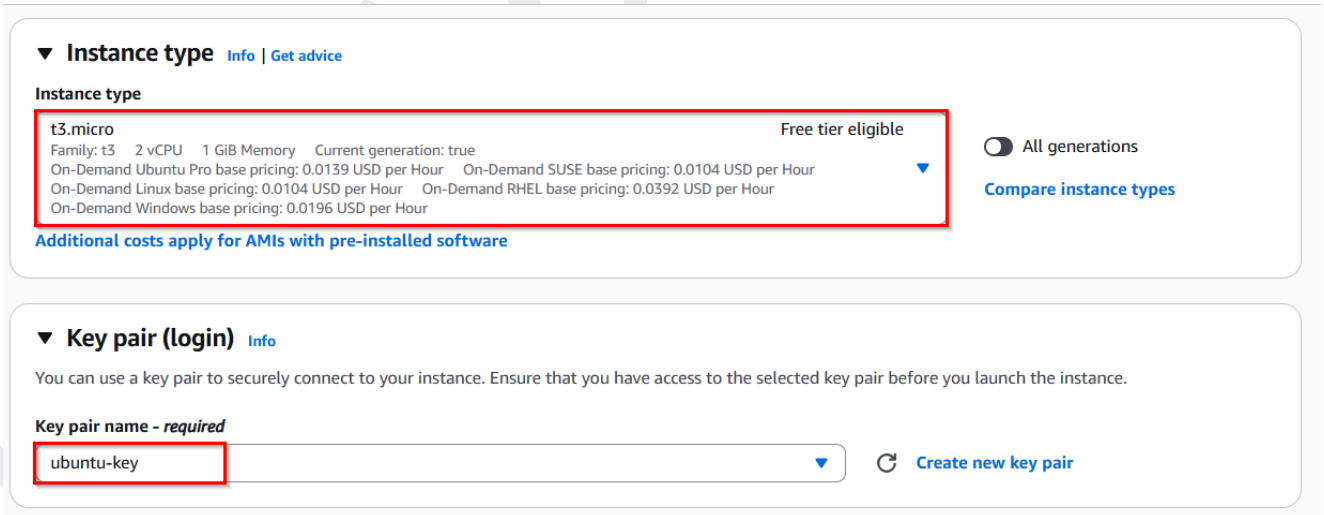


3. **Select an AMI (Amazon Linux 2023 kernel-6.1).**

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

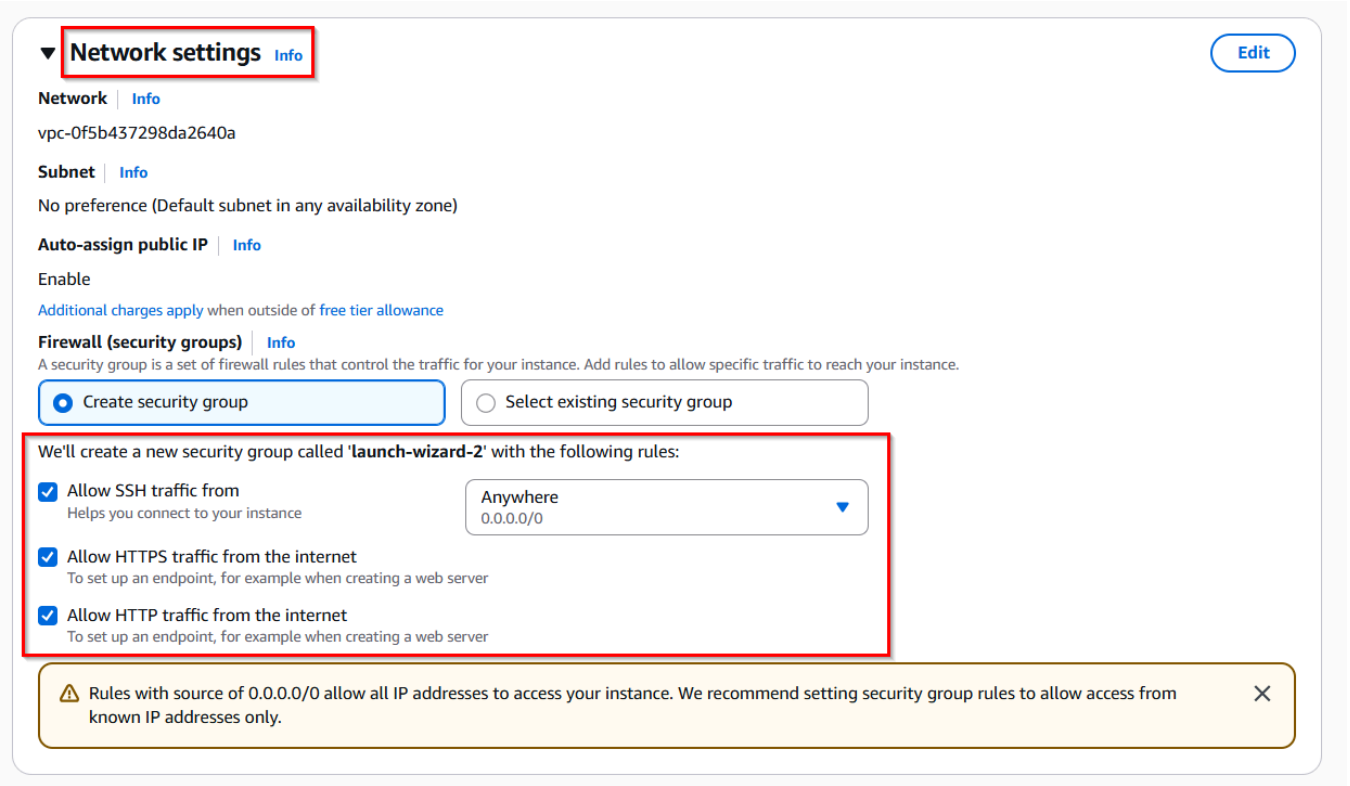


4. Select the instance type as t2.micro and key pair.

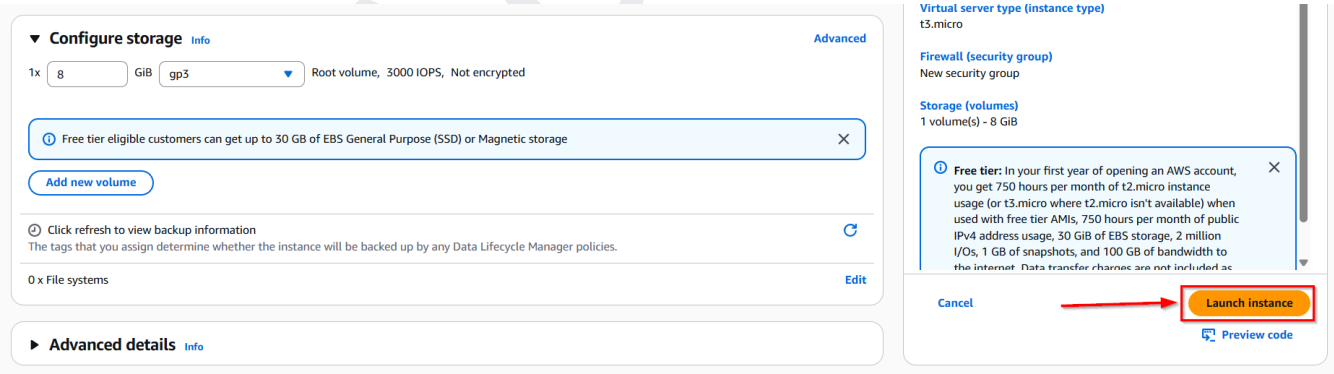


5. Under the network setting, allow http & https ports

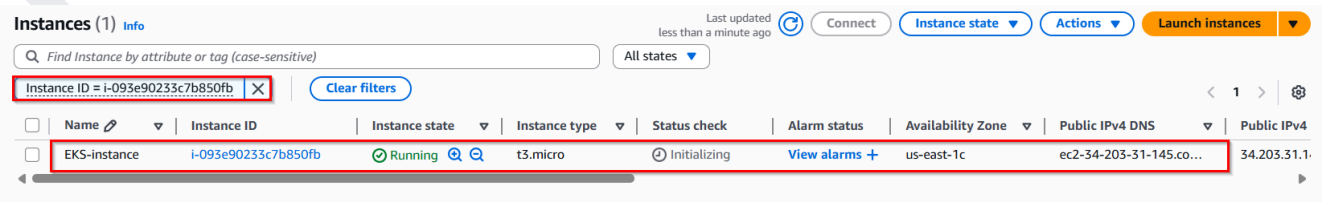
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



6. Now, click on Launch Instance.

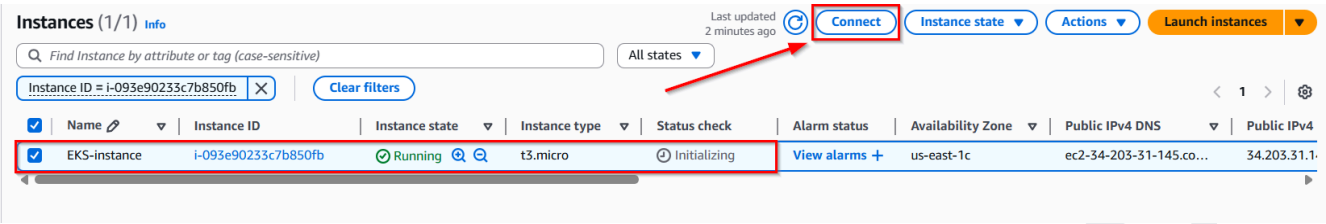


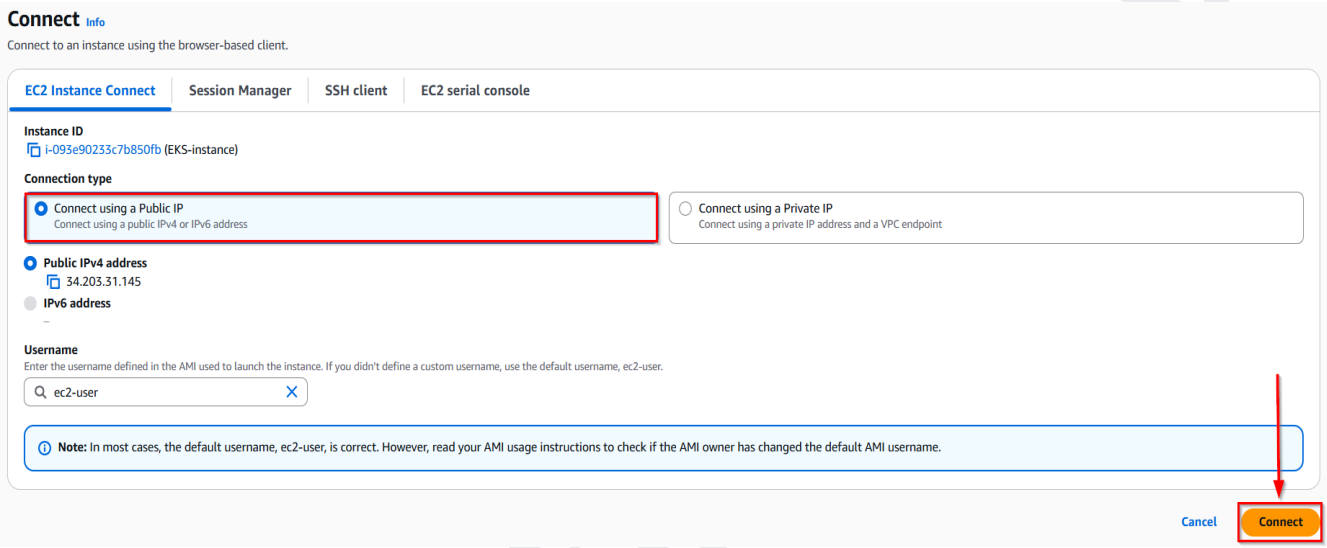
7. Now wait for 1-2 min to pass the status checks.



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

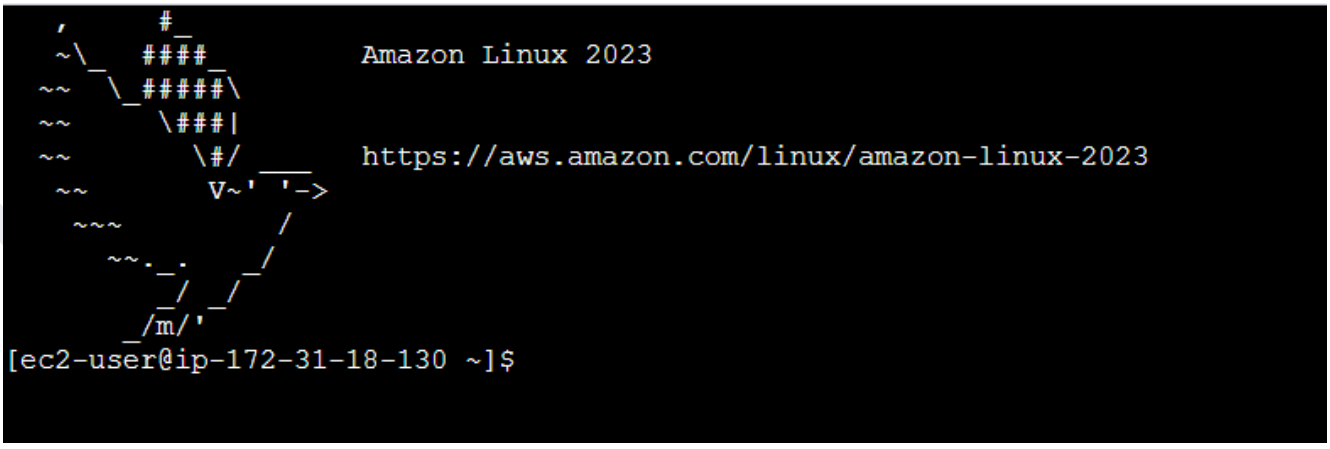
8. Select the instance and click on Connect.





9. Now we have successfully connected to the instance.





Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

7.3. [Non-Amazon Linux AMI] Install AWS CLI

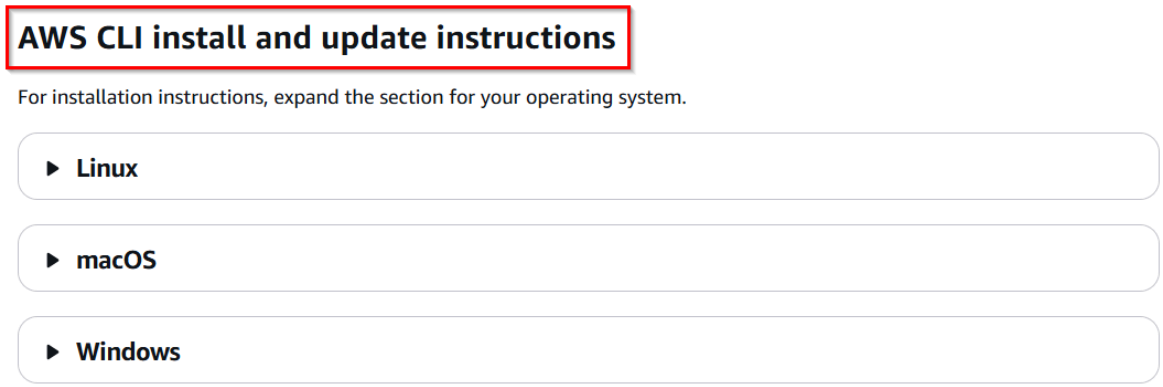
Note: This section is **optional** if you are using an **Amazon Linux machine.**

Setting up the AWS Command Line Interface (CLI) on a Linux machine is important because it makes managing Amazon Web Services (AWS) resources easier. It allows you to automate tasks and work with AWS services using simple commands.

Note: Steps in this section are optional and only required if you are using a Linux machine other than Amazon Linux. Amazon Linux comes pre-installed with AWS CLI, so if you are using Amazon Linux, you can ignore the steps in this section and move to the next to configure AWS CLI.

Follow the below links and snapshot to install and configure AWS CLI on Linux, Windows, and MAC

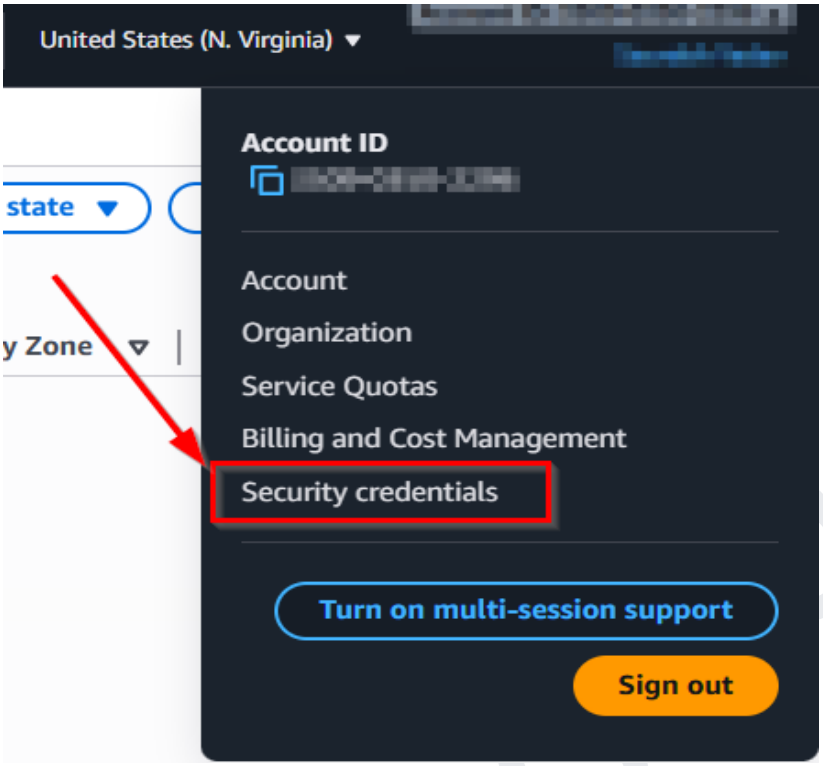
https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html



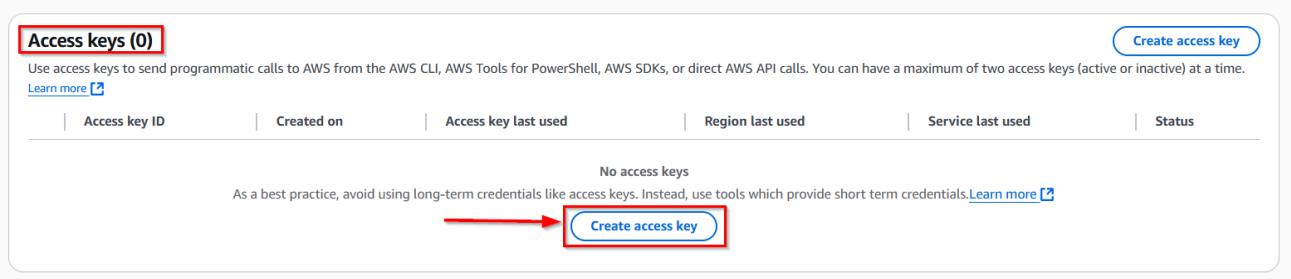
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

7.4. Configure AWS CLI

1. Create IAM Access Key

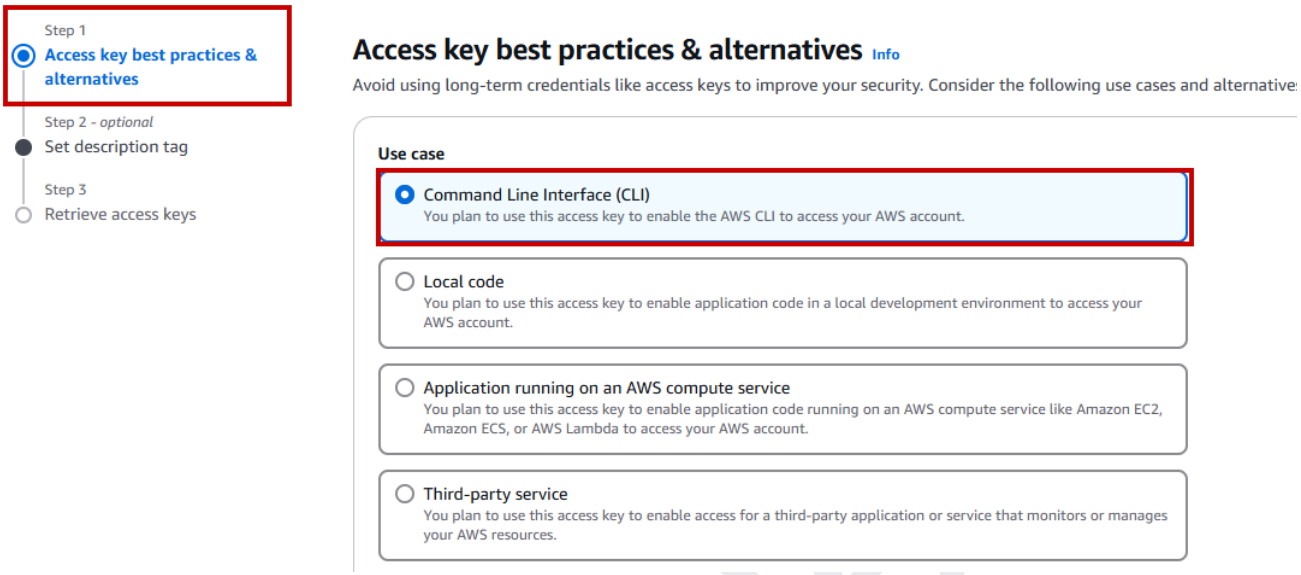


2. Scroll down to the Access key and select Create Access keys

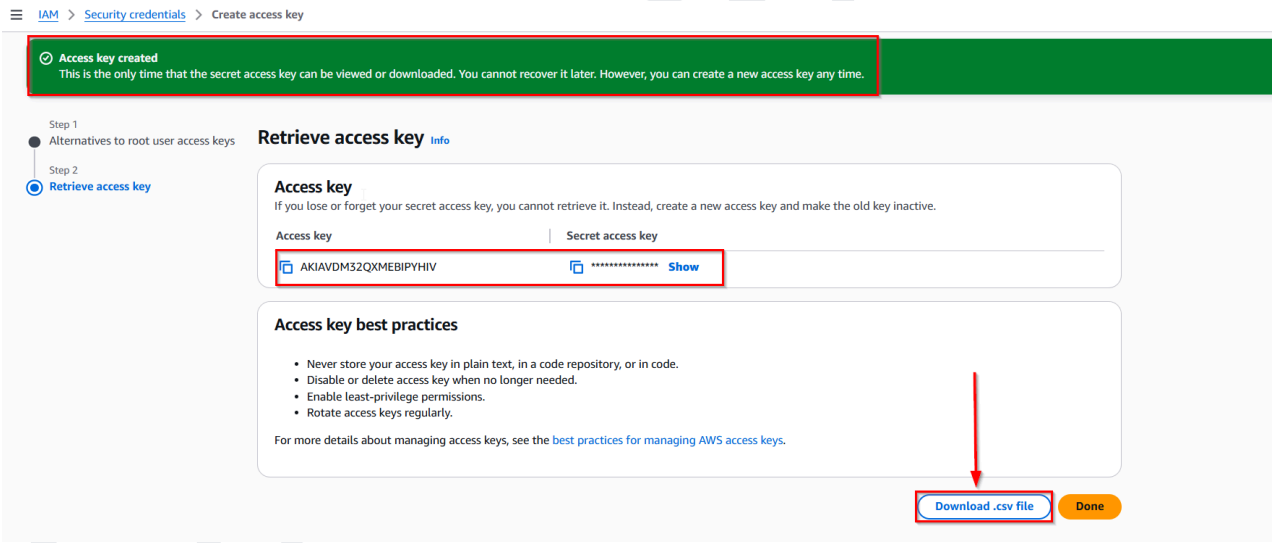


3. Create Access Keys

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



4. Download the .csv file.



5. **Go back to your EC2** and Configure AWS CLI credentials:

```
$ aws configure
AWS Access Key ID [None]: AKIAYA4LY735EQIBSBEQ
AWS Secret Access Key [None]: riAqMejzTOyoJp076fLLrTsEjyjEyYS34ajP9nug
Default region name [None]: <region-code>
Default output format [None]: <Keep it default. Press Enter Button>
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

```
        _/m/'
[ec2-user@ip-172-31-18-130 ~]$
[ec2-user@ip-172-31-18-130 ~]$ aws configure
AWS Access Key ID [None]: AKIAVDM32QXMEBIPYHIV
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]:
[ec2-user@ip-172-31-18-130 ~]$
```

7.5. Install Kubectl on an Amazon Linux Machine

Please follow the link below to install kubectl on other operating systems:

https://docs.aws.amazon.com/eks/latest/userguide/install-kubectl.html

1. Download & Install Kubectl on an Amazon Linux Machine

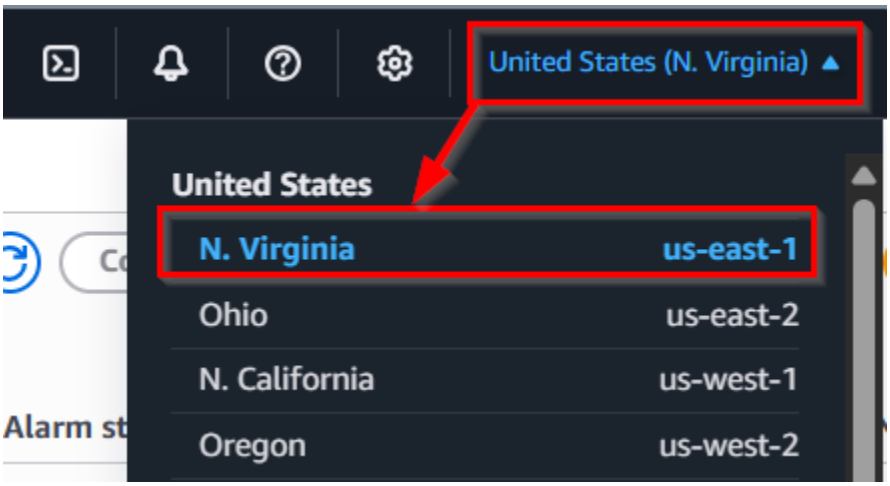
```
curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.29.0/2024-01-04/bin/linux/amd64/kubectl
chmod +x ./kubectl
mkdir -p $HOME/bin && cp ./kubectl $HOME/bin/kubectl && export PATH=$HOME/bin:$PATH
echo 'export PATH=$HOME/bin:$PATH' >> ~/.bashrc
kubectl version --client
```

```
[ec2-user@ip-172-31-18-130 ~]$ curl -O https://s3.us-west-2.amazonaws.com/amazon-eks/1.29.0/2024-01-04/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 47.4M  100 47.4M    0     0  22.3M      0  0:00:02  0:00:02 --:--:-- 22.3M
[ec2-user@ip-172-31-18-130 ~]$
[ec2-user@ip-172-31-18-130 ~]$ chmod +x ./kubectl
[ec2-user@ip-172-31-18-130 ~]$ mkdir -p $HOME/bin && cp ./kubectl $HOME/bin/kubectl && export PATH=$HOME/bin:$PATH
[ec2-user@ip-172-31-18-130 ~]$ echo 'export PATH=$HOME/bin:$PATH' >> ~/.bashrc
[ec2-user@ip-172-31-18-130 ~]$ kubectl version --client
Client Version: v1.29.0-eks-5e0fdde
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
[ec2-user@ip-172-31-18-130 ~]$
```

7.6. Set Cluster Context & Verify the Status of the Cluster on the Client Machine

1. Check your **Cluster Name** & **Region Name** where the EKS Master node is running from the console

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



2. Check the status of the Cluster as Active

Syntax: aws eks --region [EKS_Region] describe-cluster --name [EKS_Cluster_Name] --query cluster.status

$ aws eks --region us-east-1 describe-cluster --name k21eks01 --query cluster.status

```
[ec2-user@ip-172-31-18-130 ~]$
[ec2-user@ip-172-31-18-130 ~]$ aws eks --region us-east-1 describe-cluster --name k21eks01 --query cluster.status
"ACTIVE"
[ec2-user@ip-172-31-18-130 ~]$
```

3. Configure kubectl with EKS API Server credential

Syntax: aws eks --region [EKS_Region] update-kubeconfig --name [EKS_Cluster_Name]

$ aws eks --region us-east-1 update-kubeconfig --name k21eks01

```
[ec2-user@ip-172-31-18-130 ~]$
[ec2-user@ip-172-31-18-130 ~]$ aws eks --region us-east-1 update-kubeconfig --name k21eks01
Added new context arn:aws:eks:us-east-1:350903502296:cluster/k21eks01 to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-18-130 ~]$
```

4. Verify the status of nodes

$ kubectl get nodes

$ kubectl get svc

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

```
[ec2-user@ip-172-31-18-130 ~]$ kubectl get nodes
NAME                            STATUS   ROLES    AGE   VERSION
ip-172-31-32-163.ec2.internal   Ready    <none>   45m   v1.32.8-eks-99d6cc0
ip-172-31-85-187.ec2.internal   Ready    <none>   45m   v1.32.8-eks-99d6cc0
[ec2-user@ip-172-31-18-130 ~]$
[ec2-user@ip-172-31-18-130 ~]$ kubectl get svc
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   57m
[ec2-user@ip-172-31-18-130 ~]$
```

Note: If you encounter any issue related to **the error Unable to connect to the server,** please refer to the **troubleshooting section**.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

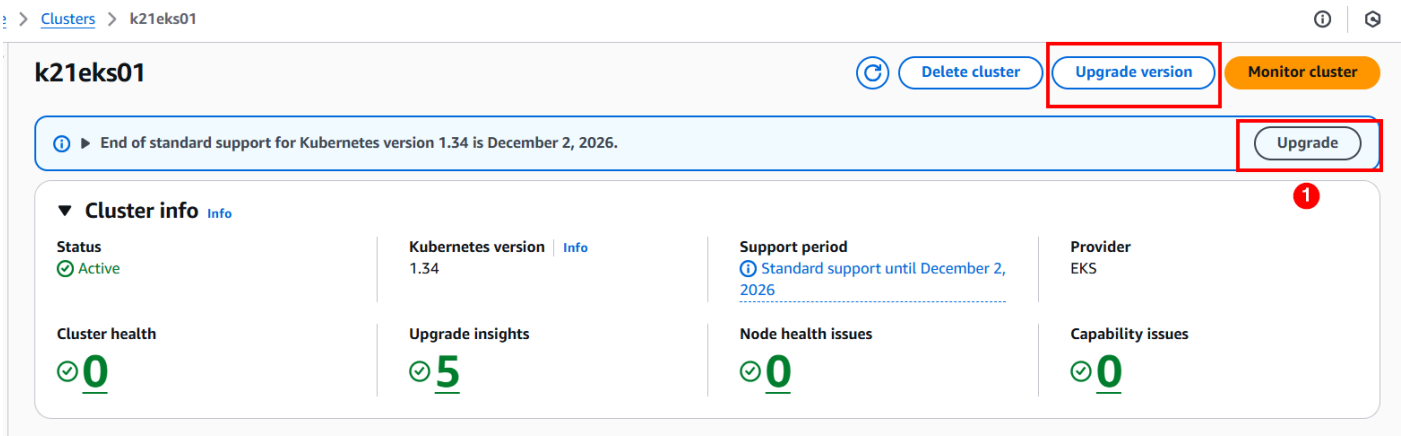
8. Upgrade Amazon EKS Cluster

Upgrading an Amazon EKS (Elastic Kubernetes Service) cluster helps keep your applications secure, stable, and up-to-date.

New versions provide **essential security fixes, better performance,** and access to the **latest Kubernetes features and tools.**

By upgrading regularly, you ensure your cluster remains compatible with current software, runs efficiently, and stays protected against vulnerabilities. This makes it easier to manage workloads and improves overall system reliability.

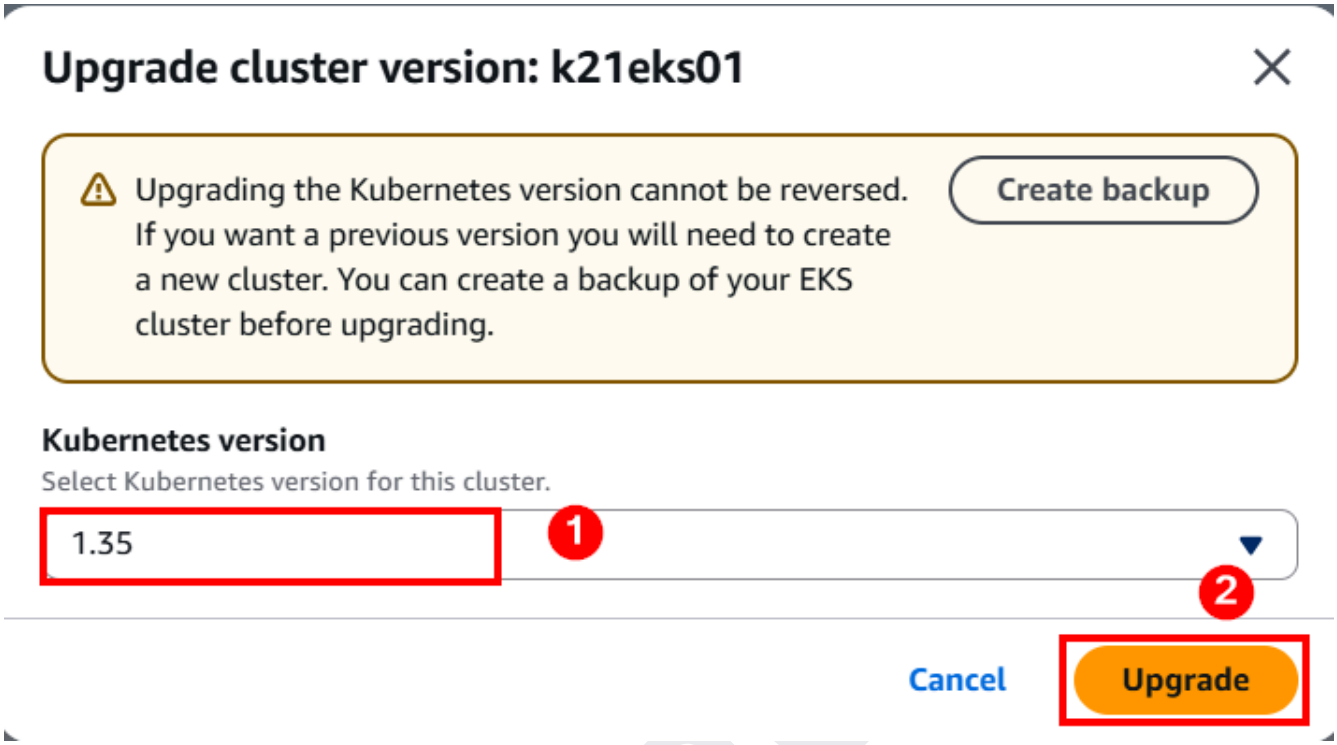
1. In the **AWS Console**, search for **EKS**.
2. Click on the **Cluster** you recently created.
3. Select the **Upgrade Version** option.



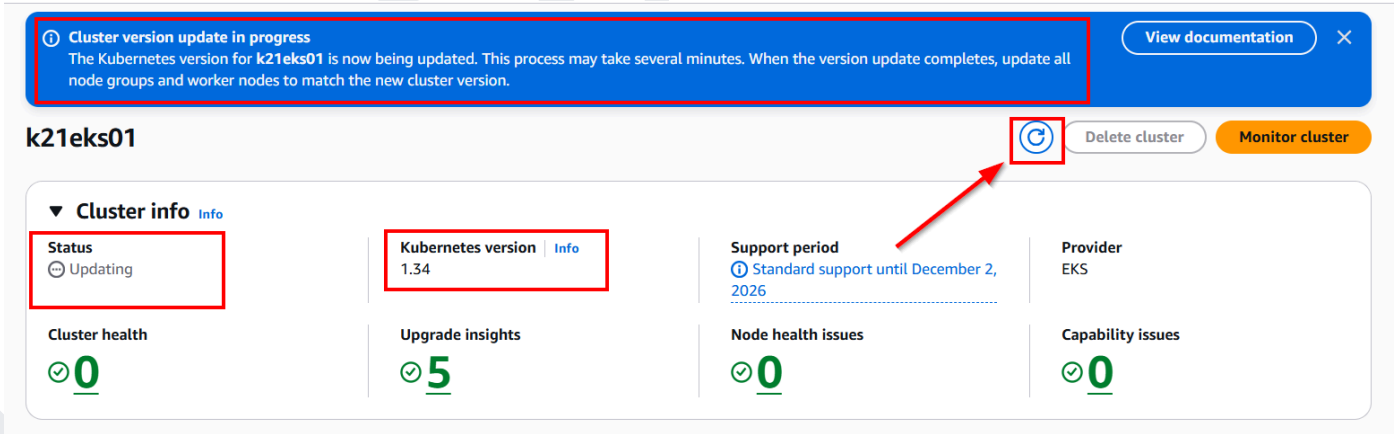
(Note: We are using version 1.34. And the latest version 1.35, you can proceed with this step or just refer to the step on how we do that.)

4. Choose **Kubernetes version 1.35** from the available options.
   - **Note**: Since the current cluster is on version 1.34, it can be upgraded to one version higher, 1.35.
5. Click on **Upgrade** to initiate the process.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

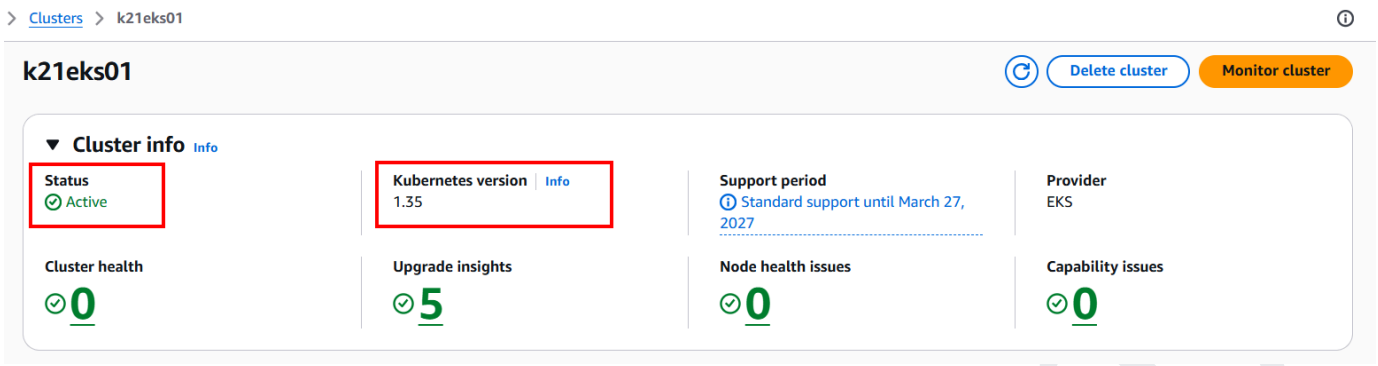


6. Click the **Refresh** button to monitor the status. It will change to **Updating**.
   - **Note**: The upgrade process may take **10-15 minutes**.

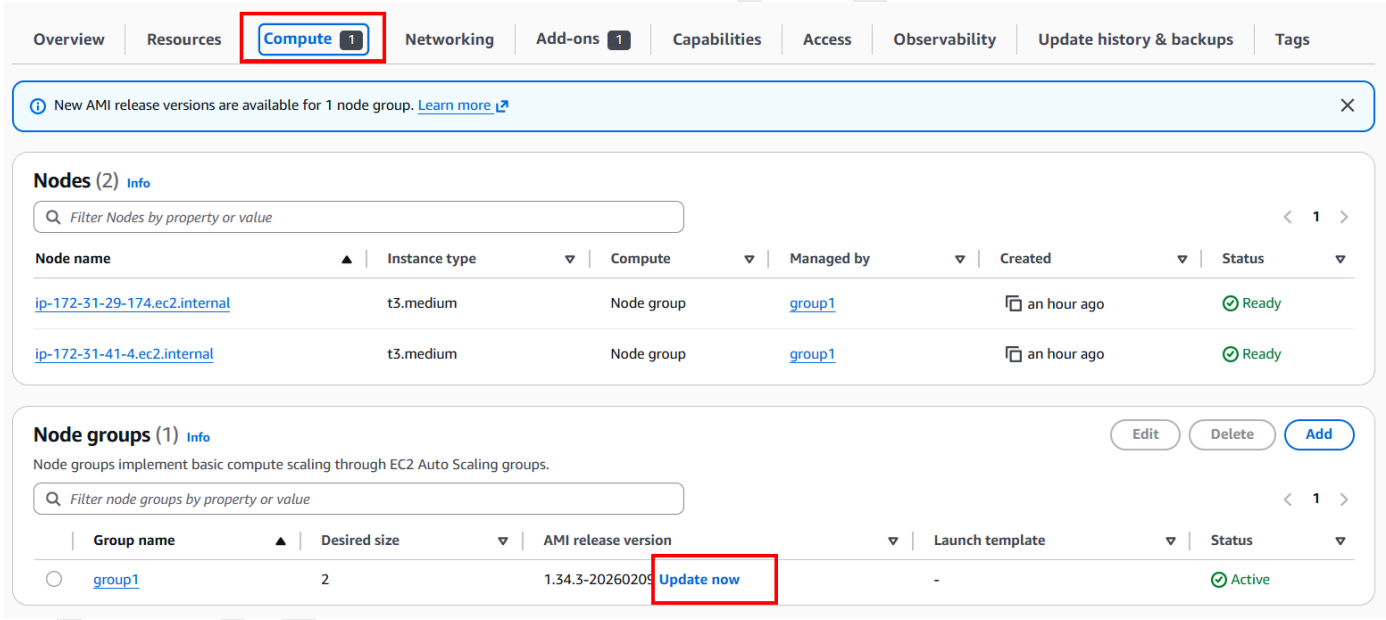


7. Once complete, click **Refresh** again, and the status will update to **Active**, with the version showing **1.35**.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

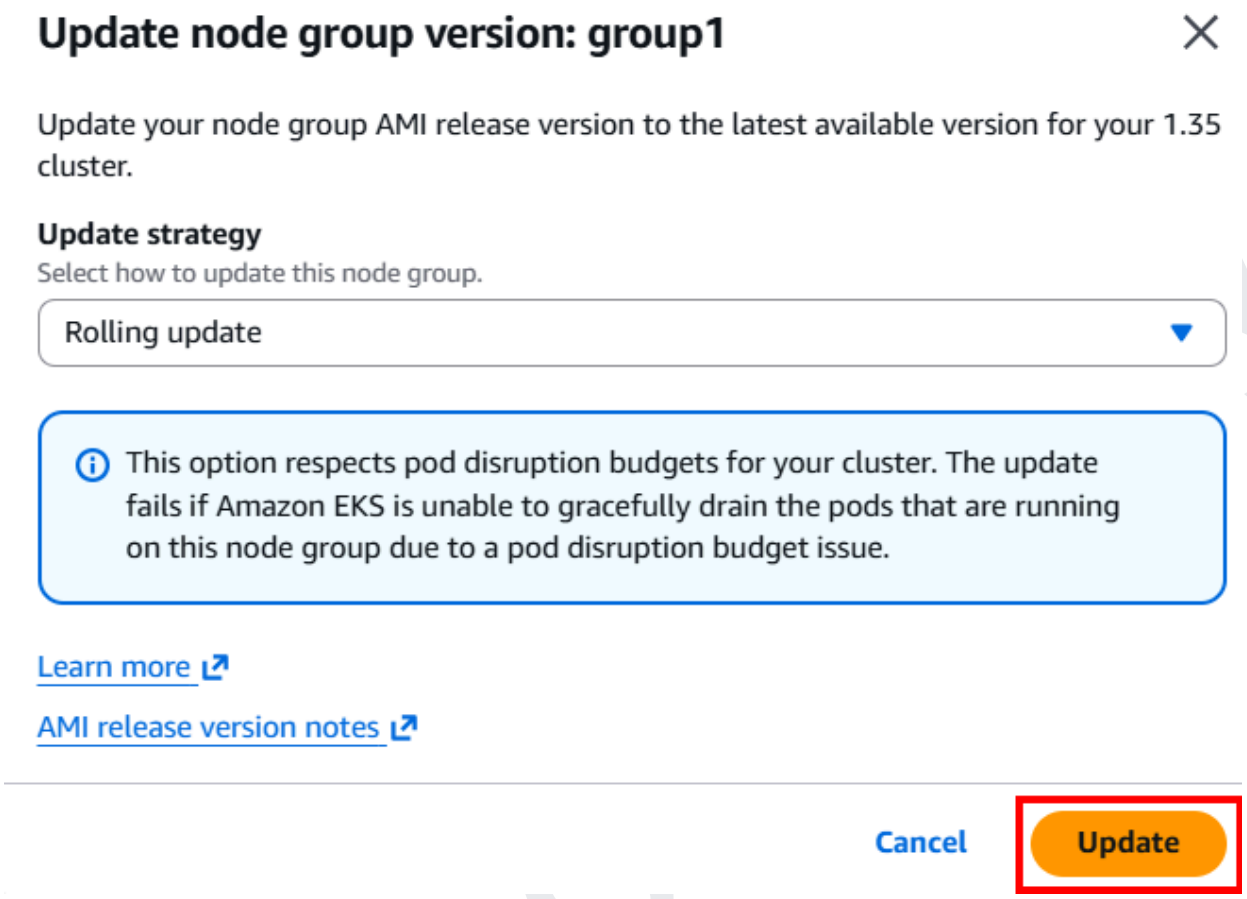


8. Since the cluster has been updated, other components, such as **Node Groups** and **Add-ons**, need to be upgraded as well.
9. In the **k21eks01** cluster, click on the **Compute** option.
10. In the **Node Group** section, select **Update Now** to upgrade the **AMI release version**.



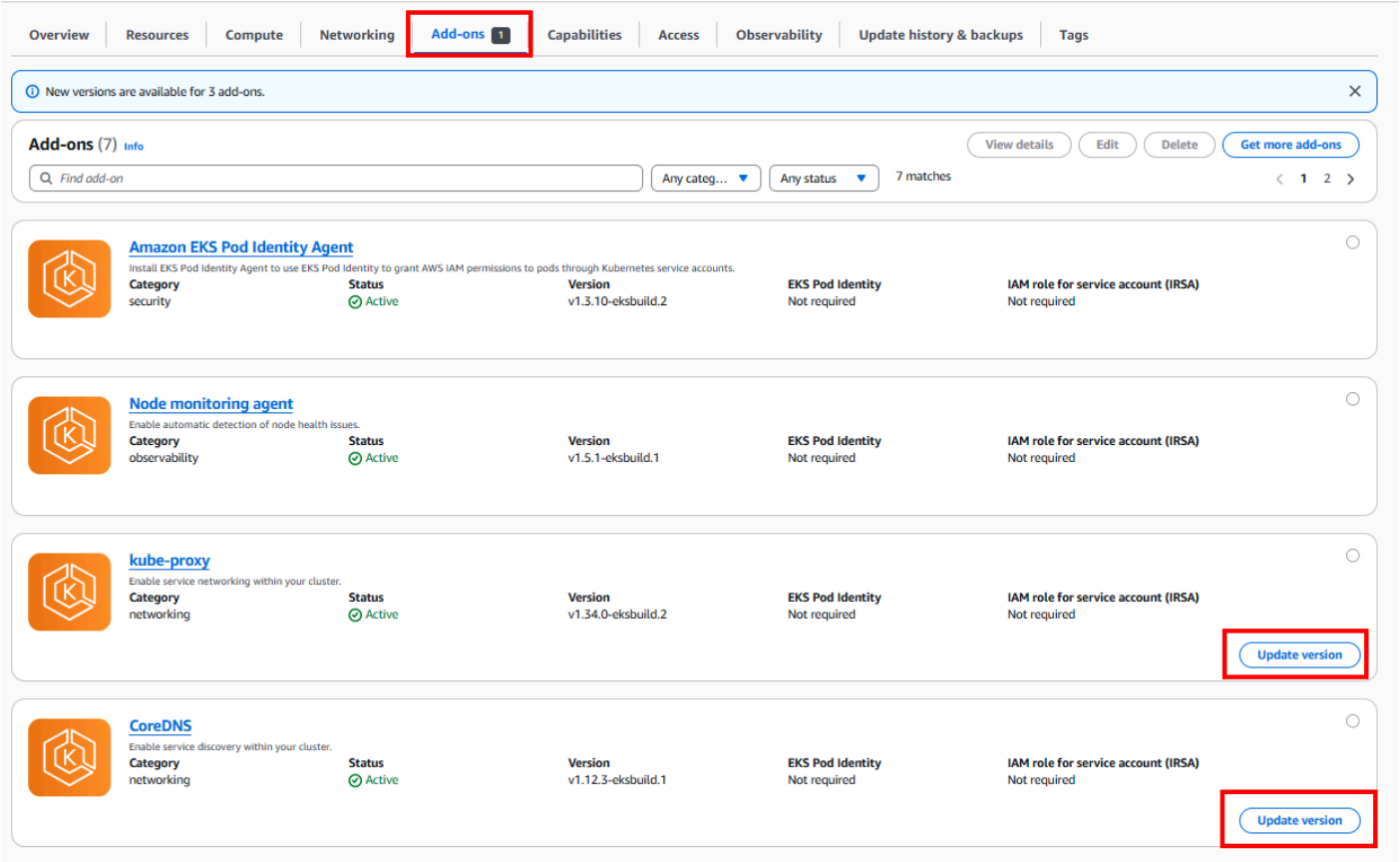
11. Select **Update Strategy** as **Rolling Update**, then click **Update**

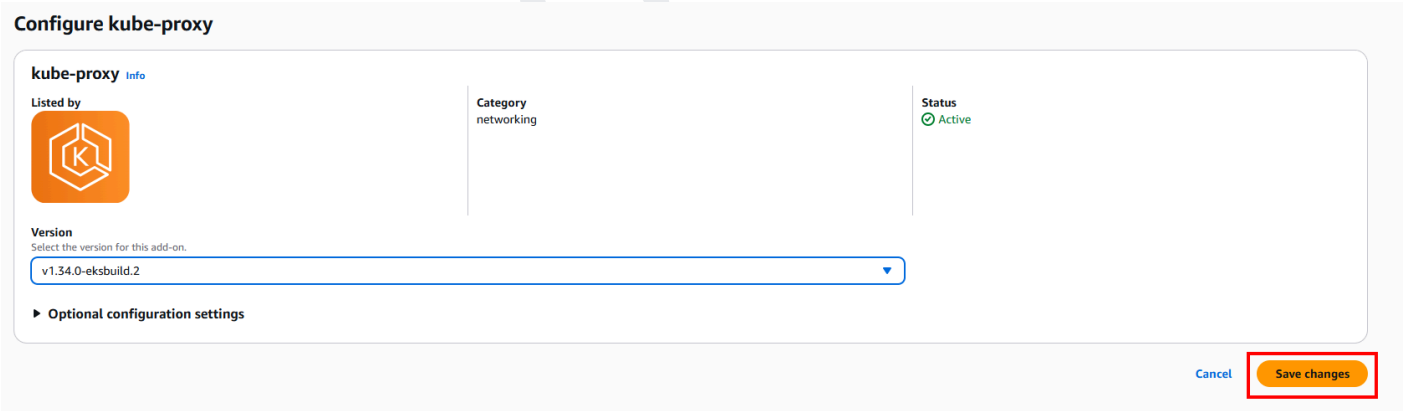
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



12. Switch to the **Add-ons** section, where you'll need to upgrade all components—such as **Amazon VPC CNI**, **CoreDNS**, **Amazon EKS Pod Identity Agent**, **Node monitoring agent** and **kube-proxy**—to their latest versions.
    - Click **Update** for each add-on where an update is available.
    - Select the latest version from the dropdown, then click **Save Changes**.

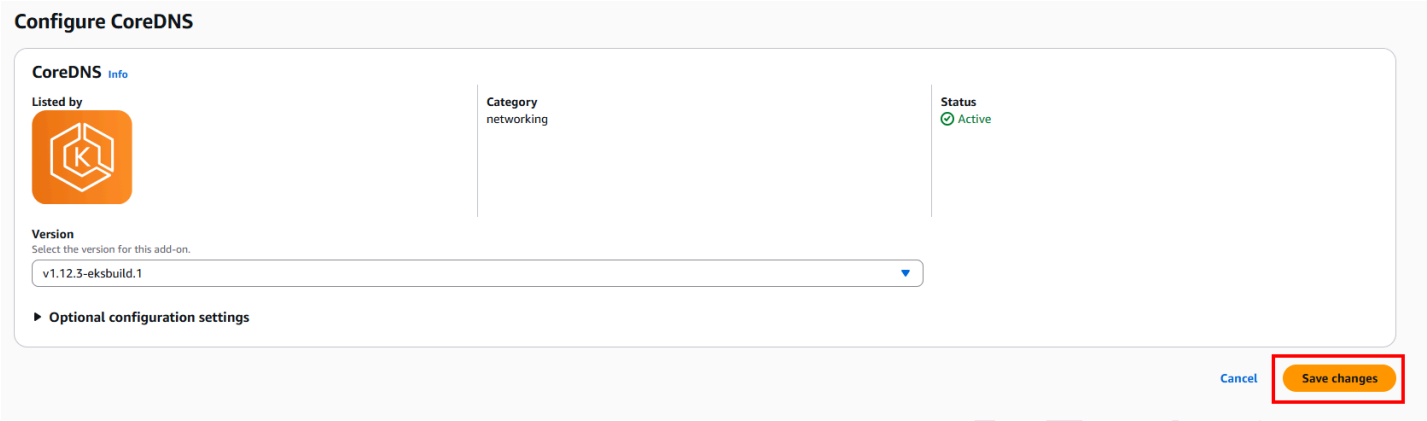
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



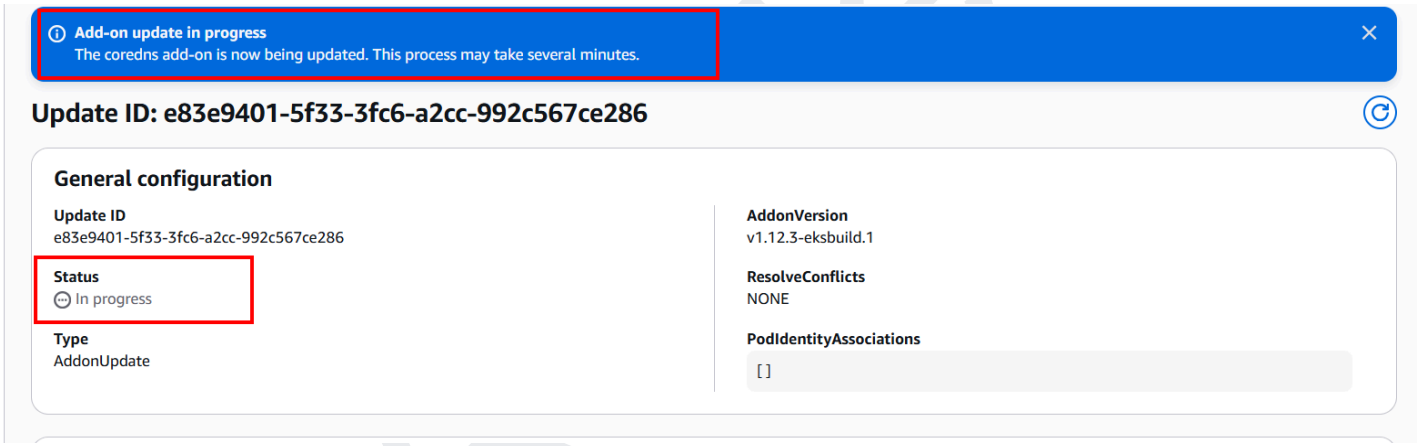


Go back and do the same for coreDNS

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



13. As of now, the add-on is in an **In-Progress state**, so we need to wait for 5-10 minutes.



14. The add-on update is now successful.



Congratulations! You have successfully upgraded the Kubernetes cluster.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

9. CLI Method: Setup EKS Cluster (Master & Worker Nodes) Using EKSCTL

Amazon EKS (Elastic Kubernetes Service):

Amazon Elastic Kubernetes Service (EKS) is a managed Kubernetes service provided by Amazon Web Services (AWS). It allows you to easily deploy, manage, and scale containerized applications using Kubernetes on AWS infrastructure. With EKS, you can run Kubernetes without needing to install, operate, and maintain your own Kubernetes control plane.

eksctl:

eksctl is a command-line tool provided by Weaveworks and officially recommended by AWS for creating and managing EKS clusters. It simplifies the process of creating, managing, and scaling Kubernetes clusters on AWS. With eksctl, you can easily create clusters, manage node groups, and perform various other cluster operations using simple commands.

To create an EKS Cluster using EKSCTL, ensure you have configured the following:

- ➢ AWS Cloud Shell

(or)

- ➢ Amazon Linux EC2 Instance on AWS
  - Install & Configure AWS CLI
  - Install Kubectl

9.1. Install EKSCTL on an Amazon Linux Machine

1. Install & Configure EKSCTL by following the link for any OS: https://eksctl.io/installation/
2. Install eksctl on Unix [Any Linux OS]

```
# for ARM systems, set ARCH to: `arm64`, `armv6` or `armv7`
$ ARCH=amd64
$ PLATFORM=$(uname -s)_$ARCH
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

```
$ curl -sLO
"https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_$PLATFORM.tar.gz"

$ tar -xzf eksctl_$PLATFORM.tar.gz -C /tmp && rm eksctl_$PLATFORM.tar.gz

$ sudo mv /tmp/eksctl /usr/local/bin

$ eksctl version
```



9.2. Create EKS Master & Worker Nodes

1. Create your AWS EKS cluster with nodes (this will take 10-20 minutes)

   Example command

   eksctl create cluster \

   --name [cluster-name] \

   --version [eks-cluster-version] \

   --region [region-name] \

   --nodegroup-name [node-name] \

   --node-type [shape_of_node] \

   --managed

   Note: *Replace value as per your configuration.*

```
eksctl create cluster \
--name eks-cluster-cli \
--version 1.34 \
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

```
--region us-east-1 \
--nodegroup-name linux-nodes \
--node-type t3.medium \
--managed
```

```
[ec2-user@ip-172-31-26-27 ~]$ eksctl version
0.223.0
[ec2-user@ip-172-31-26-27 ~]$ eksctl create cluster \
--name eks-cluster-cli \
--version 1.34 \
--region us-east-1 \
--nodegroup-name linux-nodes \
--node-type t3.medium \
--managed
2026-02-24 07:34:29 [ℹ]  eksctl version 0.223.0
2026-02-24 07:34:29 [ℹ]  using region us-east-1
2026-02-24 07:34:29 [ℹ]  setting availability zones to [us-east-1a us-east-1c]
2026-02-24 07:34:29 [ℹ]  subnets for us-east-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-02-24 07:34:29 [ℹ]  subnets for us-east-1c - public:192.168.32.0/19 private:192.168.96.0/19
2026-02-24 07:34:29 [ℹ]  nodegroup "linux-nodes" will use "" [AmazonLinux2023/1.34]
2026-02-24 07:34:29 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed n
be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn mor
ode/
2026-02-24 07:34:29 [ℹ]  using Kubernetes version 1.34
2026-02-24 07:34:29 [ℹ]  creating EKS cluster "eks-cluster-cli" in "us-east-1" region with managed nodes
2026-02-24 07:34:29 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-02-24 07:34:29 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 -
2026-02-24 07:34:29 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "eks-cluste
2026-02-24 07:34:29 [ℹ]  CloudWatch logging will not be enabled for cluster "eks-cluster-cli" in "us-east-1"
2026-02-24 07:34:29 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)}
-cluster-cli'
2026-02-24 07:34:29 [ℹ]  default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2026-02-24 07:34:29 [ℹ]
2 sequential tasks: { create cluster control plane "eks-cluster-cli",
```

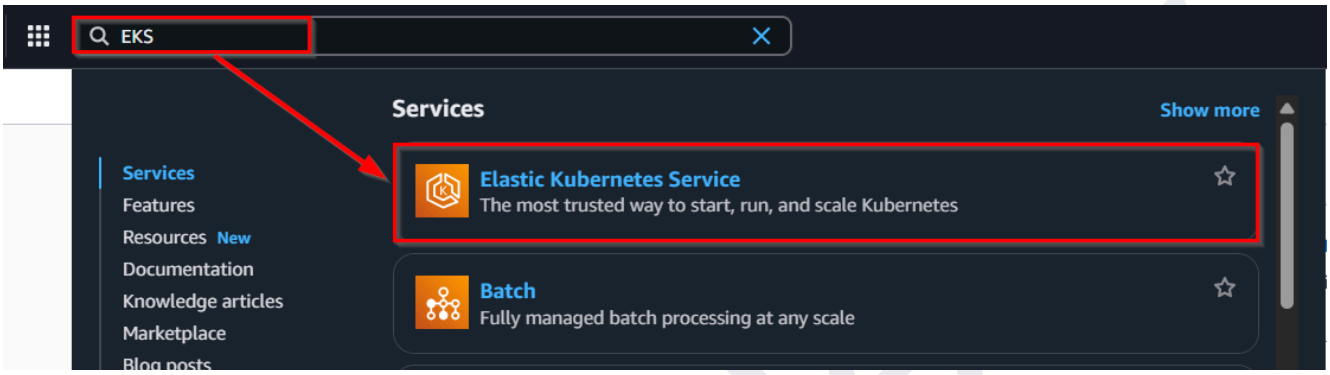
```
2025-09-12 07:28:37 [ℹ]  successfully created addon: metrics-server
2025-09-12 07:28:38 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster,
 way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed,
dentityAssociations`, and run `eksctl update addon`
2025-09-12 07:28:38 [ℹ]  creating addon: vpc-cni
2025-09-12 07:28:38 [ℹ]  successfully created addon: vpc-cni
2025-09-12 07:28:38 [ℹ]  creating addon: kube-proxy
2025-09-12 07:28:38 [ℹ]  successfully created addon: kube-proxy
2025-09-12 07:28:39 [ℹ]  creating addon: coredns
2025-09-12 07:28:39 [ℹ]  successfully created addon: coredns
2025-09-12 07:30:39 [ℹ]  building managed nodegroup stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:30:39 [ℹ]  deploying stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:30:40 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:31:10 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:31:49 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:33:18 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:33:18 [ℹ]  waiting for the control plane to become ready
2025-09-12 07:33:19 [✓]  saved kubeconfig as "/home/ec2-user/.kube/config"
2025-09-12 07:33:19 [ℹ]  no tasks
2025-09-12 07:33:19 [✓]  all EKS cluster resources for "eks-cluster-cli" have been created
2025-09-12 07:33:19 [ℹ]  nodegroup "linux-nodes" has 2 node(s)
2025-09-12 07:33:19 [ℹ]  node "ip-192-168-21-155.us-west-2.compute.internal" is ready
2025-09-12 07:33:19 [ℹ]  node "ip-192-168-55-167.us-west-2.compute.internal" is ready
2025-09-12 07:33:19 [ℹ]  waiting for at least 2 node(s) to become ready in "linux-nodes"
2025-09-12 07:33:19 [ℹ]  nodegroup "linux-nodes" has 2 node(s)
2025-09-12 07:33:19 [ℹ]  node "ip-192-168-21-155.us-west-2.compute.internal" is ready
2025-09-12 07:33:19 [ℹ]  node "ip-192-168-55-167.us-west-2.compute.internal" is ready
2025-09-12 07:33:19 [✓]  created 1 managed nodegroup(s) in cluster "eks-cluster-cli"
2025-09-12 07:33:20 [ℹ]  kubectl command should work with "/home/ec2-user/.kube/config", try 'kubectl get nodes'
2025-09-12 07:33:20 [✓]  EKS cluster "eks-cluster-cli" in "us-west-2" region is ready
[ec2-user@ip-172-31-40-137 ~]$
```

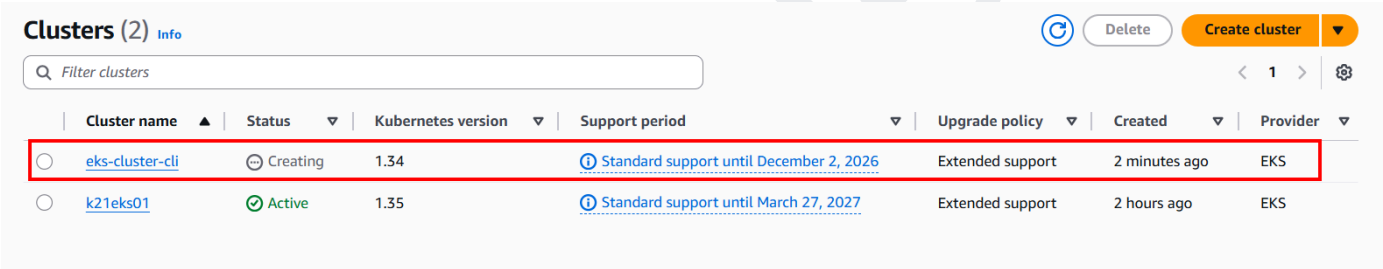
Note: If you encounter any **CloudFormation Stack Already Exists issue,** please refer to the **troubleshooting section**.

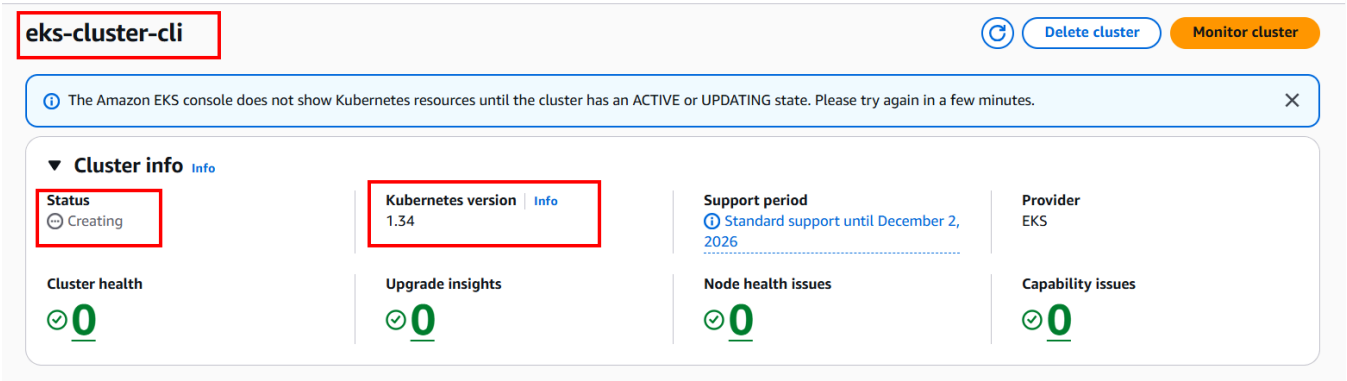
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

9.3. Verify Master & Worker Node from Console

1. Verify from the EKS Cluster using the Console

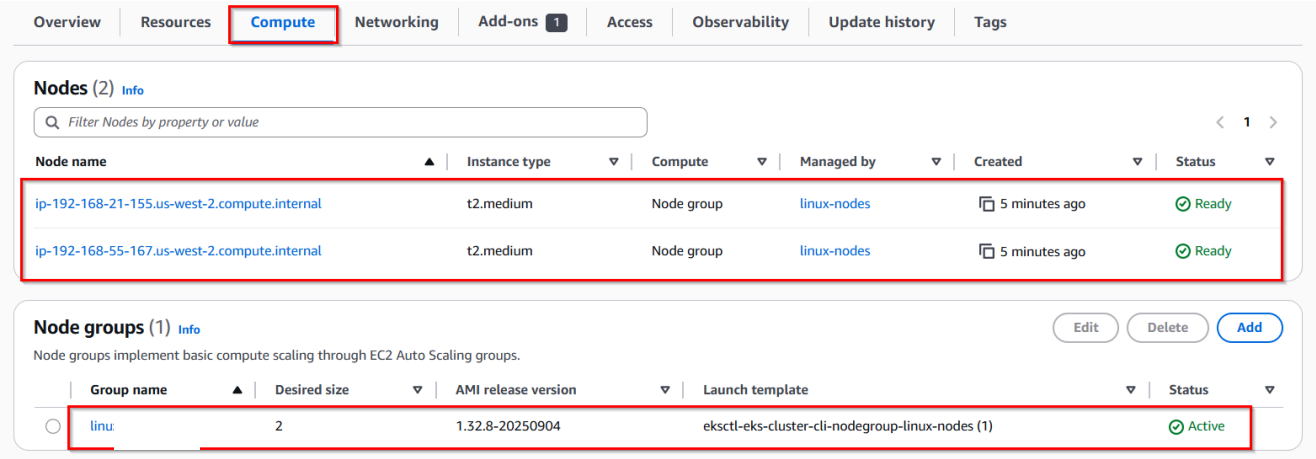






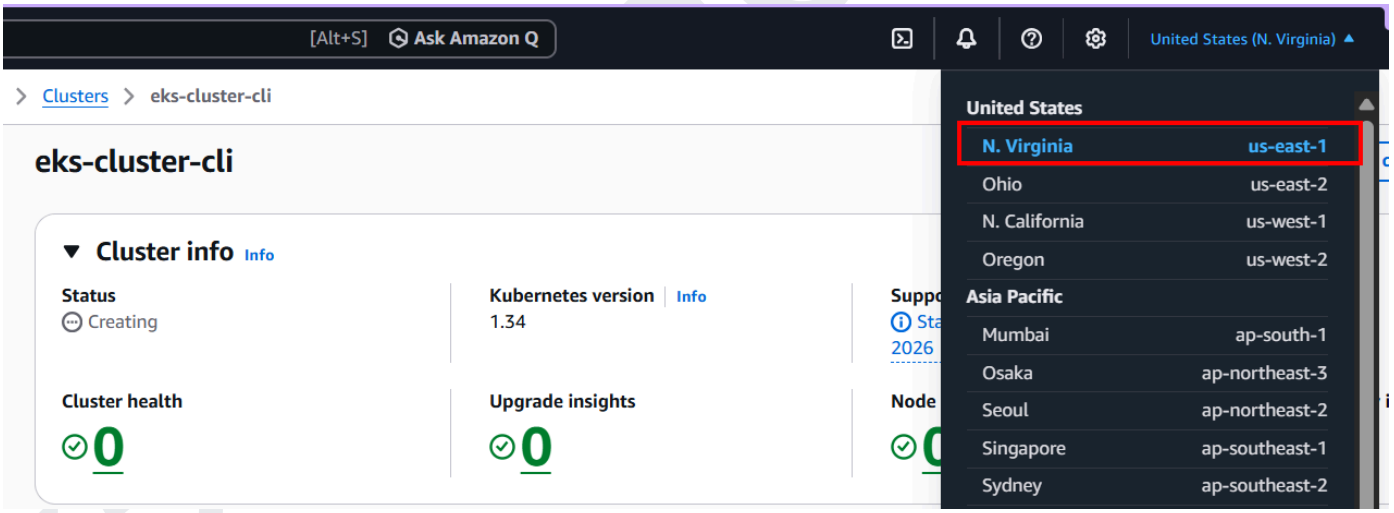
2. Verify the worker nodes
   - Click On Compute
   - Click on Node Group

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



9.4. Set up the kubectl configuration

1. Check your **Cluster Name** & **Region Name** where the EKS Master node is running from the console



2. Check status of Cluster as: aws eks --region [EKS_Region] describe-cluster --name [EKS_Cluster_Name] --query cluster.status

```
aws eks --region us-east-1 describe-cluster --name eks-cluster-cli --query cluster.status
```

```
[ec2-user@ip-172-31-40-137 ~]$ aws eks --region us-west-2 describe-cluster --name eks-cluster-cli --query cluster.status
"ACTIVE"
[ec2-user@ip-172-31-40-137 ~]$
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

3. [Optional] If you have ever configured/used this EC2 Instance with Kubernetes server i.e. if you already have a config file ($HOME/.kube/config) then move the existing file from the EC2 instance.

```
mv $HOME/.kube/config  $HOME/.kube/config.old
```

4. Configure kubectl with EKS API Server credential: aws eks --region [EKS_Region] update-kubeconfig --name [EKS_Cluster_Name]

```
aws eks --region us-east-1 update-kubeconfig --name eks-cluster-cli
```

```
[ec2-user@ip-172-31-40-137 ~]$ mv $HOME/.kube/config  $HOME/.kube/config.old
[ec2-user@ip-172-31-40-137 ~]$
[ec2-user@ip-172-31-40-137 ~]$ aws eks --region us-west-2 update-kubeconfig --name eks-cluster-cli
Added new context arn:aws:eks:us-west-2:350903502296:cluster/eks-cluster-cli to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-40-137 ~]$
```

5. Validate the kubectl configuration on the master node

```
kubectl get svc
```

```
[ec2-user@ip-172-31-40-137 ~]$ kubectl get svc
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   16m
[ec2-user@ip-172-31-40-137 ~]$
[ec2-user@ip-172-31-40-137 ~]$
```

6. Validate worker nodes

```
kubectl get nodes
```

```
[ec2-user@ip-172-31-40-137 ~]$ kubectl get nodes
NAME                                        STATUS   ROLES    AGE   VERSION
ip-192-168-21-155.us-west-2.compute.internal   Ready    <none>   12m   v1.32.8-eks-99d6cc0
ip-192-168-55-167.us-west-2.compute.internal   Ready    <none>   12m   v1.32.8-eks-99d6cc0
[ec2-user@ip-172-31-40-137 ~]$
[ec2-user@ip-172-31-40-137 ~]$
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

9.5. Delete EKS Cluster using EKSCTL

1. Delete your AWS EKS cluster with nodes (this will take 10-20 minutes)

   Example:

   eksctl delete cluster \

   --name [cluster-name] \

   --region [region-name]

```
eksctl delete cluster \
--name eks-cluster-cli \
--region us-east-1
```

```
[ec2-user@ip-172-31-40-137 ~]$ eksctl delete cluster \
--name eks-cluster-cli \
--region us-west-2
2025-09-12 07:46:31 [ℹ]  deleting EKS cluster "eks-cluster-cli"
2025-09-12 07:46:31 [ℹ]  will drain 0 unmanaged nodegroup(s) in cluster "eks-cluster-cli"
2025-09-12 07:46:31 [ℹ]  starting parallel draining, max in-flight of 1
2025-09-12 07:46:31 [ℹ]  deleted 0 Fargate profile(s)
2025-09-12 07:46:31 [ℹ]  cleaning up AWS load balancers created by Kubernetes objects of Kind Service or Ingress
2025-09-12 07:46:32 [ℹ]
2 sequential tasks: { delete nodegroup "linux-nodes", delete cluster control plane "eks-cluster-cli" [async]
}
2025-09-12 07:46:32 [ℹ]  will delete stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:46:32 [ℹ]  waiting for stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes" to get deleted
2025-09-12 07:46:32 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:47:02 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:47:36 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:48:38 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:49:45 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:51:31 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:52:05 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:53:20 [ℹ]  waiting for CloudFormation stack "eksctl-eks-cluster-cli-nodegroup-linux-nodes"
2025-09-12 07:53:20 [ℹ]  will delete stack "eksctl-eks-cluster-cli-cluster"
2025-09-12 07:53:20 [✓]  all cluster resources were deleted
[ec2-user@ip-172-31-40-137 ~]$
```

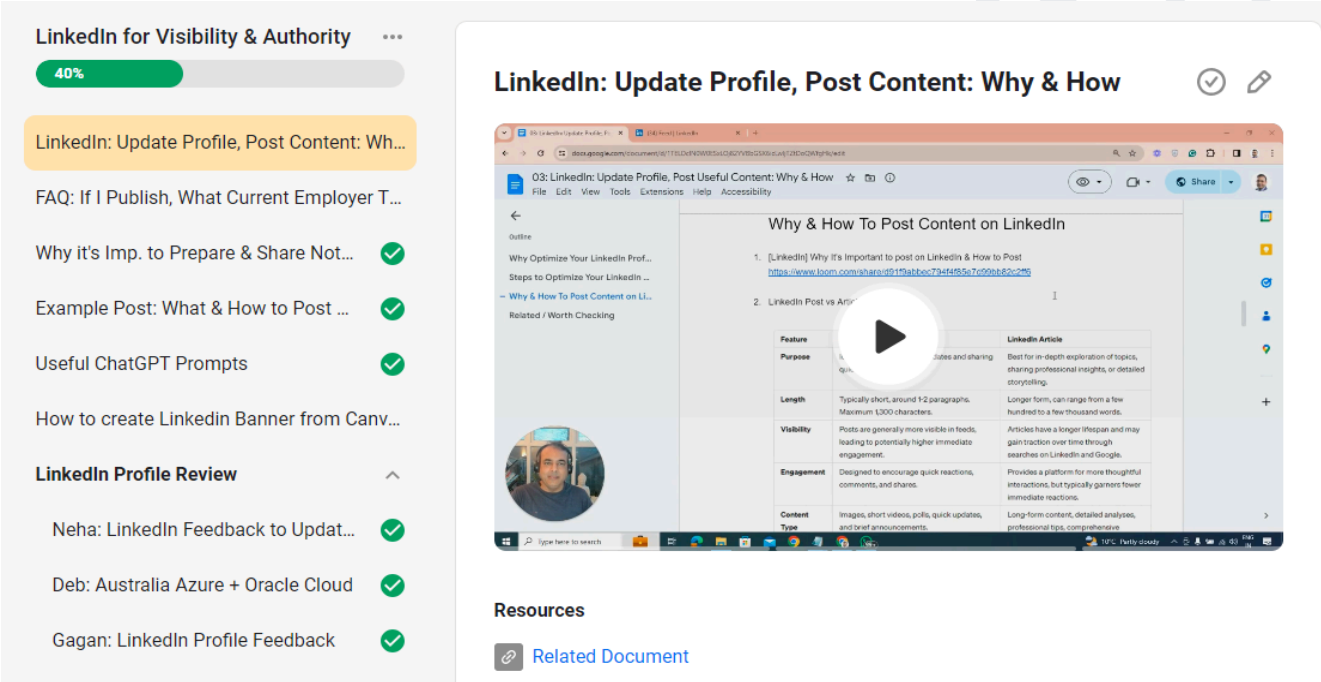
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

10. SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and in the community.

Please watch the below👇 video below to understand why it's important to post & how to post:

https://www.skool.com/k21academy/classroom/0c24b6af?md=f7975c8e525b4d5d8191e131bdef5740



👩‍💼 As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend that you create one for yourself.

📝 🖥️ By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

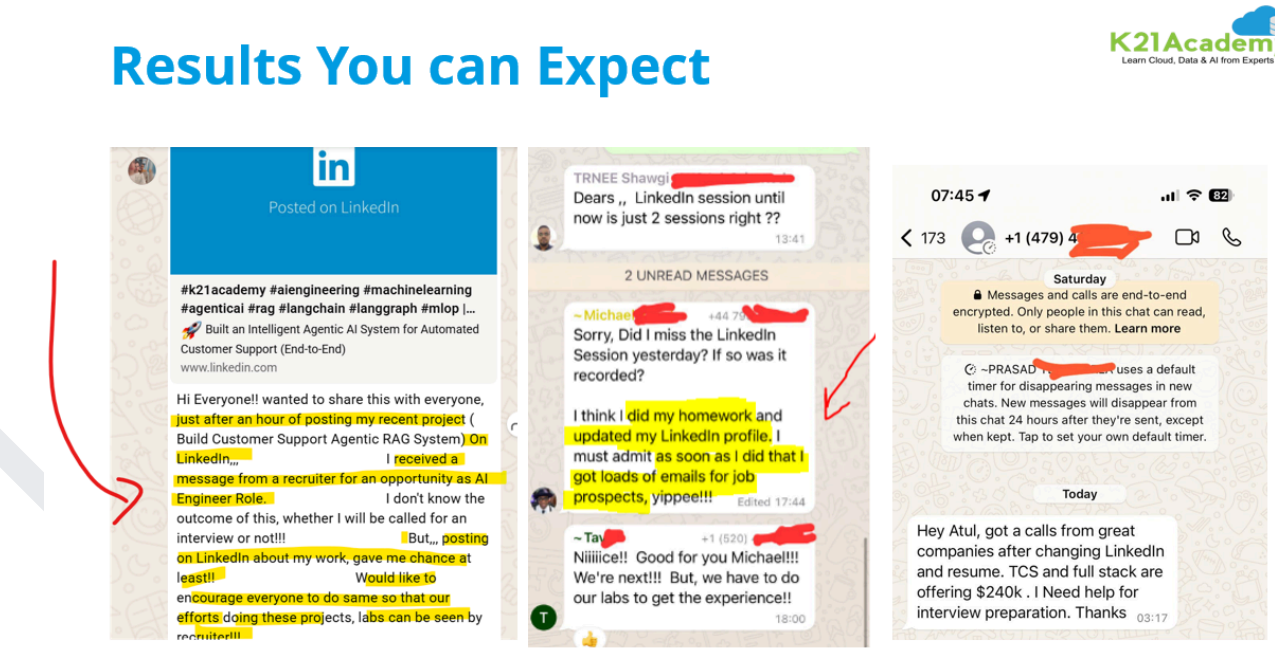
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

10.1. Share Hands-On Lab on LinkedIn to Build Authority & Job Opportunities

1. This will attract recruiters and potential employers to your profile and increase your Authority & LinkedIn reach.

2. **Do remember to tag so we can share too, and we'll ask other members to share too (this will increase your post reach)**
   a. K21Academy (https://www.linkedin.com/company/k21academy) &
   b. Atul Kumar (https://www.linkedin.com/in/atulk21academy/), as we will further circulate in our network, too, to increase your reach.

3. Here is what you can expect
   a. Jessy got a message from a recruiter for an AI Engineer Role
   b. Michael got loads of emails for Job prospects
   c. Prasad got calls from great companies that are offering $240K Salary

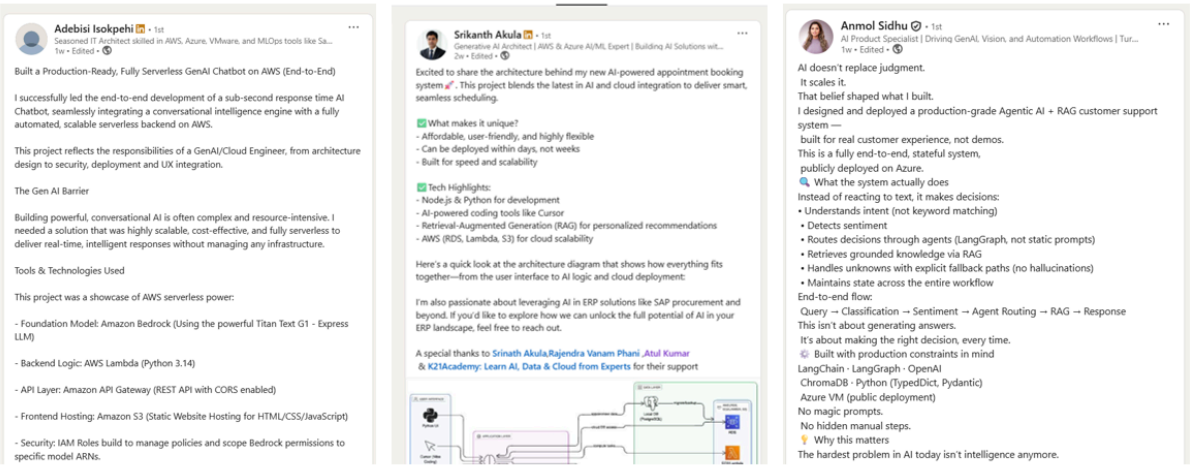


4. Example Links on How to post

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

a. Example Posts

Post on LinkedIn for Interview Calls



Adebisi Isokpehi • 1st
Seasoned IT Architect skilled in AWS, Azure, VMware, and MLOps tools like Sa...
1w • Edited •

Built a Production-Ready, Fully Serverless GenAI Chatbot on AWS (End-to-End)

I successfully led the end-to-end development of a sub-second response time AI Chatbot, seamlessly integrating a conversational intelligence engine with a fully automated, scalable serverless backend on AWS.

This project reflects the responsibilities of a GenAI/Cloud Engineer, from architecture design to security, deployment and UX integration.

The Gen AI Barrier

Building powerful, conversational AI is often complex and resource-intensive. I needed a solution that was highly scalable, cost-effective, and fully serverless to deliver real-time, intelligent responses without managing any infrastructure.

Tools & Technologies Used

This project was a showcase of AWS serverless power:

- Foundation Model: Amazon Bedrock (Using the powerful Titan Text G1 - Express LLM)

- Backend Logic: AWS Lambda (Python 3.14)

- API Layer: Amazon API Gateway (REST API with CORS enabled)

- Frontend Hosting: Amazon S3 (Static Website Hosting for HTML/CSS/JavaScript)

- Security: IAM Roles build to manage policies and scope Bedrock permissions to specific model ARNs.

Srikanth Akula • 1st
Generative AI Architect | AWS & Azure AI/ML Expert | Building AI Solutions wit...
2w • Edited •

Excited to share the architecture behind my new AI-powered appointment booking system. This project blends the latest in AI and cloud integration to deliver smart, seamless scheduling.

What makes it unique?
- Affordable, user-friendly, and highly flexible
- Can be deployed within days, not weeks
- Built for speed and scalability

Tech Highlights:
- Node.js & Python for development
- AI-powered coding tools like Cursor
- Retrieval-Augmented Generation (RAG) for personalized recommendations
- AWS (RDS, Lambda, S3) for cloud scalability

Here's a quick look at the architecture diagram that shows how everything fits together—from the user interface to AI logic and cloud deployment:

I'm also passionate about leveraging AI in ERP solutions like SAP procurement and beyond. If you'd like to explore how we can unlock the full potential of AI in your ERP landscape, feel free to reach out.

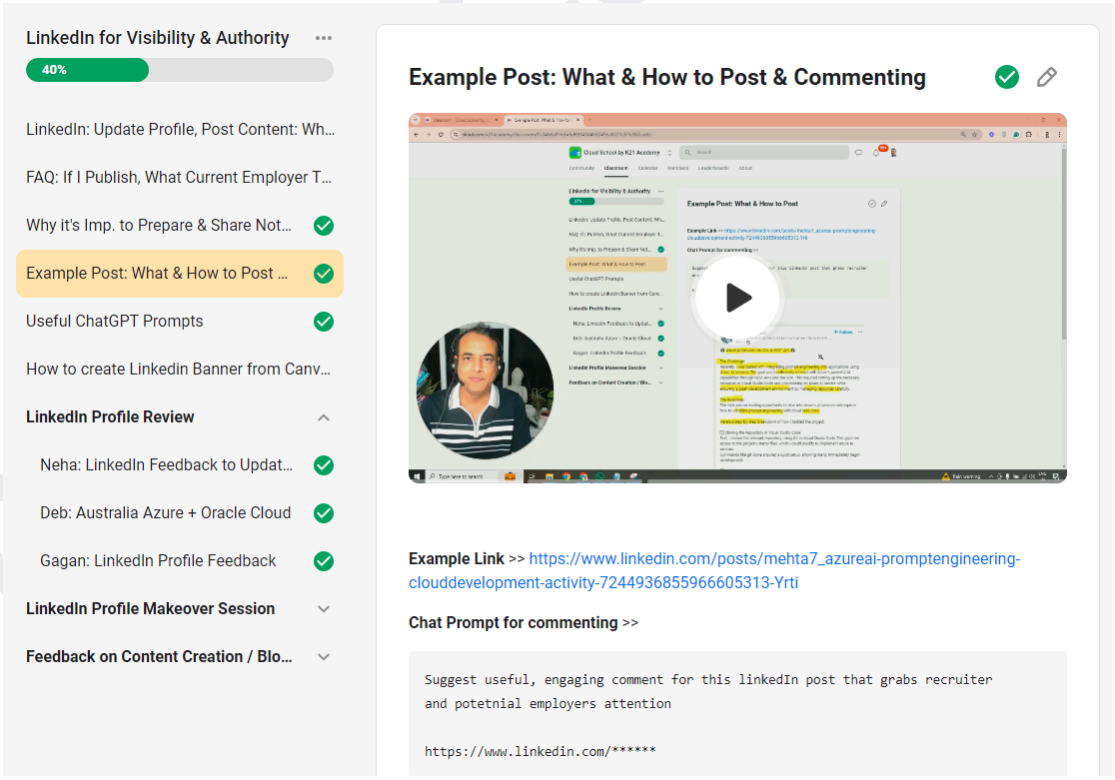
A special thanks to Srinath Akula,Rajendra Vanam Phani ,Atul Kumar & K21Academy: Learn AI, Data & Cloud from Experts for their support

Anmol Sidhu • 1st
AI Product Specialist | Driving GenAI, Vision, and Automation Workflows | Tur...
1w • Edited •

AI doesn't replace judgment.
It scales it.
That belief shaped what I built.
I designed and deployed a production-grade Agentic AI + RAG customer support system —
built for real customer experience, not demos.
This is a fully end-to-end, stateful system,
publicly deployed on Azure.
What the system actually does
Instead of reacting to text, it makes decisions:
• Understands intent (not keyword matching)
• Detects sentiment
• Routes decisions through agents (LangGraph, not static prompts)
• Retrieves grounded knowledge via RAG
• Handles unknowns with explicit fallback paths (no hallucinations)
• Maintains state across the entire workflow
End-to-end flow:
Query → Classification → Sentiment → Agent Routing → RAG → Response
This isn't about generating answers.
It's about making the right decision, every time.
Built with production constraints in mind
LangChain · LangGraph · OpenAI
ChromaDB · Python (TypedDict, Pydantic)
Azure VM (public deployment)
No magic prompts.
No hidden manual steps.
Why this matters
The hardest problem in AI today isn't intelligence anymore.

b. Example Video & Post

https://www.skool.com/k21academy/classroom/0c24b6af



LinkedIn for Visibility & Authority
40%

LinkedIn: Update Profile, Post Content: Wh...
FAQ: If I Publish, What Current Employer T...
Why it's Imp. to Prepare & Share Not...
Example Post: What & How to Post ...
Useful ChatGPT Prompts
How to create Linkedin Banner from Canv...
LinkedIn Profile Review
Neha: LinkedIn Feedback to Updat...
Deb: Australia Azure + Oracle Cloud
Gagan: LinkedIn Profile Feedback
LinkedIn Profile Makeover Session
Feedback on Content Creation / Blo...

Example Post: What & How to Post & Commenting

Example Link >> https://www.linkedin.com/posts/mehta7_azureai-promptengineering-clouddevelopment-activity-7244936855966605313-Yrti

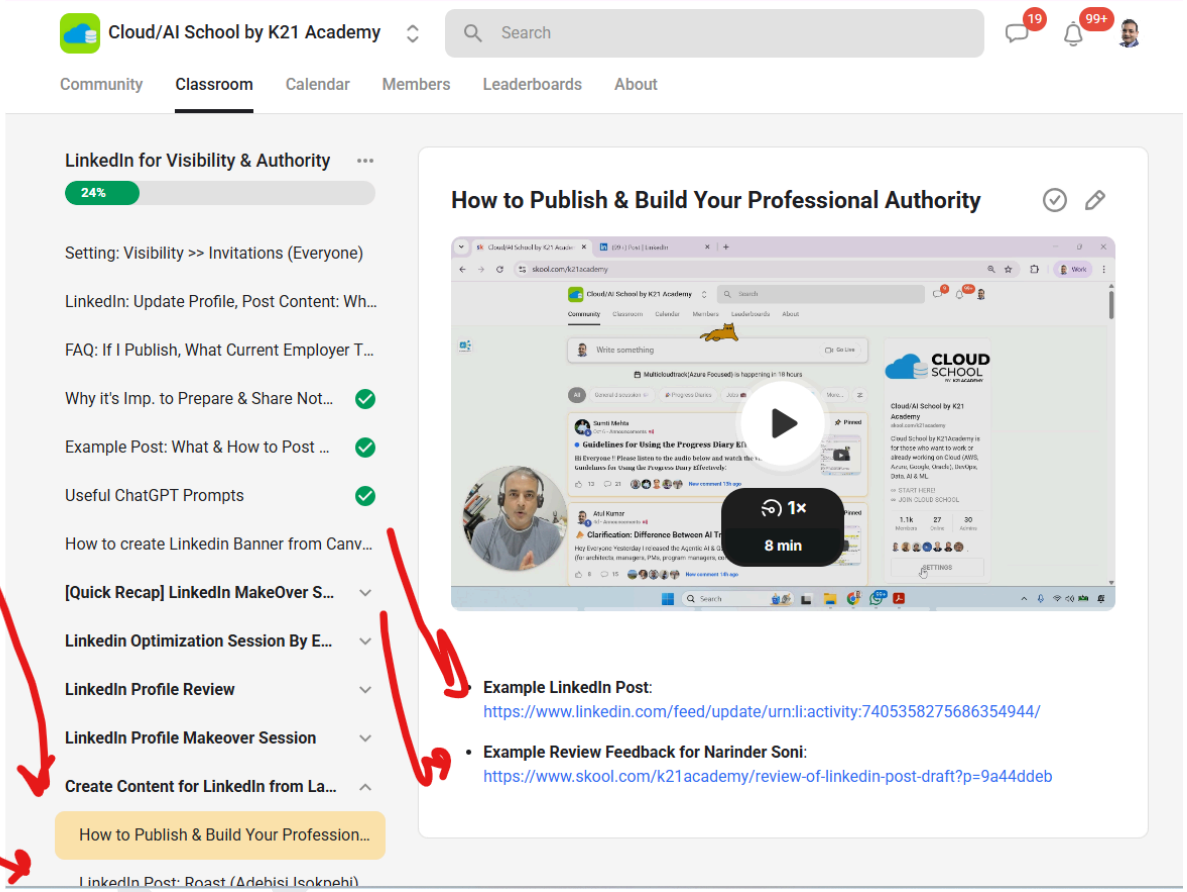
Chat Prompt for commenting >>

```
Suggest useful, engaging comment for this linkedIn post that grabs recruiter
and potetnial employers attention

https://www.linkedin.com/******
```

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

c. Check few more linkedIn post examples (do's and don'ts): https://www.skool.com/k21academy/classroom/0c24b6af?md=edfe6092fbb84f6bba03ceb439d1e00a

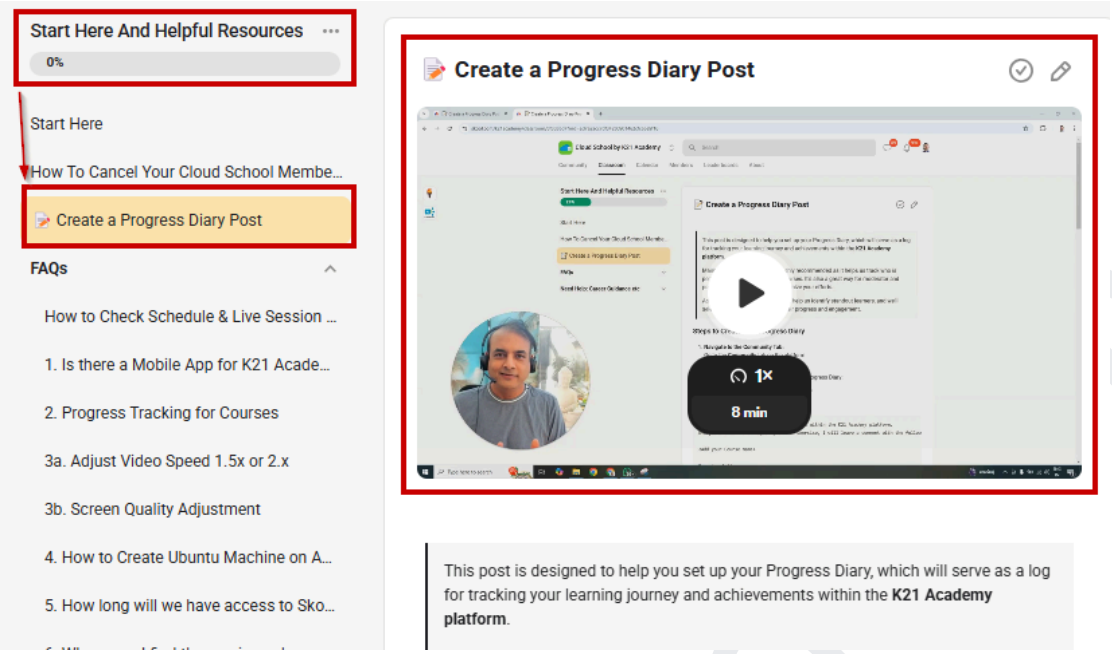


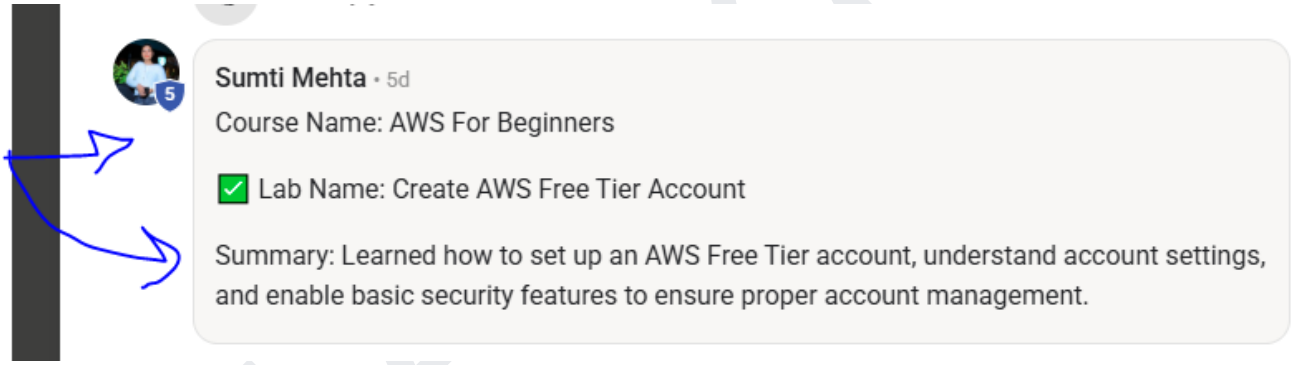
10.2. Share the win on the Community via Progress Diary

Share screenshots in Cloud School Community under your **'Progress Diary'** (If you have not created a Progress Diary yet, then check at https://k21academy.com/progressDiary

This will boost your confidence in the progressive path you are following, and encourage/inspire others to perform these hands-on labs.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.





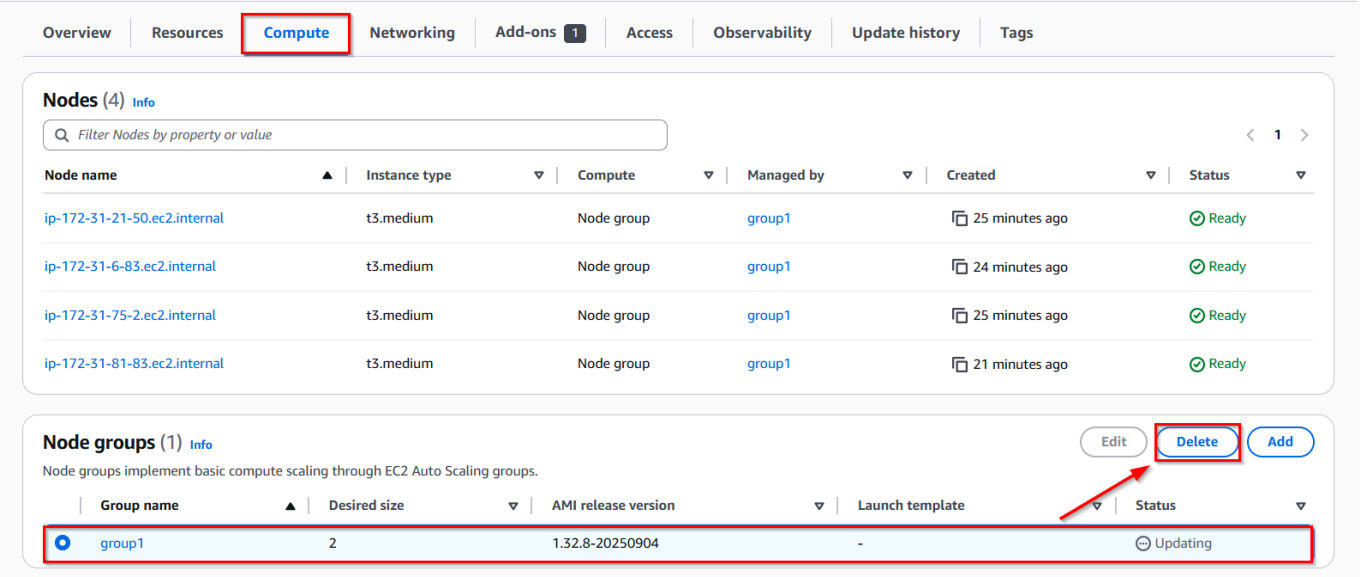
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

11. Delete EKS Cluster from the Console

Deleting the EKS cluster from the console ensures a clean environment before recreating it with the CLI. This process allows for automation, consistency, and better control over cluster configuration.

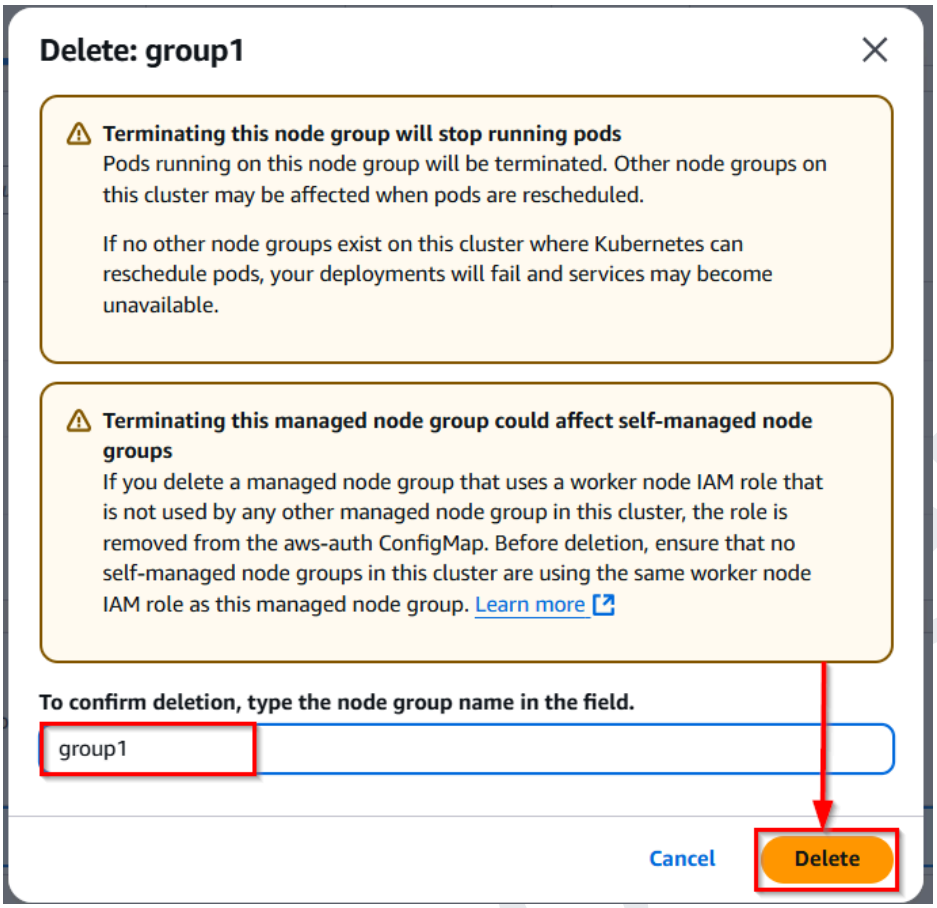
11.1. Delete Worker Nodes

1. Delete Node Group

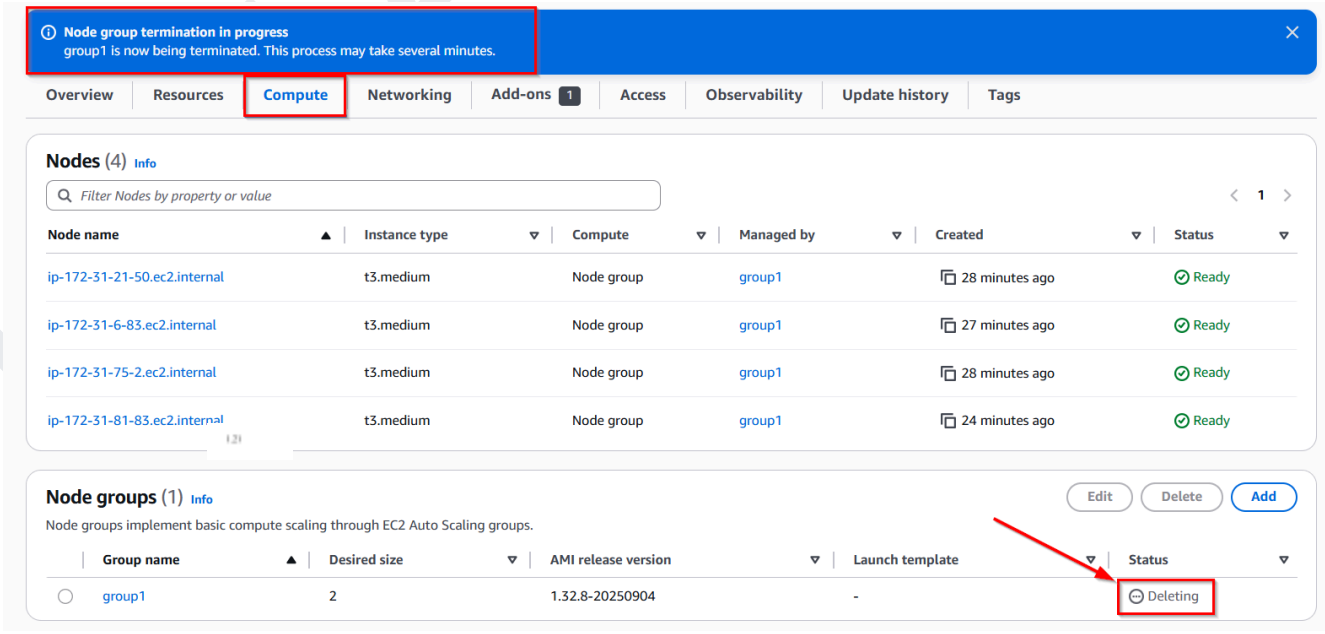


2. Now confirm the deletion.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



3. Now Deletion is starting. It will take a few minutes to delete it completely.

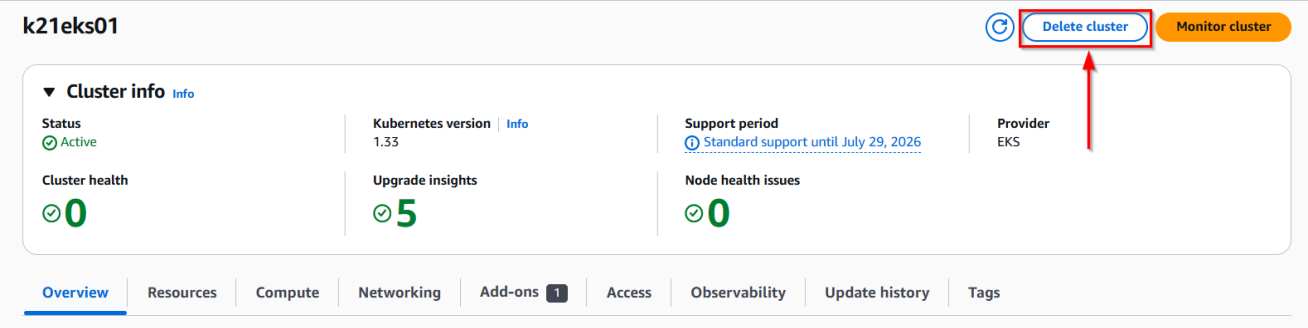


***Note**: This process will take approx. 10-15 minutes*

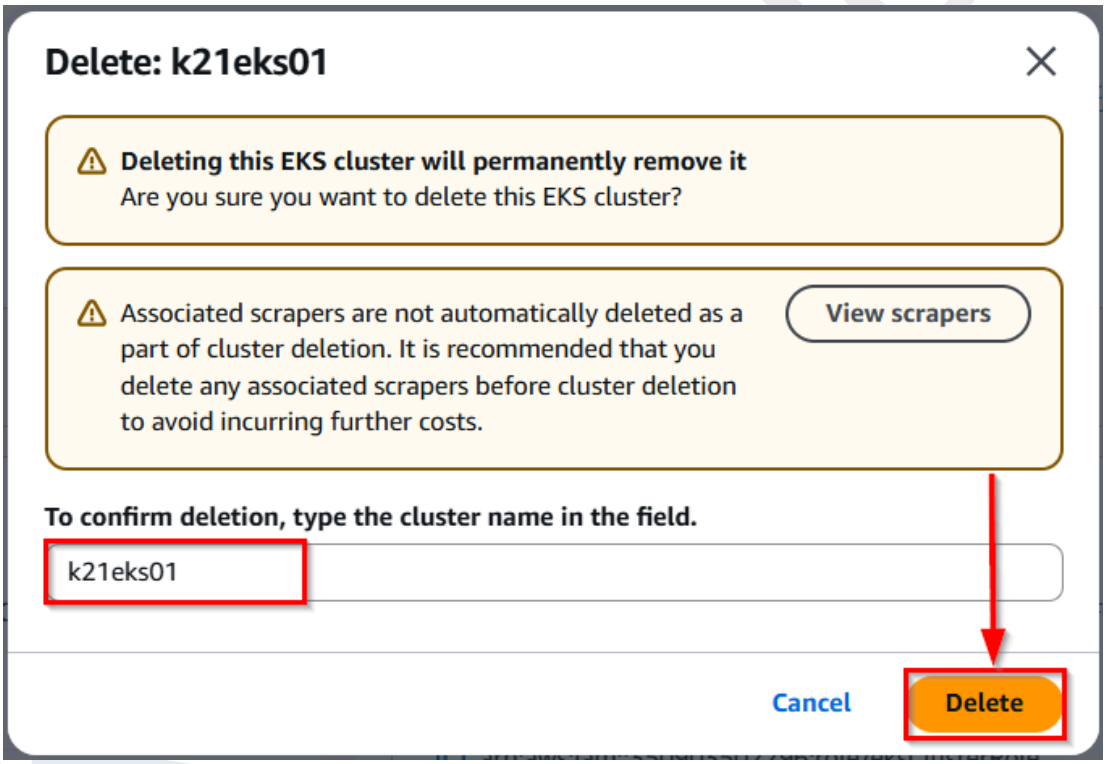
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

11.2. Delete Master Node

1. Now, go to the master node and click on Delete cluster.



2. Now enter the Cluster name and click on Delete.

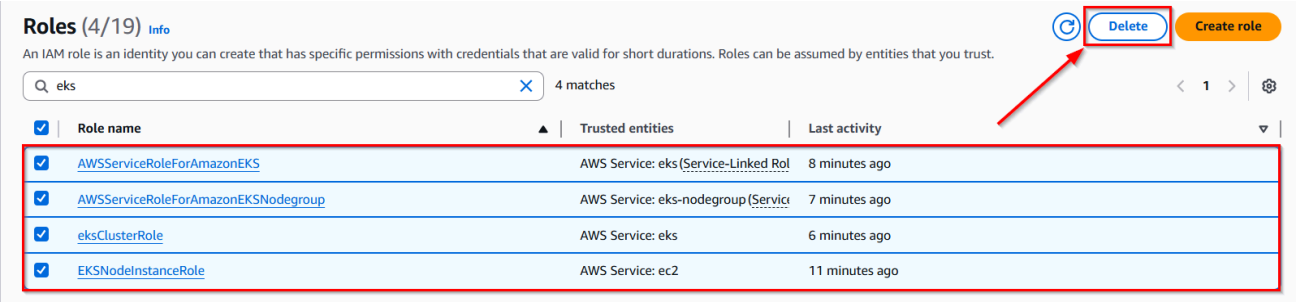


Note: if you encounter any issue related to **not being able to delete the Cluster,** please refer to the **troubleshooting section**.

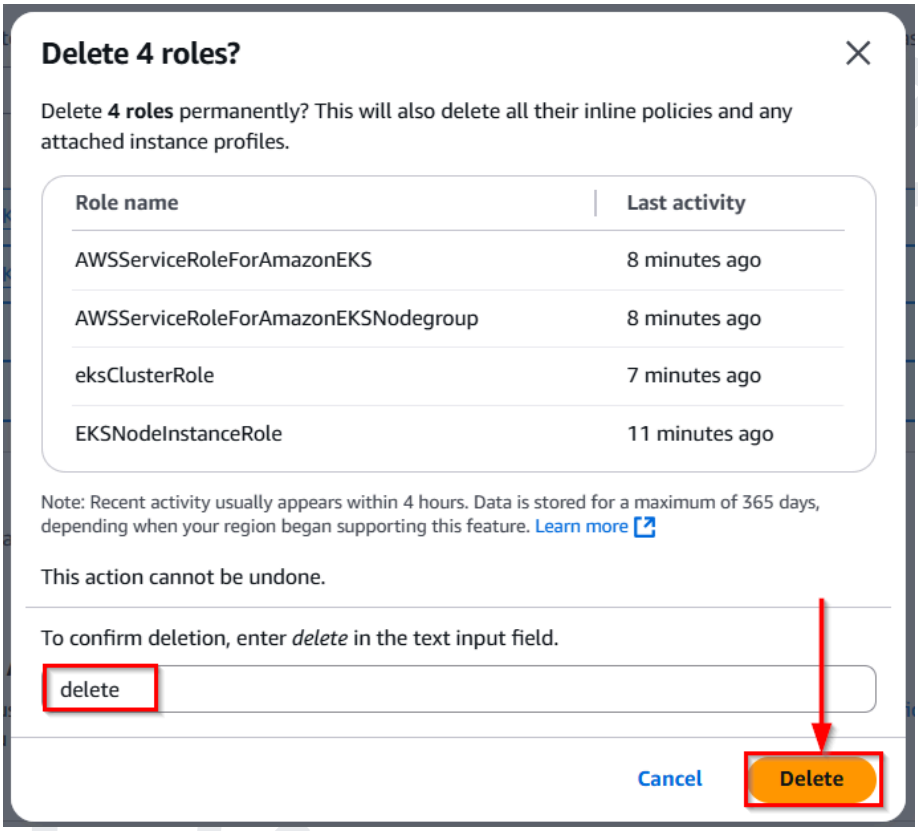
11.3. Delete IAM Roles

1. Go to the IAM dashboard, select all roles that you created, and Click on delete.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



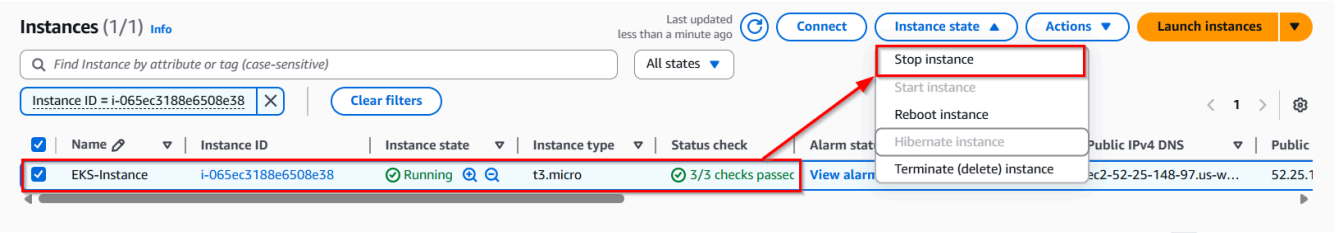
2. Now confirm the deletion of roles by clicking on Delete.

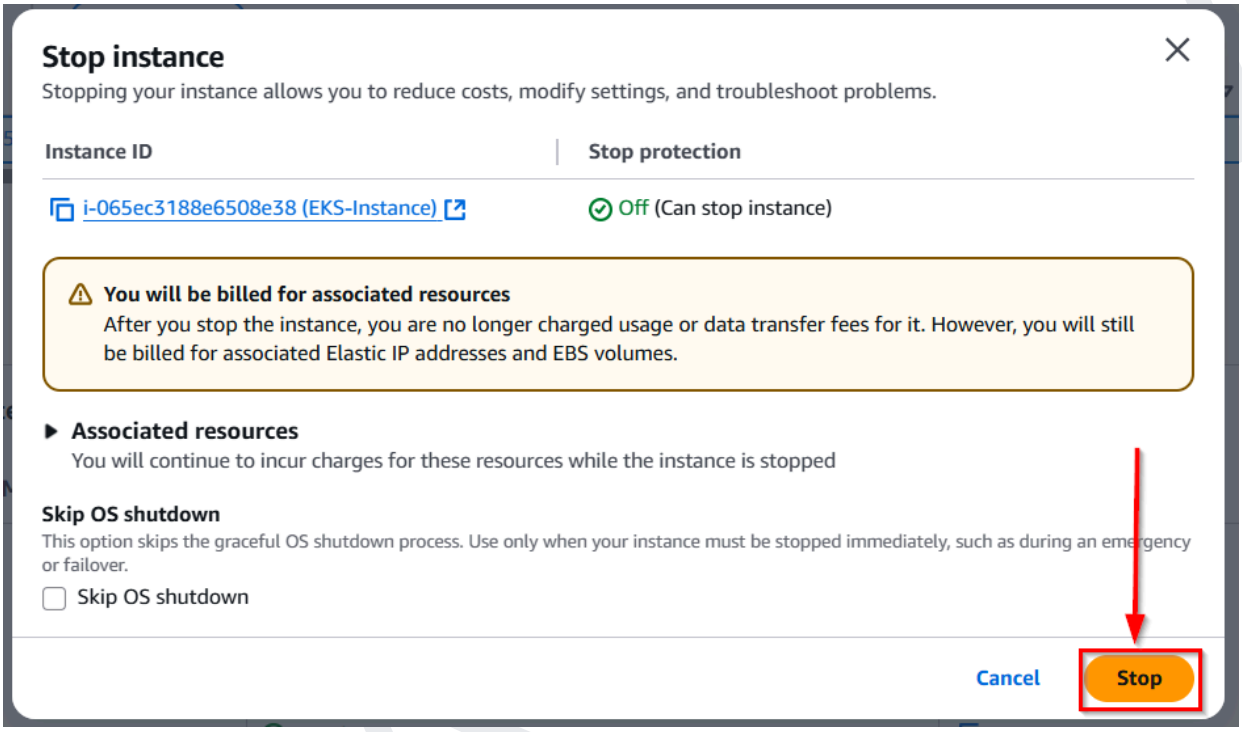


11.4. Stop/Terminate EC2 instance

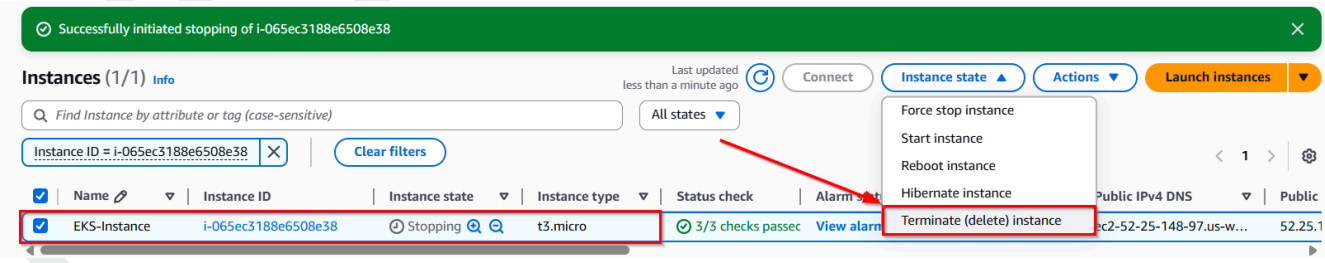
1. Select the instance and go to the instance state If you want to use the same instance again, then you can preserve the instance by **stopping** the instance.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

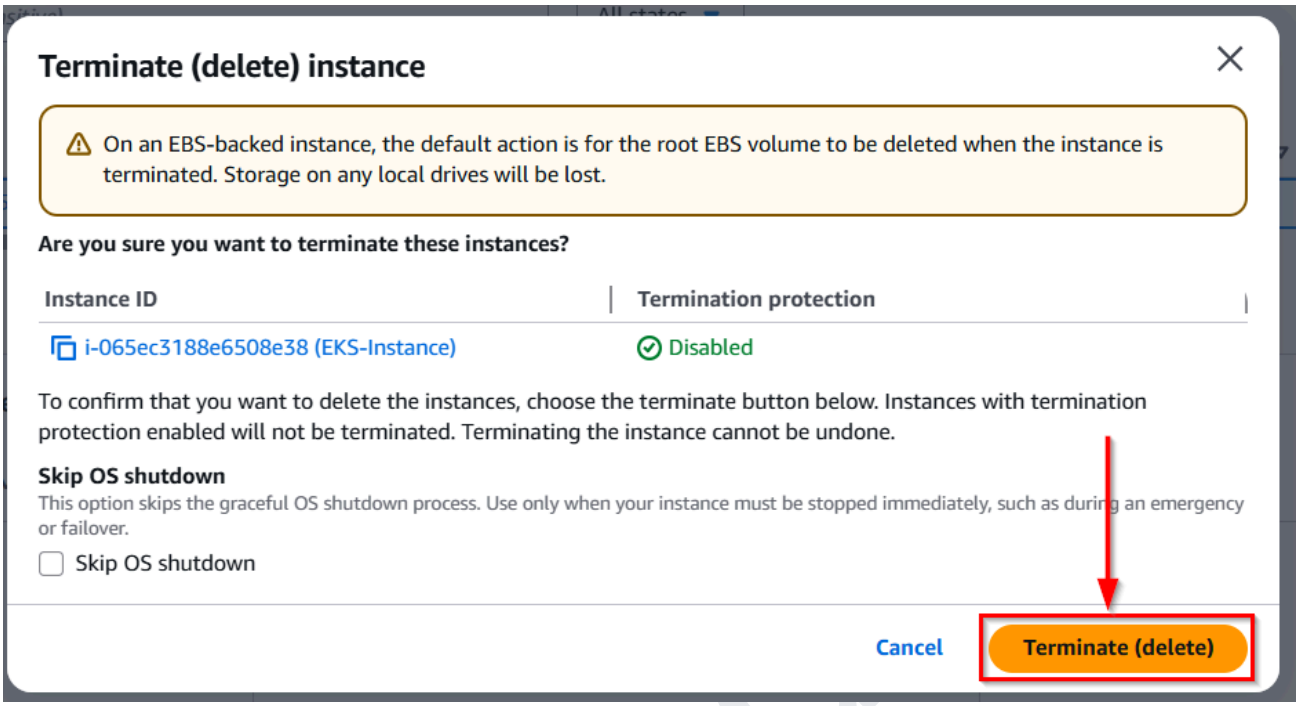




2. If you do not want to keep the instance, you can terminate the instance by clicking on Terminate Instance.



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

12. TROUBLESHOOTING

Note: In this section, we will document any issues and their fixes that may arise while performing this lab. If you encounter any issues, please let us know so that we can update this section with the relevant information and help others who may face similar problems

12.1. Error on kubectl 'Unable to connect to the server: dial tcp: lookup *. eks.amazonaws.com on IP:53: no such host

12.1.1. ISSUE

When executing a command like kubectl get svc, getting the following error

Unable to connect to the server: dial tcp: lookup 092122AC5DBA16FEF19320503E4DF5B8.gr7.eu-west-2.eks.amazonaws.com on 172.31.0.2:53: no such host

```
[ec2-user@ip-172-31-34-236 ~]$ kubectl get svc
Unable to connect to the server: dial tcp: lookup 092122AC5DBA16FEF19320503E4DF5B8.gr7.eu-west-2.eks.amazonaws.com on 172.31.0.2:53: no such host
[ec2-user@ip-172-31-34-236 ~]$
```

12.1.2. CAUSE:

EKS Cluster was deleted, and the old config file $HOME/.kube/config still exists

12.1.3. FIX:

1. If you already have a config file ($HOME/.kube/config) then move the existing file from the EC2 instance.

```
mv $HOME/.kube/config  $HOME/.kube/config.old
```

```
[ec2-user@ip-172-31-34-236 ~]$ mv $HOME/.kube/config  $HOME/.kube/config.old
[ec2-user@ip-172-31-34-236 ~]$
```

2. And then Configure kubectl with EKS API Server credential: aws eks --region [EKS_Region] update-kubeconfig --name [EKS_Cluster_Name]

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

```
aws eks --region us-east-1 update-kubeconfig --name k21-eks-cluster-cli
```

```
[ec2-user@ip-172-31-53-18 ~]$ aws eks --region us-east-1 update-kubeconfig --name k21-eks-cluster-cli
Added new context arn:aws:eks:us-east-1:532661433831:cluster/k21-eks-cluster-cli to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-53-18 ~]$
```

3. Now run the ***kubectl get svc*** command again.

12.2. CloudFormation Stack Already Exists

12.2.1. Issue:

When we are creatingan EKSCTL cluster using EKSCTL and getting CloudFormation already exists error.

```
eksctl create cluster --name asif-eks-cli --version 1.18 --region us-east-1 --zones "us-east-1a us-east-1b us-east-1c" --nodegroup-name linux-nodes --node-type t3.medium --managed
```

12.2.2. Cause:

The "CloudFormation already exists" typically occurs when you're trying to create an Amazon EKS cluster using eksctl, but there's an existing CloudFormation stack with the same name in the specified AWS region.

12.2.3. Fix:

We got this type of error because of the command we entered. Please check your command that you have entered; there should not be any extra spaces in the command.

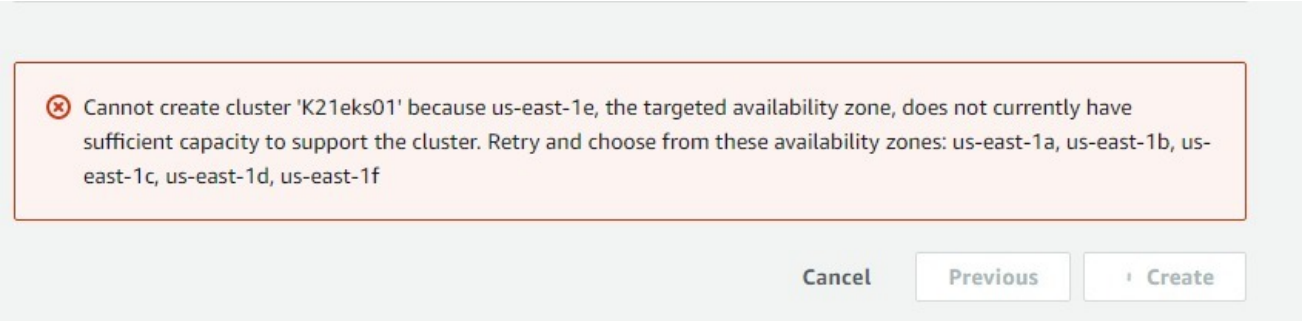
```
eksctl create cluster --name asif-eks-cli --version 1.18 --region us-east-1 --zones "us-east-1a,us-east-1b,us-east-1c" --nodegroup-name linux-nodes --node-type t3.medium --managed
```

12.3. Not able to create the Cluster

12.3.1. Issue:

Not Able to Create the EKS Cluster

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



12.3.2. CAUSE:

The error message "Cannot create cluster in availability zone because the targeted availability zone does not currently have sufficient capacity to support the cluster" is typically encountered when trying to create an Amazon EKS cluster using eksctl in a specific availability zone, but that zone doesn't have the necessary capacity for creating the cluster's control plane.

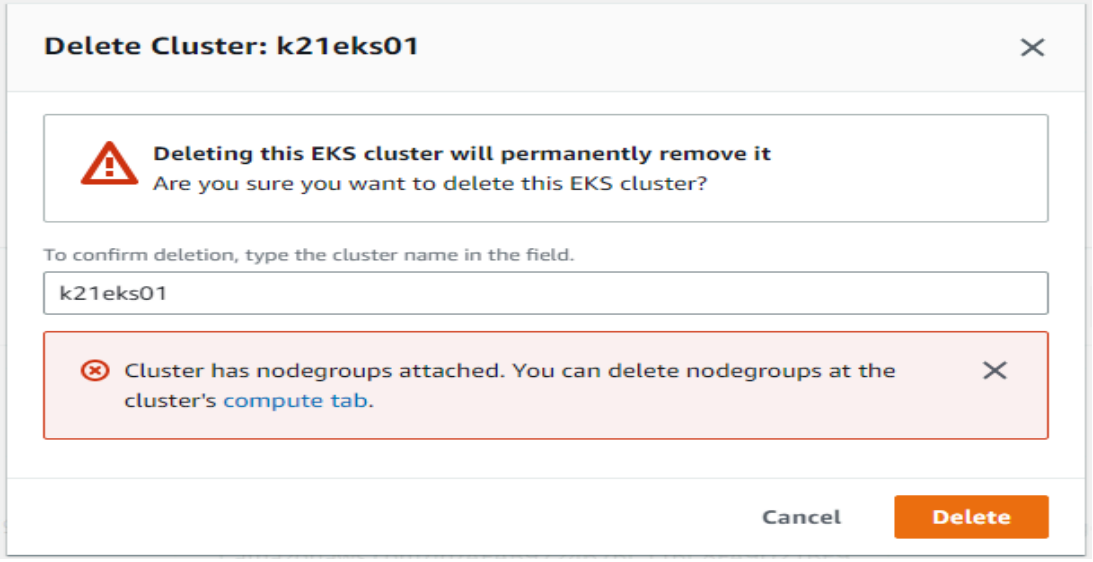
12.3.3. FIX:

Go back to Networking under Step 2 and remove that particular subnet.

12.4. Not able to delete the Cluster

12.4.1. ERROR:

Not Able to Delete the EKS Cluster



Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

12.4.2. CAUSE:

This error is caused by the fact that you cannot delete an EKS cluster until all associated node groups have been removed. Node groups are groups of worker nodes that provide compute capacity to your EKS cluster.

12.4.3. FIX:

You need to make sure that you delete the Worker Node before the Master cluster.

12.5. Couldn't get the current server API group list

Get "https://FB6AEAFA661DC621695835AF1B4FD5E5.gr7.eu-central-2.eks.amazonaws.com/api?timeout=32s": dial tcp 172.31.2.198:443: i/o timeout

12.5.1. ERROR:

Encountering the below error when trying to connect to the EKS Cluster from the Client Machine

```
ubuntu@ip-172-31-4-214:~$ kubectl get nodes
E0424 09:38:56.121608   17247 memcache.go:265] couldn't get current server API group list: Get "https://FB6AEAFA661DC621695835AF1B4FD5E5.gr7.eu-central-2.eks.amazonaws.com/api?timeout=32s": dial tcp 172.31.2.198:443: i/o timeout
```

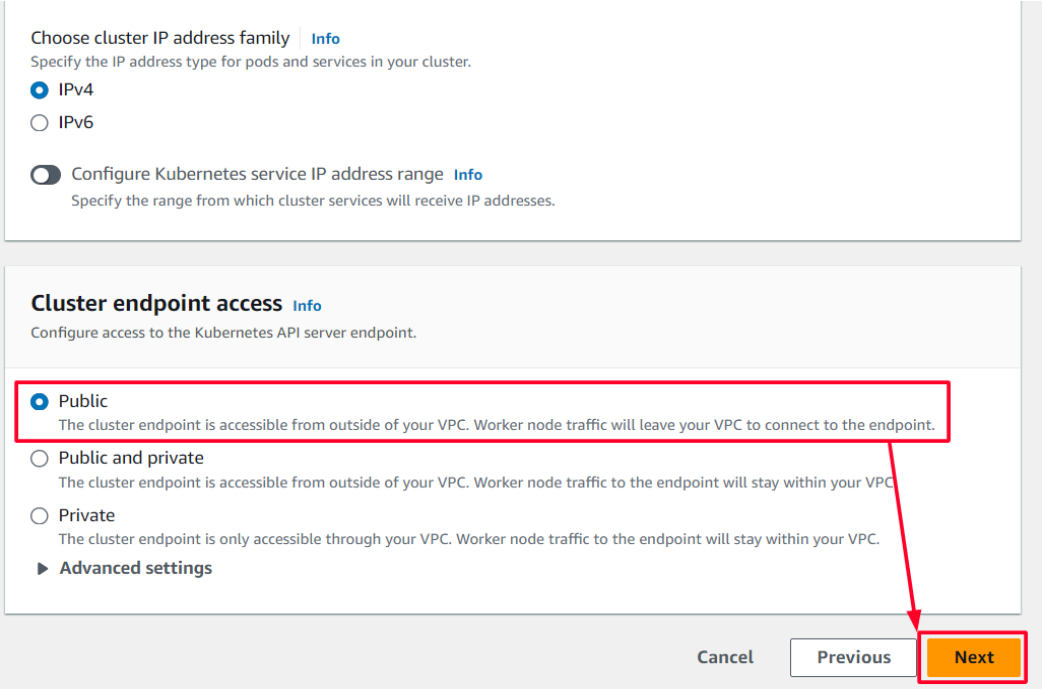
12.5.2. CAUSE:

The EKS cluster cannot be accessed from the client machine because the EKS cluster and the client machine's VPC are different.

12.5.3. FIX:

Under Networking settings, select **"Public"** for Cluster Endpoint Access to resolve this issue.

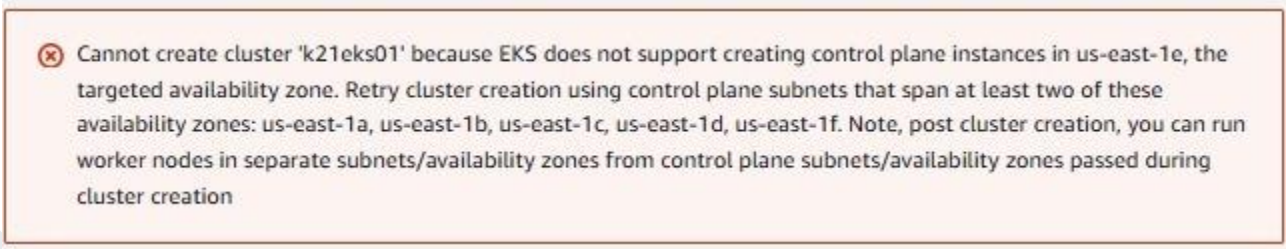
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



12.6. Cannot create cluster 'k21eks01' because EKS does not support creating control plane instances in us-east-1e, the targeted availability zone.

12.6.1. Issue

I encountered the following error while creating an EKS cluster.



12.6.2. Reason:

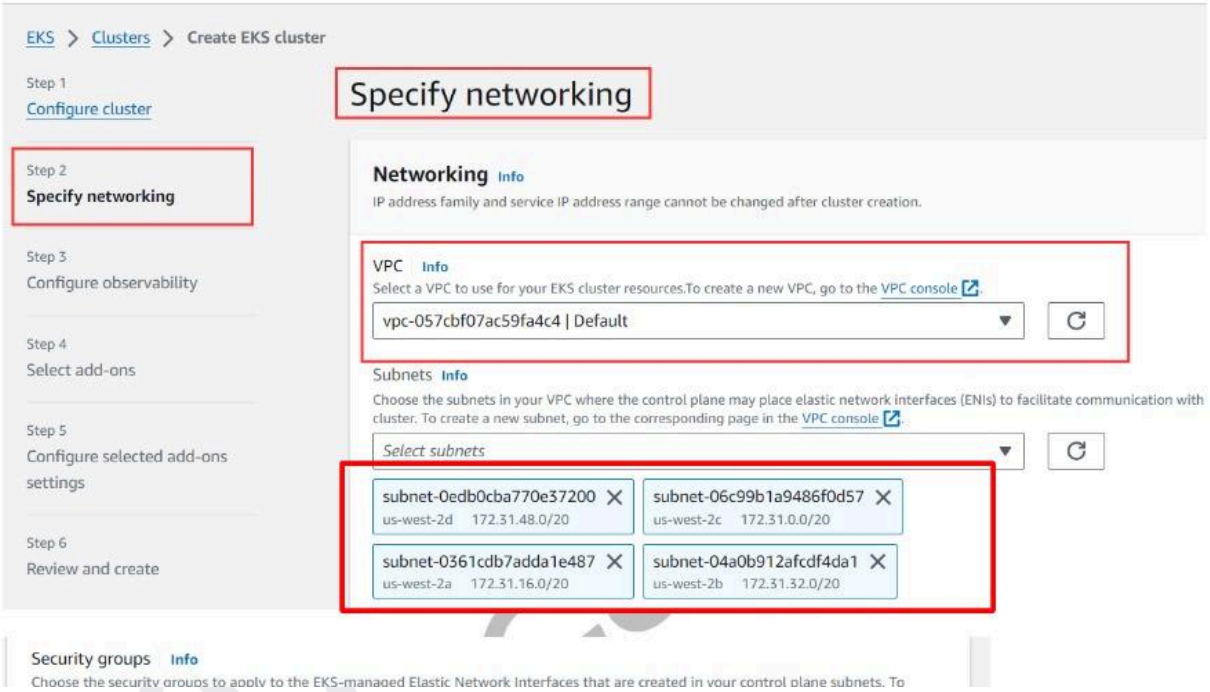
The error message in the image indicates that Amazon EKS (Elastic Kubernetes Service) does not support creating control plane instances in the us-east-1e availability zone, which was selected during the cluster creation process.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

12.6.3. Fix:

In the networking section, you should leave the subnet settings as the default. This will automatically select available subnets within the region.

2. In the **Step 2: Specify Networking** section, **do not make any changes to the VPC, subnet, or security groups; leave them as default.** In the **Cluster Endpoint Access** section, select **Public** and click on **Next.**



12.7. EKS Node Group Creation Failure

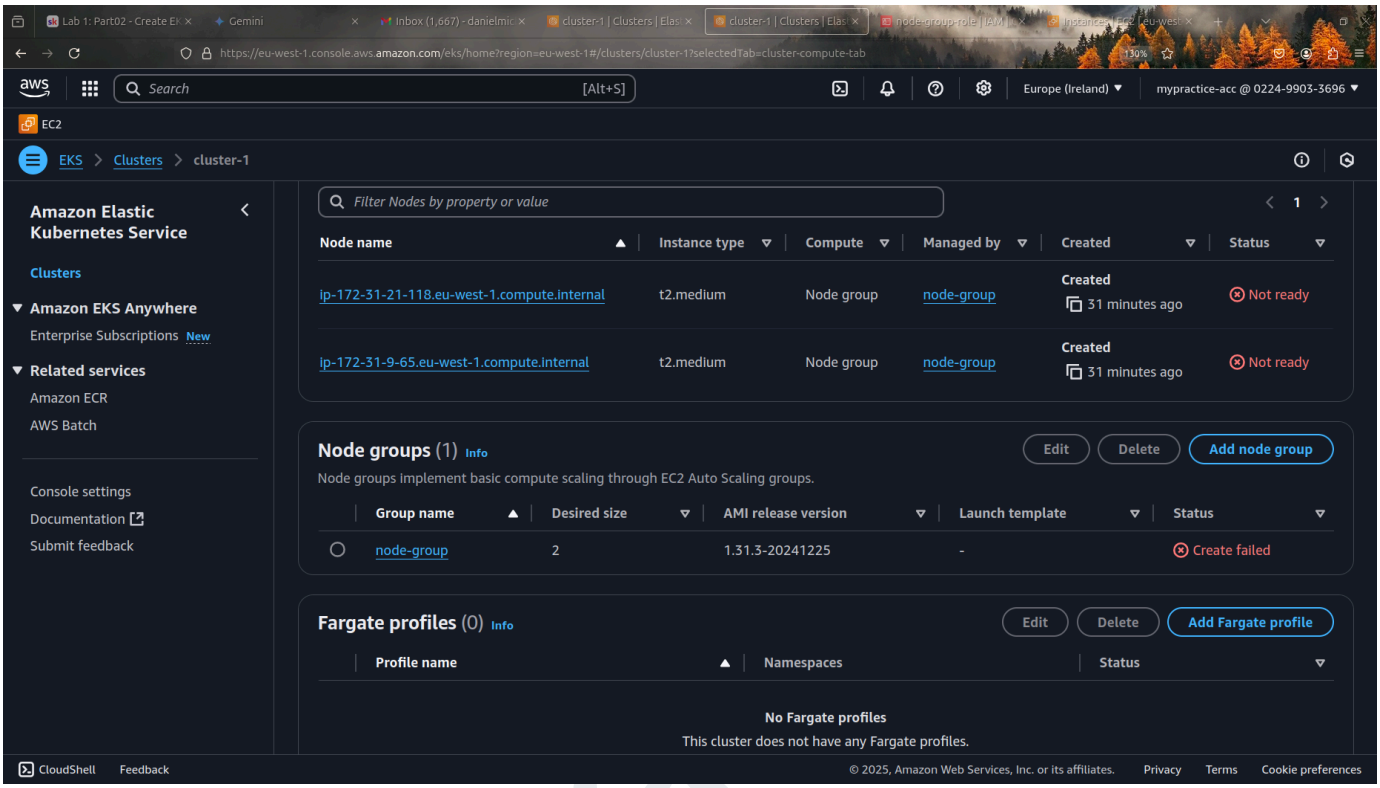
12.7.1. Issue:

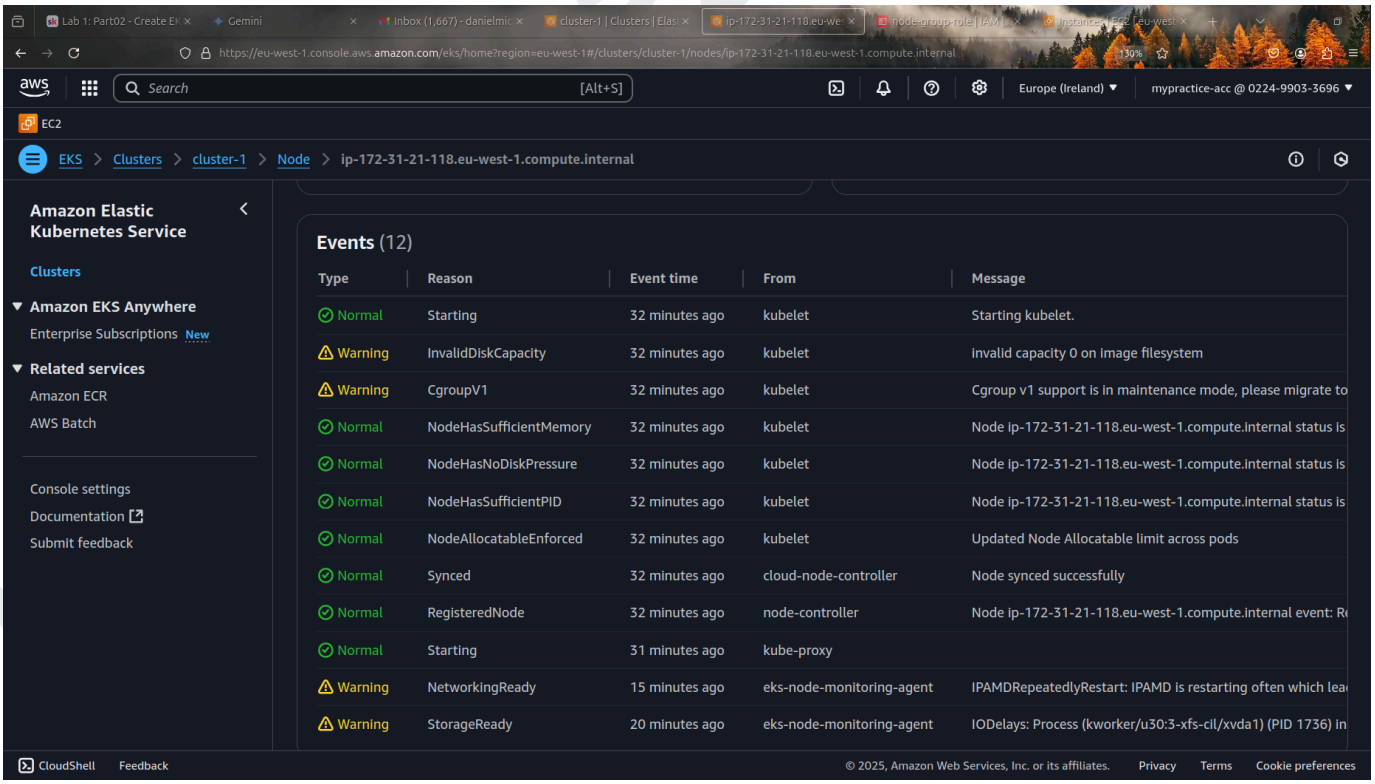
The EKS node group creation fails with errors indicating potential misconfiguration or resource-related issues. The specific errors highlighted include:

- `NetworkPluginNotReady`: Suggesting the CNI plugin is not initialized.
- `InvalidDiskCapacity`: Potential storage allocation issue for worker nodes.
- Node health issues: Nodes are in `Not Ready` state due to networking or disk capacity issues.

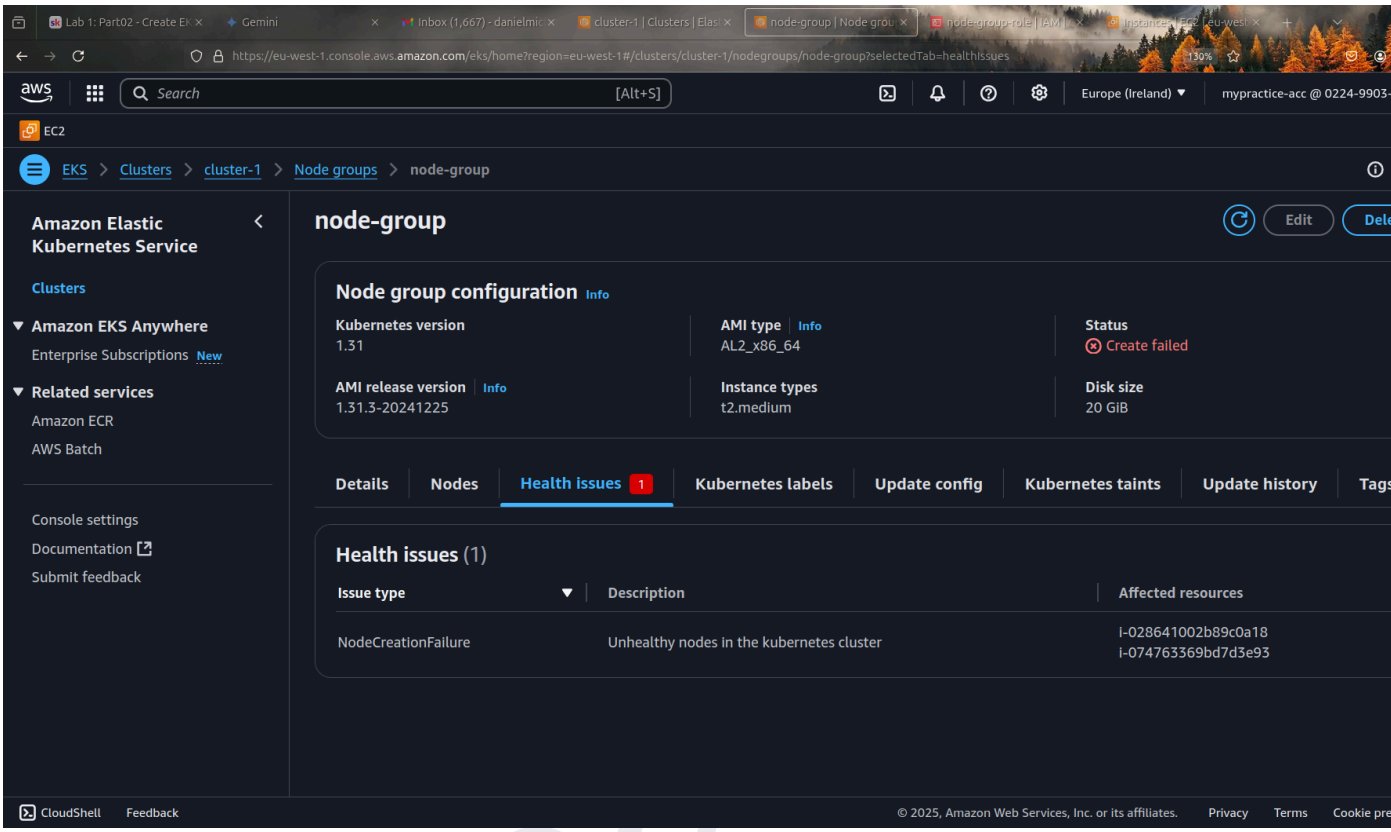
Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

K21Academy
Learn Cloud, Data & AI from Experts





Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.



12.7.2. CAUSE:

The error message in the image indicates that the likely causes for the failure are:

- **IAM Role Permissions:**

  The assigned IAM role for the node group may lack necessary policies, or the permissions may not be properly configured.

- **Networking Configuration:**

  The VPC CNI plugin (`aws-node`) might not be running or configured correctly.

  Subnets associated with the node group might not have sufficient IPs available or are improperly tagged.

- **Node Configuration:**

  The selected AMI version might not be compatible with the Kubernetes version of the cluster.

  Insufficient disk space or unsupported instance types (e.g., `t2.medium`) for the workload.

- **Cluster and Security Group Misconfigurations:**

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

- Security groups might not allow the required inbound/outbound traffic for node communication.
- Route tables and VPC endpoints may not be correctly set up.

12.7.3. Fix:

Step 1: Validate IAM Roles and Policies

Attach the following policies to the IAM role assigned to the node group:

- AmazonEKSWorkerNodePolicy
- AmazonEC2ContainerRegistryReadOnly
- AmazonEKS_CNI_Policy

If the issue persists, temporarily attach AdministratorAccess to test permissions.

Step 2: Check Node Group Configuration

Instance Type: Change the instance type to t3.medium or higher for better performance.

Disk Size: Increase the disk size of worker nodes to at least **50 GiB**.

AMI Version: Verify that the AMI version matches the Kubernetes version (e.g., 1.31 for your cluster).

Step 3: Fix Networking Issues

Run the following command to check if the aws-node pod is running in the kube-system namespace:

```
kubectl get pods -n kube-system
```

If it is not running, apply the latest CNI configuration:

```
kubectl apply -f https://raw.githubusercontent.com/aws/amazon-vpc-cni-k8s/release-1.12/config/master/aws-k8s-cni.yaml
```

Verify that the subnets associated with the node group are properly tagged:

- **Key**: kubernetes.io/cluster/<your-cluster-name>
- **Value**: shared or owned

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

Step 4: Verify Cluster and Security Group Settings

- Check the security group rules for the nodes:
  - **Inbound**: Allow HTTPS (443), kubelet (10250), and any custom ports required.
  - **Outbound**: Allow all traffic or the specific ports required for communication.
- Ensure the VPC route table has appropriate routes for internet connectivity.

Step 5: Inspect Logs and Node Readiness

Get detailed node status:

```
kubectl describe nodes
```

Check logs for failing pods:

```
kubectl logs <pod-name> -n kube-system
```

Inspect node health and errors in the AWS Console under the **Node Group Health Issues** section.

Step 6: Recreate the Node Group

- If the issue persists, delete the failed node group and recreate it with updated settings:
  - Choose a compatible instance type and AMI.
  - Ensure subnets, security groups, and roles are properly configured.

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.

13. SUMMARY

In this guide, we covered:

- ➢ Log in to the AWS Portal & Sign in using your account credentials.
- ➢ GUI Method: Create EKS Cluster using AWS Console:
  - Create EKS Cluster Master Node (Control Plane).
  - Create EKS Cluster Worker Nodes (Data Plane).
- ➢ Connect & Manage EKS Cluster:
  - Method 1: Connect & Manage the EKS Cluster on AWS Cloud Shell.
  - Method 2: Connect & Manage the EKS Cluster on the Linux Client Machine.
- ➢ Upgrade Amazon EKS Cluster
- ➢ Delete an EKS Cluster that was created using AWS Console.
- ➢ CLI Method: Create EKS Cluster using EKSCTL
  - Install EKSCTL.
  - Create EKS Cluster (Master & Worker Nodes) using EKSCTL.
- ➢ Delete an EKS Cluster that was created using EKSCTL.

K21Academy

Copyright© K21Academy | All Rights Reserved
Sharing, Reselling, or duplication of this content is strictly prohibited without K21Academy's written permission.